MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR MONOLITHIC AND MULTICHIP, SILICON 12 BIT DIGITAL-TO-ANALOG CONVERTERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic and multichip 12 bit digital-to-analog converters. Two product assurance classes and a choice of lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	D/A converter, 12 bit, external reference
02	D/A converter, 12 bit, internal reference
03	D/A converter, 12 bit, internal reference high speed
04	D/A converter, 12 bit, external reference high speed

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
J	D-3 (24-lead, 1/2" x 1 1/4"), dual-in-line package

1.3 Absolute maximum ratings.

	01, 02	03, 04
Supply voltage ($\pm V_S$)	±18 V dc	±18 V dc
Reference voltage, Hi input	$\pm V_S$	±12 V dc
Reference voltage, Lo input	$\pm V_S$	
Digital inputs	$\pm V_S$	-1, +7 V dc
Bipolar offset to analog common	$\pm V_S$	±12 V dc
10 V span R to analog common	$\pm V_S$	±12 V dc
20 V span R to analog common	$\pm V_S$	±24 V dc
DAC output voltage	$\pm V_S$, -5 V	-3, 12 V dc
Power ground to reference ground	---	±1 V dc
Lead temperature (soldering - 10 V seconds)	+300°C	
Junction temperature (T_J)	+175°C	
Storage temperature range	-65°C to +150°C	

1.4 Recommended operating conditions.

Supply voltage range (V_{CC}):	
Device type 01	4.5 V to 16.5 V
Device type 02	4.75 V to 15.75 V
Device type 03	13.5 V to 16.5 V
Device type 04	Not required.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Supply voltage range (V_{EE}):	
Device types 01, 03, and 04	-15 V ±1.5 V
Device type 02	-15 V ±.75 V
Reference voltage:	
Device types 01, 03, and 04	10.000 V
Device type 02	2.500 V
Output compliance voltage	0 V dc
Logic thresholds:	
TTL logic "0"	0.8 V dc maximum
TTL logic "1"	2.0 V dc minimum
CMOS logic "0" (devices 01 and 02)	30% V_{CC} maximum
CMOS logic "1" (devices 01 and 02)	70% V_{CC} minimum
Ambient operating temperature range (T_A)	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{J-C}	Maximum θ_{J-A}
Dual-in-line	J	1 W @ T_A = +125°C	40°C/W	50°C/W

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification For.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and 1.2.3 herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Functional schematic circuit. The schematic circuit shall be as specified on figure 2.

3.2.3 Case outline. The case outline shall be as specified in 1.2.3.

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TABLE I. Electrical performance characteristics.

Characteristic	Symbol	Conditions: See 3.4, figure 5, and table III, unless otherwise specified 1/ 2/	Device type	Limits Min	Limits Max	Unit
Monotonicity	---	Guaranteed by the bit linearity and major carry error tests	01,02, 03,04	12 12		Bits Bits
Supply current from V_{CC}	I_{CC}	V_{CC} = 15 V, All input bits = 10.5 V	01	+3	+18	mA
			02	+3	+20	mA
		All input bits = 2.4 V 8/	03	1	5	mA
Supply current from V_{EE}	I_{EE}	V_{CC} = 15 V, All input bits = 4.5 V	01,02, 03,04	-25 -18	-5 -5	mA mA
		All input bits = 0.8 V				
Logic "1" input current	I_{IH}	V_{CC} = 15 V, V_{in} (logic "1") = 10.5 V, Each input measured separately	01,02	-1	+100	µA
		V_{in} (logic "1") = 5.5 V	03,04	145	300	µA
Logic "0" input current	I_{IL}	V_{CC} = 15 V, V_{in} (logic "0") = 0 V, Each input measured separately	01,02, 03,04	-200 5	+1 100	µA µA
Full scale current	I_{FS}	All inputs logic "1", V_O = 0 V	01,02, 03,04	-2.7	-1.6	mA
Zero scale current (TTL)	I_{ZS1}	All inputs logic "0", V_O = 0 V, 7/ T_A = 25°C	01,02, 03,04	-2	2	LSB
		-55°C ≤ T_A ≤125°C	01,02, 03,04	-3	3	LSB
Zero scale current (CMOS)	I_{ZS2}	All input bits = 4.5 V, V_O = 0 V, 7/ V_{CC} = 15 V, T_A = 25°C	01,02	-2	2	LSB
		-55°C ≤ T_A ≤125°C	01,02	-3	3	LSB
Zero scale current drift	$\frac{dI_{ZS}}{dT}$	25°C to 125°C and 25°C to -55°C 7/	01,02, 03,04	-0.01	0.01	LSB/°C
Gain error (TTL)	A_{E1}	All inputs = Logic "1" 3/ 7/ T_A = 25°C	01,03, 04	-.20	+.20	%
			02	-.16	+.16	%
Gain error (CMOS)	A_{E2}	Logic "1" = 10.5 V, V_{CC} = 15 V; T_A = 25°C 3/ 7/ 10/	01	-.20	+.20	%
			02	-.16	+.16	%
Bipolar gain error	B_{PAE}	Over ±10 V output range T_A = 25°C 3/ 7/	01,03, 04	-.20	+.20	%
			02	-.16	+.16	%
Gain drift	$\frac{dA_{E1}}{dT}$	From 25°C to 125°C and 25°C to -55°C 7/	01	-5	+5	PPM /C°
			02,03	-30	+30	
			04	-10	10	
Reference output voltage	V_{REF}	All logic bits logic "0" V_{CC} = 15 V, V_{EE} = -15 V	02	2.45	2.55	V
			03	9.90	10.1	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions: See 3.4, figure 5, and table III, unless otherwise specified 1/ 2/	Device type	Limits Min	Limits Max	Unit
Bipolar offset error	B_{POE}	All inputs logic "0" 4/ 7/ T_A = 25°C	01,03, 04	-8	+8	LSB
			02	-6.5	6.5	LSB
Bipolar offset drift	$\frac{dB_{POE}}{dT}$	25°C to 125°C and 7/ 25°C to -55°C	01	-0.02	0.02	LSB/°C
			02,03	-0.12	0.12	LSB/°C
			04	-0.04	0.04	LSB/°C
Bipolar zero error (TTL)	B_{ZE}	Input bits = 4,000 (octal), 7/ T_A = 25°C	01	-6.5	6.5	LSB
			02	-6.5	6.5	LSB
			03,04	-6.0	6.0	LSB
Bipolar zero drift	$\frac{dB_{ZE}}{dT}$	25°C to 125°C and 7/ 25°C to -55°C	01	-0.02	0.02	LSB/°C
			02,03, 04	-0.04	0.04	LSB/°C
Power supply sensitivity from V_{CC} at full scale (TTL)	$+P_{SS1}$	V_{CC} = 5 ±0.5 V, T_A = 25°C 5/ V_{CC} = 5 ±0.25 V	01	-0.04	0.04	LSB/%PS
			02	-0.15	0.15	"
		V_{CC} = 5 ±0.5 V, -55°C < T_A <125°C 5/ V_{CC} = 5 ±0.25 V	01	-0.07	0.07	"
			02	-0.25	0.25	"
Power supply sensitivity from V_{CC} at full scale (CMOS)	$+P_{SS2}$	V_{CC} = 15 ±1.5 V, T_A = 25°C 5/ V_{CC} = 15 ±.75 V	01	-0.04	0.04	"
			02	-0.15	0.15	"
		V_{CC} = 15 ±1.5 V, 12 ±.6 V	03	-0.04	0.04	"
		V_{CC} = 15 ±1.5 V, -55°C < T_A <125°C, 5/, V_{CC} = 15 ±.75 V	01	-0.07	0.07	"
			02	-0.25	0.25	"
		V_{CC} = 15 ±1.5 V, 12 ±.6 V	03	-0.08	0.08	"
Power supply sensitivity from V_{EE} at full scale	$-P_{SS}$	V_{EE} = -15 ±1.5 V, T_A = 25°C 5/ V_{EE} = -15 ±.75 V	01	-0.07	0.07	"
			02	-0.25	0.25	"
		V_{EE} = -15 ±1.5 V, -12 ±.6 V	03,04	-0.1	0.1	"
		V_{EE} = -15 ±1.5 V, -55°C < T_A <125°C 5/ V_{EE} = -15 ±.75 V	01	-0.15	0.15	"
			02	-0.5	0.5	"
		V_{EE} = -15 ±1.5 V, -12 ±.6 V	03,04	-0.2	0.2	"
Bit linearity error	LE	Subgroups 1, 2, and 3 abbrev. codes test Subgroups 7 and 8 all codes test 6/ 7/	01,02, 03,04	-0.5	0.5	LSB
Major carry error (differential linearity)	MCE	The difference between adjacent codes at all major transitions (i.e. 0111 1111 1111 to 1000 0000 0000). 5/ 7/	01,02, 03,04	-1.0	+1.0	LSB
Output current settling time, zero to full-scale	t_{SLH}	All inputs switched simultaneously. Measure the time to settle to within 1/2 LSB of the final value T_A = 25°C, figures 6 and 7	01,02	---	3.0	µs
			03	---	400	ns
			04	---	600	ns
Output current settling time, full-scale to zero	t_{SHL}	All inputs switched simultaneously. Measure the time to settle to within 1/2 LSB of the final value. T_A = 25°C, figures 6 and 7	01,02	---	3.0	µs
			03	---	400	ns
			04	---	600	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions: See 3.4, figure 5, and table III, unless otherwise specified 1/ 2/	Device type	Limits Min	Limits Max	Unit
Reference input resistance	Ri	V_{REF} = 10 V, T_A = 25°C figure 8	01,03, 04	15	25	kΩ
Output noise voltage	N_O	V_{CC} = 15 V, BW = 10 Hz to 100 kHz T_A = 25°C, figure 9	01	---	60	µVrms
			02	---	100	"
			03	---	80	"
			04	---	60	"

1/ The output compliance voltage range (see 6.4) may vary for each vendor. Devices with a definite output resistance will draw additional current that is equal to the output compliance voltage divided by the device output resistance.

2/ See paragraph 6.4 for symbols and definitions.

3/ The gain error of a 12 bit D/A converter in % corresponds to a zero scale to full-scale range error in LSB units. (i.e. ±0.2% x 4096 LSB/100% = 8.2 LSB)

4/ B_{POE} is the error in the -10 V output voltage when the device is operated in the bipolar mode with all bits off.

5/ This test is performed in the unipolar mode over a 0 to 10 V range. The unipolar scale factor is 4096 LSB/10 V = 410 LSB/V. The bipolar factor is 4096 LSB/20 V = 205 LSB/ V.

6/ The manufacturer has the option of testing devices at all digital input codes or an abbreviated series of codes. The abbreviated test should have a sufficient number of codes with or without limit guard banding such that an all codes test will not fail a device which passed the abbreviated test.

7/ These parameters are to be tested in a power stable temperature condition (see MIL-STD-883, 4.5.8.1.3). All other parameters will be tested using low duty cycle pulse testing (i.e., $T_A = T_J = T_C$ (see MIL-STD-883, 4.5.8.1.1 and 4.5.8.1.2)).

8/ V_{CC} not used for device 04.

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3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. The electrical performance characteristics of table I apply over the full recommended ambient temperature range unbless otherwise specified. The following conditions shall also apply unless otherwise specified:

V_{CC} = 5.0 V for device types 01 and 02; 15 V for device type 03
V_{EE} = -15.0 V
V_{ref} = 10.000 V
Logic "0" = 0.8 V
Logic "1" = 2.0 V
Temperature range = -55°C to +125°C.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in a microcircuit group number 56 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*, 2, 3, 4	1*, 2, 3, 4
Group A test requirements (method 5005)	1,2,3,4,5,6, 7, 8, 12	1,2,3,4,5,6, 7, 8
Group C end-point and group B, class S, electrical parameters (method 5005)	1, 2, 3, 4, 5, 6, and table IV delta limits	1, 4, and table IV delta limits
Additional electrical subgroups for group C periodic inspections	N/A	12
Group D end-point electrical parameters (method 5005)	1, 2, 3, 4, 5, 6	1,4

*PDA applies to subgroups 1 (see 4.2d).

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Device type 01

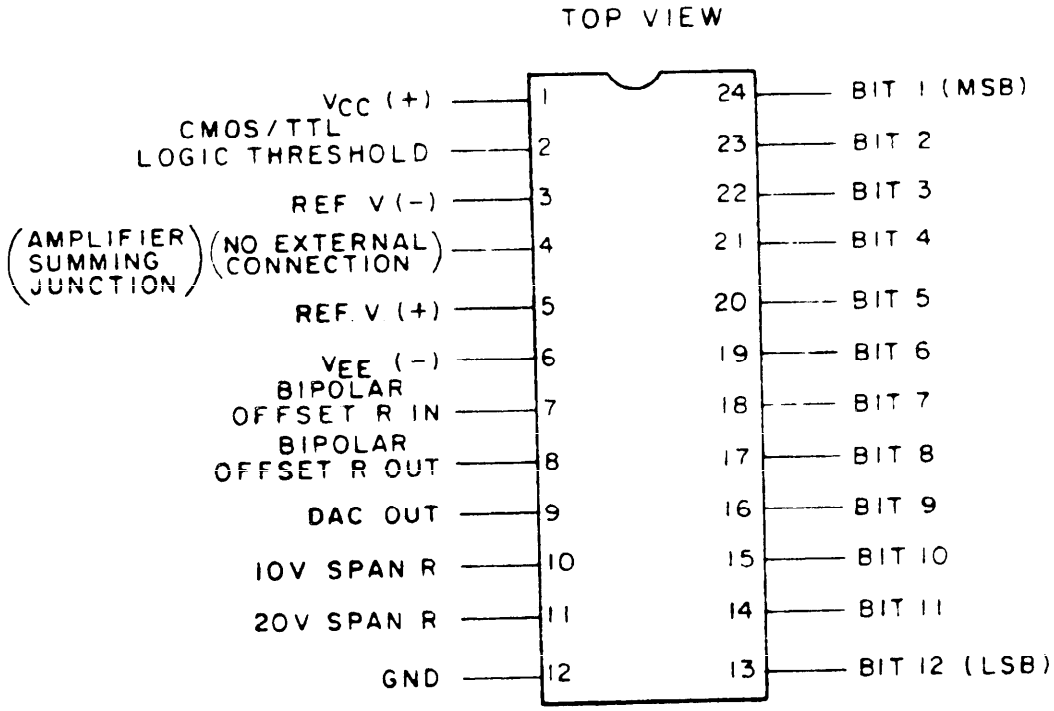


FIGURE 1 Terminal connections.

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Device type 02

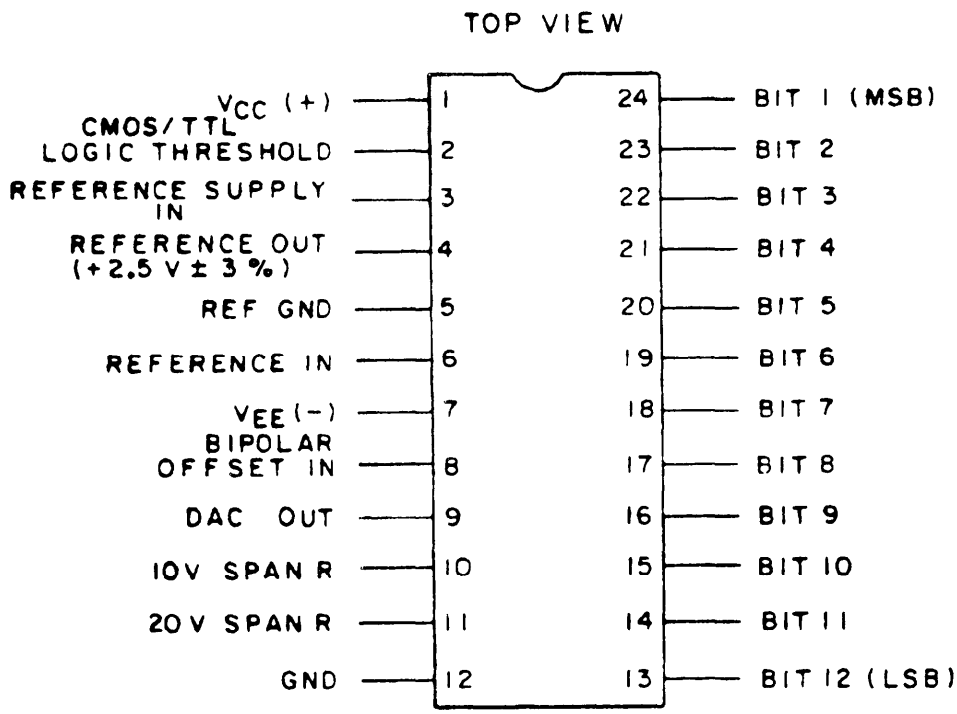


FIGURE 1 Terminal connections - Continued.

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Device type 03

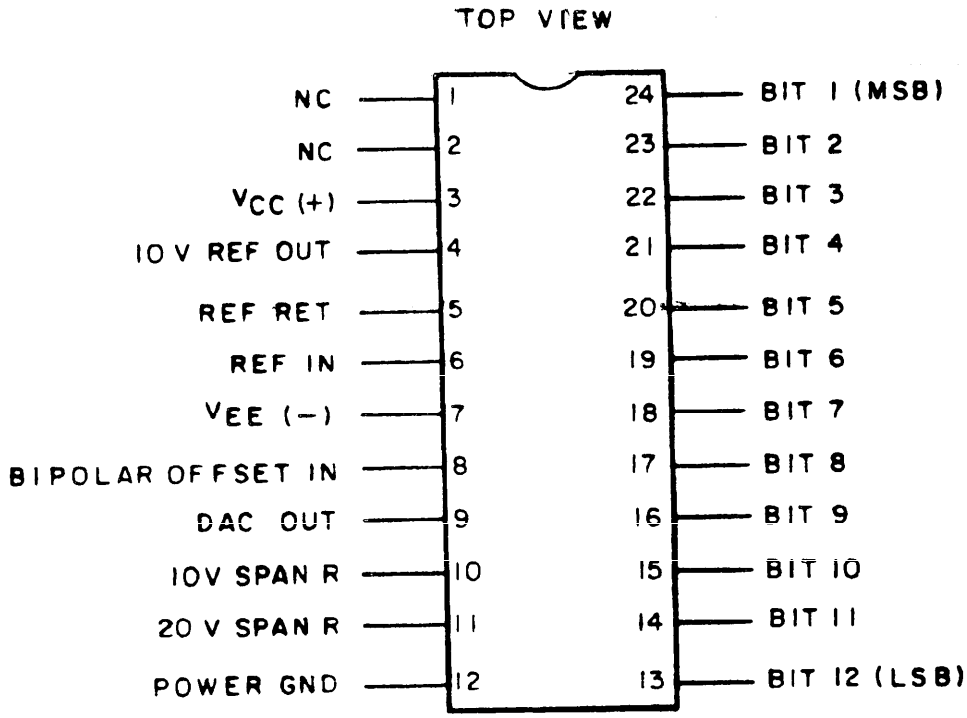


FIGURE 1. Terminal connections - Continued.

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Device type 04

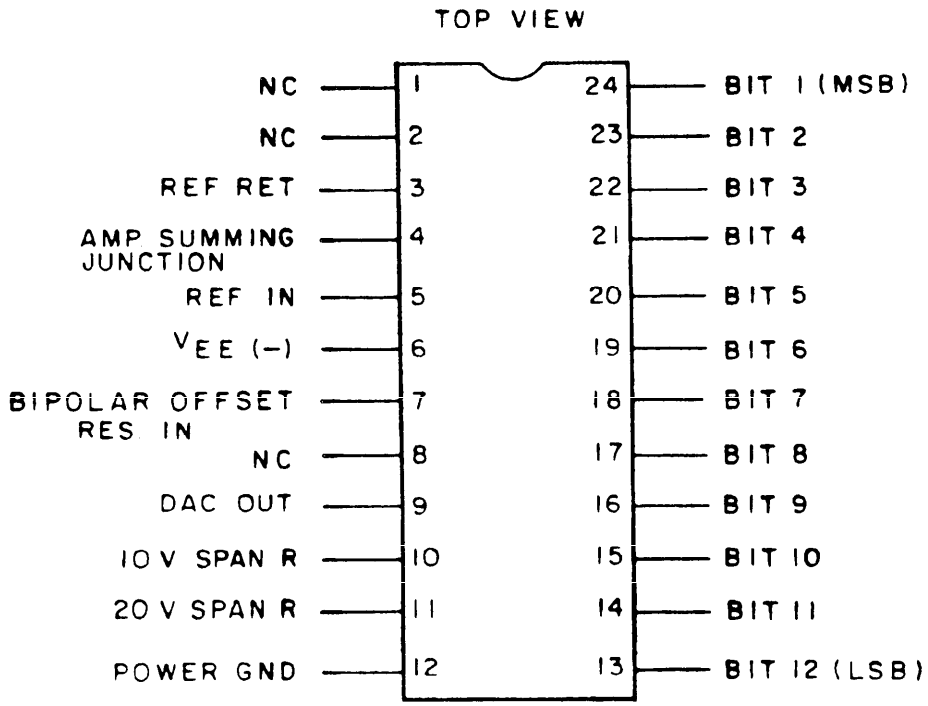
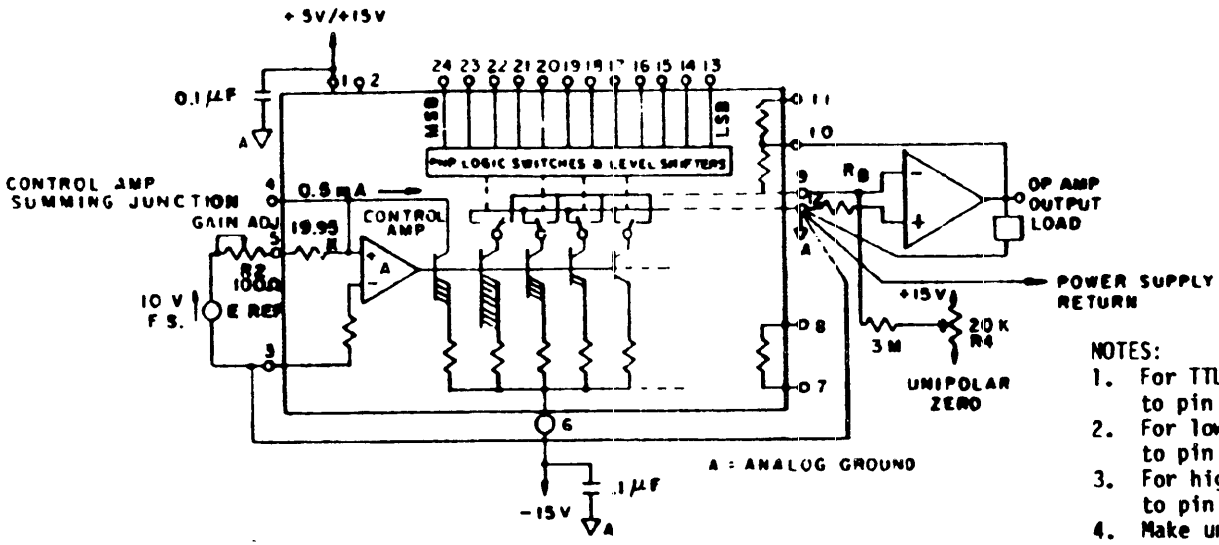


FIGURE 1. Terminal connections - Continued.

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NOTES:
1. For TTL and DTL compatibility, connect +5 volts to pin 1 and leave pin 2 open.
2. For low voltage CMOS compatibility, connect +5 volts to pin 1 and short pin 2 to pin 1.
3. For high voltage CMOS compatibility, connect +15 volts to pin 1 and short pin 2 to pin 1.
4. Make unipoint ground at device pin 12.

Step I. Unipolar zero adjust

Turn all bits OFF. Adjust R4 until op amp is 0 volts. R1 and connection between pins 8 and 9 are not needed.

Step II. Gain adjust

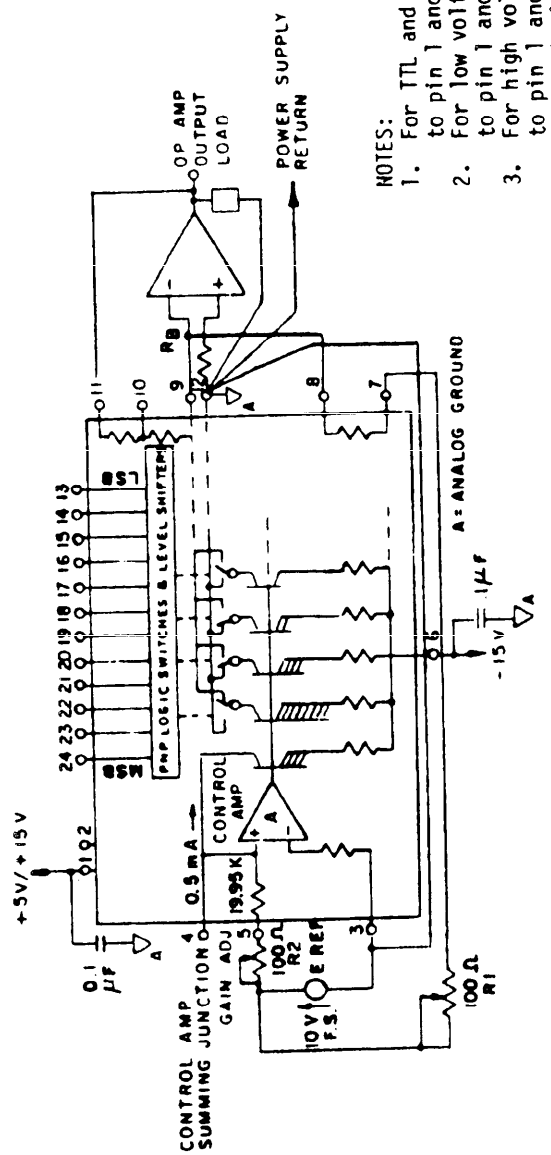
Turn all bits ON. Adjust R2 for an output of:
+4.9988 V for 0 to +5.0 V range
+9.9976 V for 0 to +10.0 V range

Output range	Connections	R_B
0 V to +5.0 V	Short pin 11 to pin 9	2.49 kΩ
0 V to +10.0 V	N.C. to pin 11	4.99 kΩ

Unipolar mode for device type 01

FIGURE 2. Functional schematics with application notes.

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NOTES:
1. For TTL and DTL compatibility, connect +5 volts to pin 1 and leave pin 2 open.
2. For low voltage CMOS compatibility, connect +5 volts to pin 1 and short pin 2 to pin 1.
3. For high voltage CMOS compatibility, connect +15 volts to pin 1 and short pin 2 to pin 1.
4. Make unipoint ground at device pin 12.

Step I. Bipolar offset adjust

Turn all bits OFF. Adjust R1 until op amp output is:
-2.5 V for ±2.5 V range
-5.0 V for ±5.0 V range
-10.0 V for ±10.0 V range

Output range	Connections	OP amp output to:
-2.5 to +2.5 V	Short pin 11 to pin 9	Pin 10
-5.0 to +5.0 V	N.C. to pin 11	Pin 10
-10.0 to +10.0 V	N.C. to pin 10	Pin 11

Step II. Gain adjust (bipolar zero)

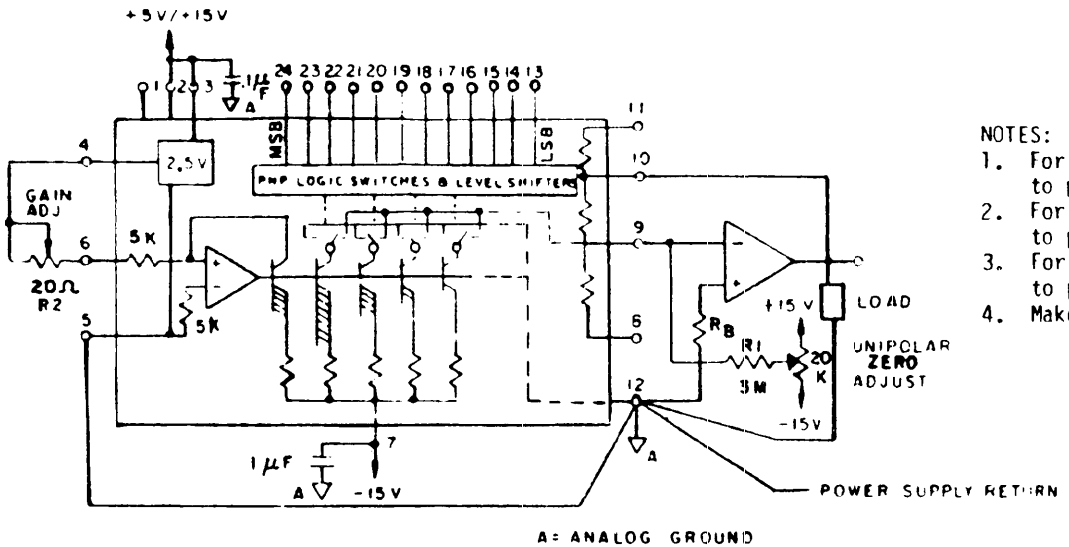
Turn bit 1 (MSB) ON; bits 2 through 12 OFF. Adjust R2 until op amp output is 0 V.

R_B
1.24 kΩ
1.69 kΩ
2.00 kΩ

Bipolar mode for device type 01

FIGURE 2. Functional schematics with application notes - Continued.

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NOTES:
1. For TTL and DTL compatibility, connect +5 volts to pin 1 and leave pin 2 open.
2. For low voltage CMOS compability, connect +5 volts to pin 1 and short pin 2 to pin 1.
3. For high voltage CMOS compatibility, connect +15 volts to pin 1 and short pin 2 to pin 1.
4. Make unipoint ground connection at device pin 12.

Step I. Unipolar zero adjust

Turn all bits OFF and adjust R_1 until op amp output is 0 V.

Step II. Gain adjust

Turn all bits ON. Adjust R_2 until op amp output is:

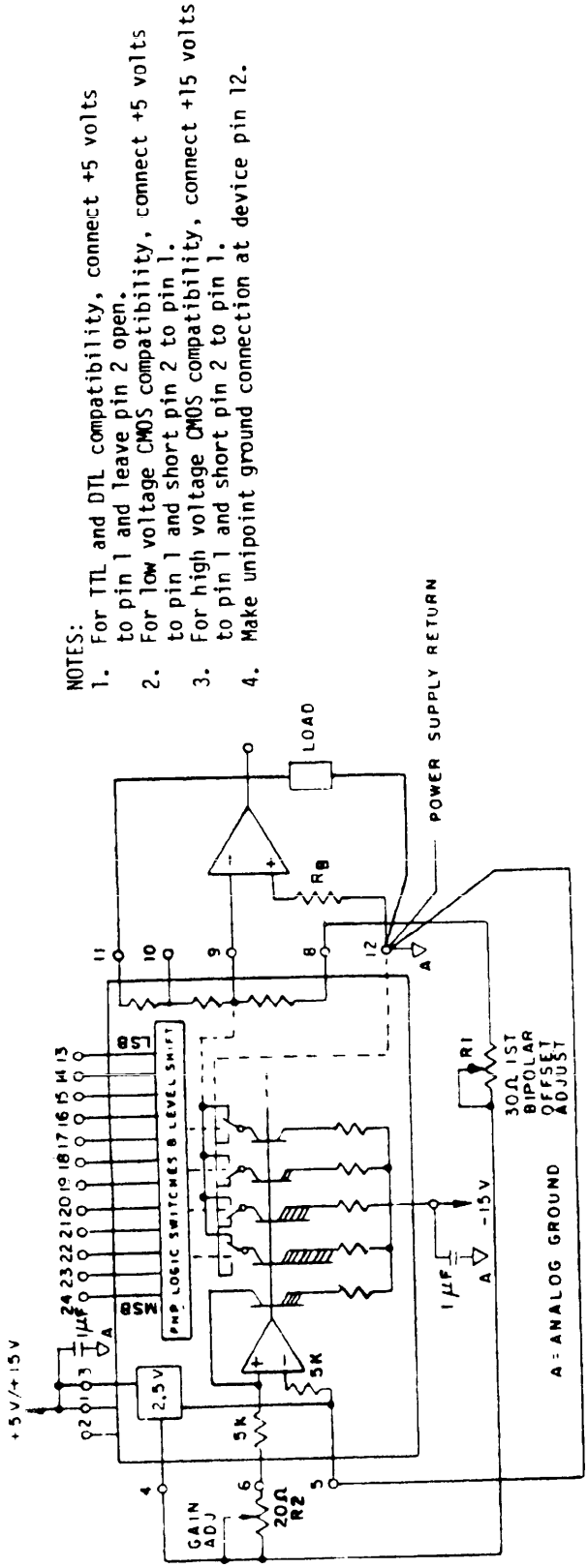
4.9988 V for +5 V range
9.9976 V for +10 V range

Output range	Connections	R_B
0 V to +5.0 V	Short pin 11 to pin 9	2.49 kΩ
0 V to +10.0 V	N.C. to pin 11	4.99 kΩ

Unipolar mode for device type 02

FIGURE 2. Functional schematics with application notes - Continued.

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NOTES:
1. For TTL and DTL compatibility, connect +5 volts to pin 1 and leave pin 2 open.
2. For low voltage CMOS compatibility, connect +5 volts to pin 1 and short pin 2 to pin 1.
3. For high voltage CMOS compatibility, connect +15 volts to pin 1 and short pin 2 to pin 1.
4. Make unipoint ground connection at device pin 12.

Step I. Bipolar offset adjust

Turn all bits OFF and adjust R_3 until op amp output is:

-2.5000 V for ±2.5 V range
-5.0000 V for ±5 V range
-10.0000 V for ±10 V range

Output range	Connections
-2.5 V to +2.5 V	Short pin 11 to pin 9
-5.0 V to +5.0 V	N.C. to pin 11
-10.0 V to +10.0 V	N.C. to pin 10

Step II. Gain adjust (bipolar zero)

Turn bit 1 ON. Adjust R_2 until op amp output is 0 V.

OP amp output to:	R_B
Pin 10	1.24 kΩ
Pin 10	1.69 kΩ
Pin 11	2.00 kΩ

Bipolar mode for device type 02

FIGURE 2. Functional schematics with application notes - Continued.

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0 V/10 V UNIPOLAR CONFIGURATION

This configuration will provide a unipolar 0 to +10 volt output range. In this mode, the bipolar terminal, pin 8, should be grounded if not used for trimming.

STEP I. ZERO ADJUST

Turn all bits OFF and adjust zero trimmer R1, until the output reads 0.000 volts (1 LSB = 2.44 mV). In most cases this trim is not needed, but pin 8 should then be connected to pin 12.

STEP II. GAIN ADJUST

Turn all bits ON and adjust 100Ω gain trimmer R2, until the output is 9.9976 volts. (Full scale is adjusted to 1 LSB less than nominal full scale of 10,000 volts.) If a 10.2375 V full scale is desired exactly 2.5 mV/bit), insert a 120Ω resistor in series with the gain resistor at pin 10 to the op amp output.

-5 V/5 V BIPOLAR CONFIGURATION

This configuration will provide a bipolar output voltage from -5.000 to +4.9976 volts, with positive full scale occurring with all bits ON (all 1's).

STEP I. OFFSET ADJUST

Turn OFF all bits. Adjust 100Ω trimmer R1 to give -5.000 volts output.

STEP II. GAIN ADJUST

Turn ON all bits. Adjust 100Ω gain trimmer R2 to give reading of +4.9976 volts.

-10 V/10 V BIPOLAR CONFIGURATION

This configuration will provide a bipolar output voltage from -10.000 to +9.9951 volts with positive full scale occuring with all bits ON (all 1's).

STEP I. OFFSET ADJUST

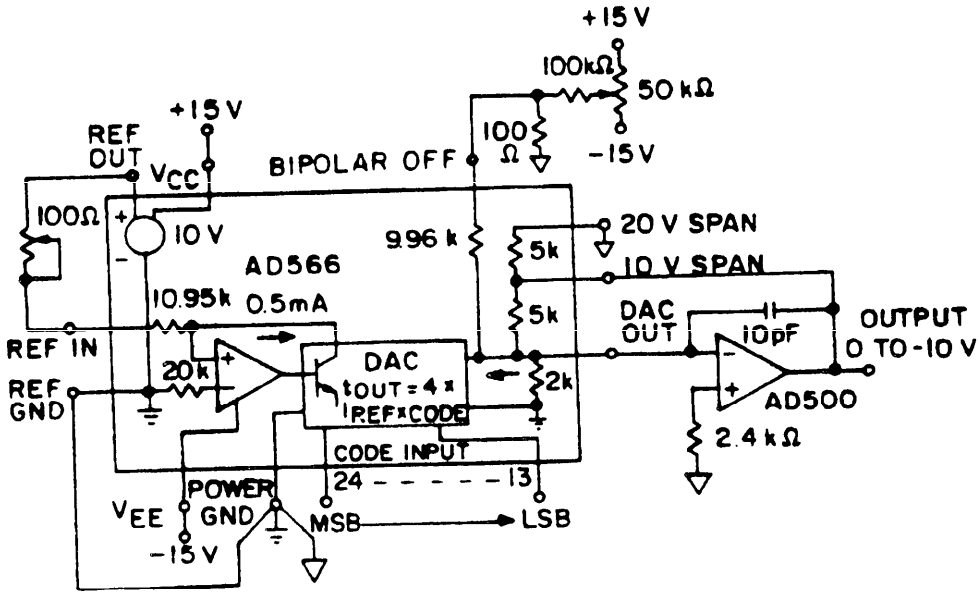
Turn OFF all bits. Adjust 100Ω trimmer R1 to give -10.000 volts output.

STEP II. GAIN ADJUST

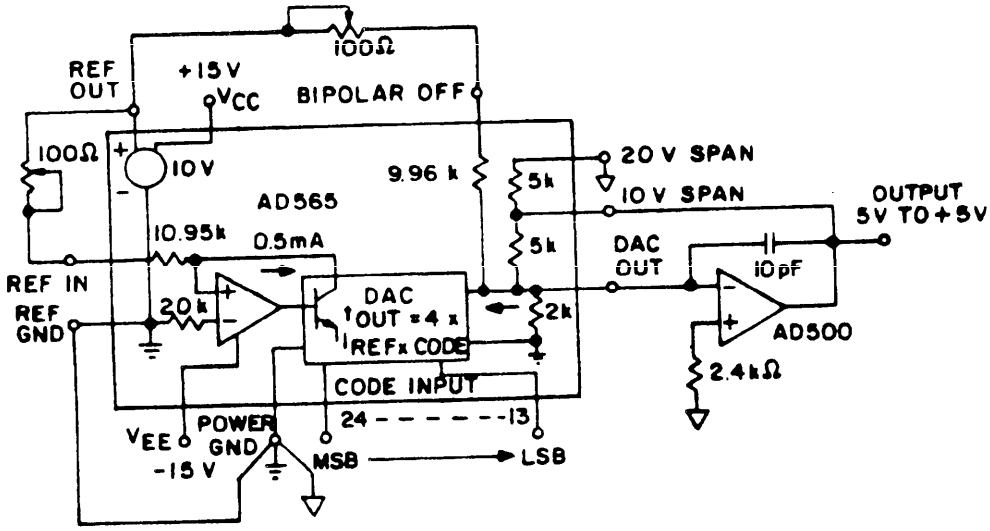
Turn ON all bits. Adjust 100Ω gain trimmer R2 to give a reading of +9.9951 volts.

FIGURE 2. Functional schematic with application notes for device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.



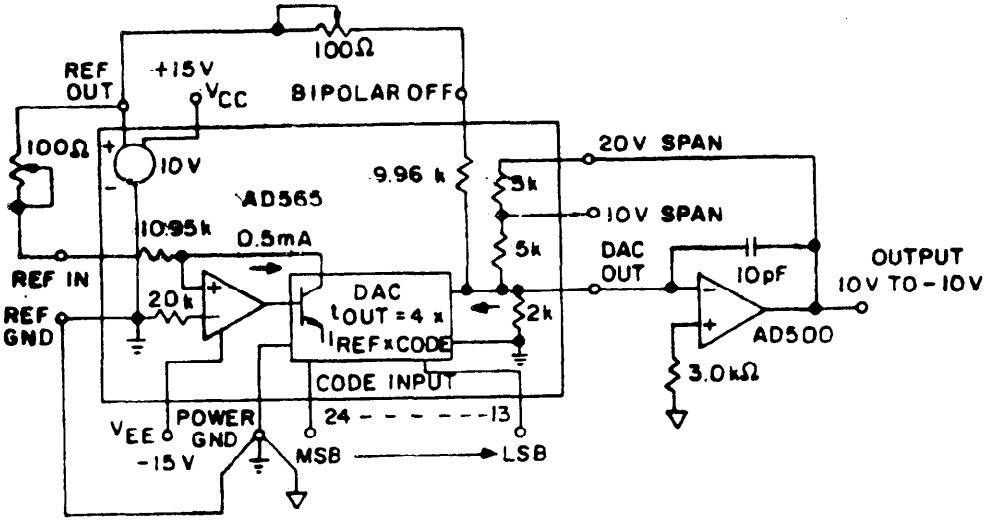
0 TO +10V UNIPOLAR VOLTAGE OUTPUT



±5 V BIPOLAR VOLTAGE OUTPUT

FIGURE 2. Functional schematic with application notes for device type 03 - Continued.

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±10V BIPOLAR VOLTAGE OUTPUT

FIGURE 2. Functional schematic with application notes for device type 03 - Continued.

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0 V/10 V UNIPOLAR CONFIGURATION

This configuration will provide a unipolar 0 to +10 volt output range. In this mode, the bipolar terminal, pin 7, should be grounded if not used for trimming.

STEP I. ZERO ADJUST

Turn all bits OFF and adjust zero trimmer R1, until the output reads 0.000 volts (1 LSB = 2.44 mV). In most cases this trim is not needed, but pin 7 should then be connected to pin 12.

STEP II. GAIN ADJUST

Turn all bits ON and adjust 100Ω gain trimmer R2, until the output is 9.9976 volts. (Full scale is adjusted to 1 LSB less than nominal full scale of 10,000 volts.) If a 10.2375 V full scale is desired exactly 2.5 mV/bit), insert a 120Ω resistor in series with the gain resistor at pin 10 to the op amp output.

-5 V/5 V BIPOLAR CONFIGURATION

This configuration will provide a bipolar output voltage from -5.000 to +4.9976 volts, with positive full scale occurring with all bits ON (all 1's).

STEP I. OFFSET ADJUST

Turn OFF all bits. Adjust 100Ω trimmer R1 to give -5.000 volts output.

STEP II. GAIN ADJUST

Turn ON all bits. Adjust 100Ω gain trimmer R2 to give reading of +4.9976 volts.

-10 V/10 V BIPOLAR CONFIGURATION

This configuration will provide a bipolar output voltage from -10.000 to +9.9951 volts with positive full scale occuring with all bits ON (all 1's).

STEP I. OFFSET ADJUST

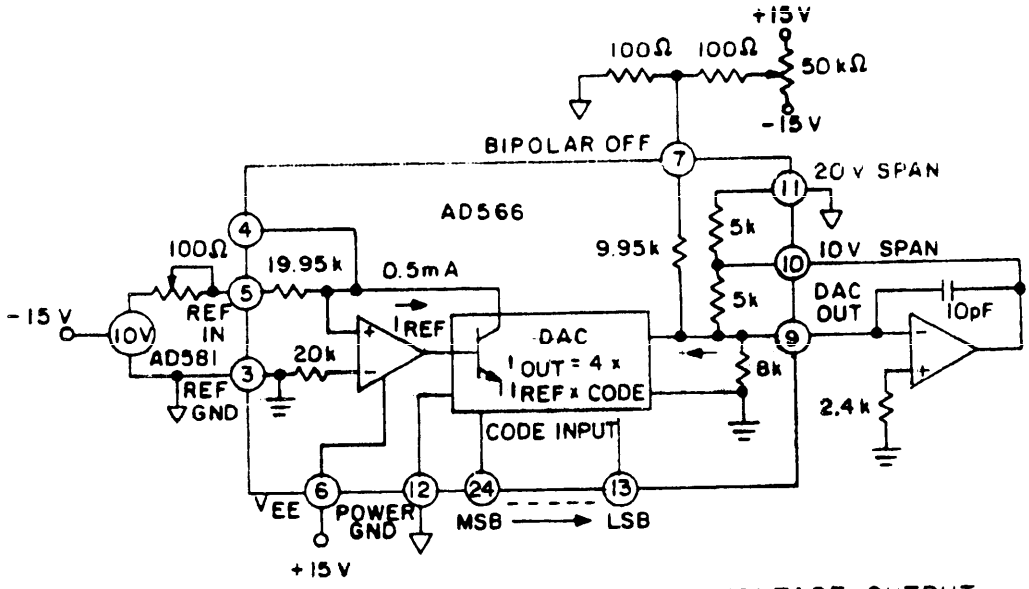
Turn OFF all bits. Adjust 100Ω trimmer R1 to give -10.000 volts output.

STEP II. GAIN ADJUST

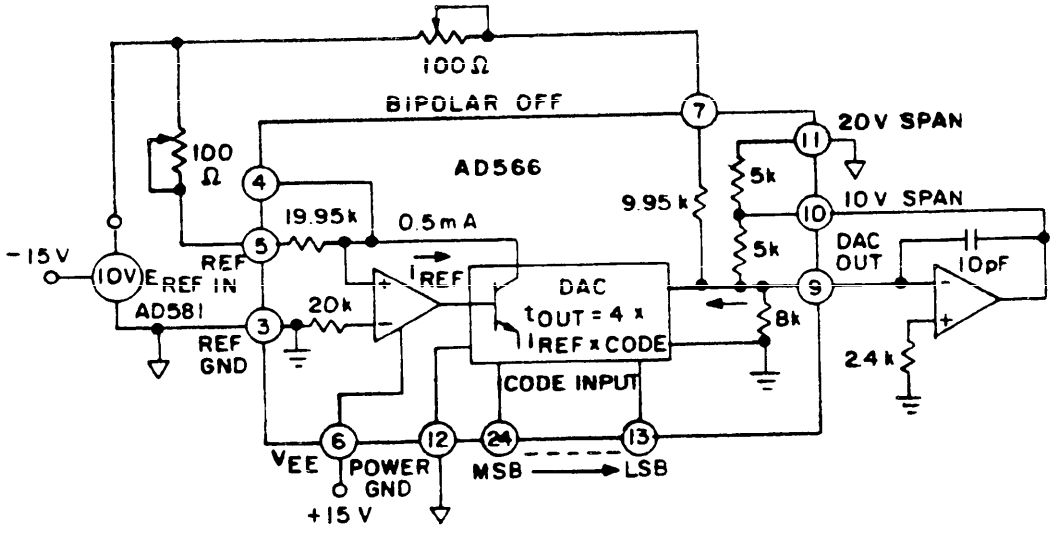
Turn ON all bits. Adjust 100Ω gain trimmer R2 to give a reading of +9.9951 volts.

FIGURE 2. Functional schematic with application notes for device type 04.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
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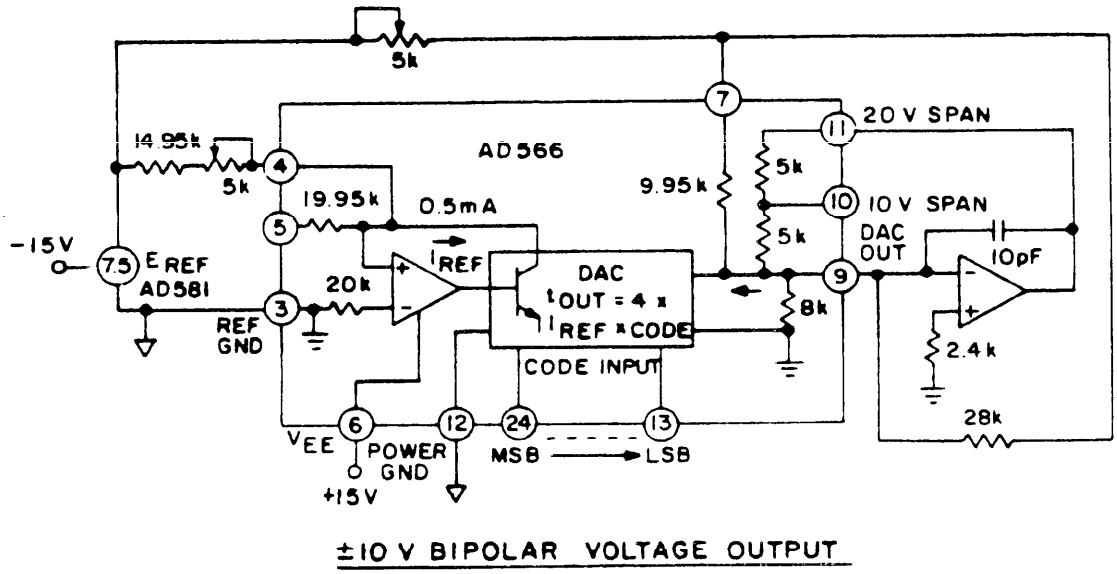
0 TO +10V UNIPOLAR VOLTAGE OUTPUT



±5 V BIPOLAR VOLTAGE OUTPUT

FIGURE 2. Functional schematic with application notes for device type 04 - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:26Z
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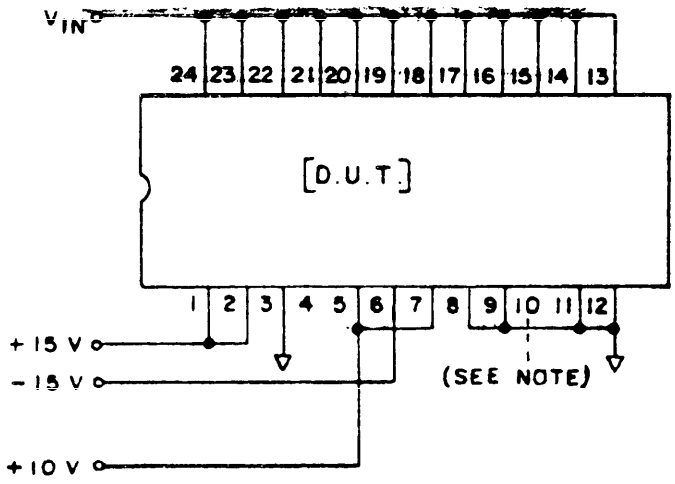


±10 V BIPOLAR VOLTAGE OUTPUT

FIGURE 2. Functional schematic with application notes for device type 04 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
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Device type 01

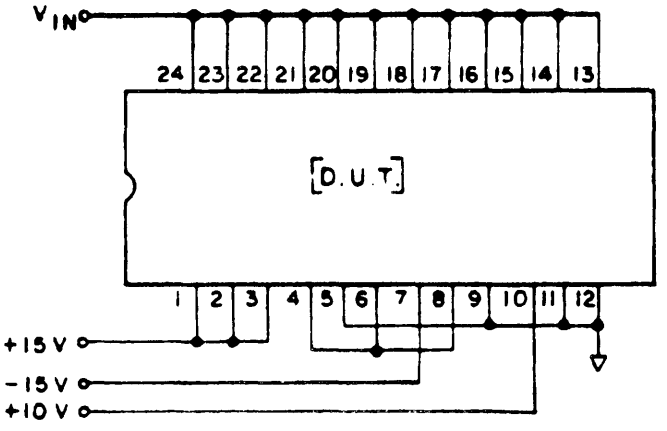


NOTE: For device type 01, V_{IN} = 0 volts to 15 volts peak to peak square wave @ fo = 5 ±1 Hz, and pin 10 shall be connected to +10 volts.

FIGURE 3. Test circuit, burn-in and steady state test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
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Device type 02

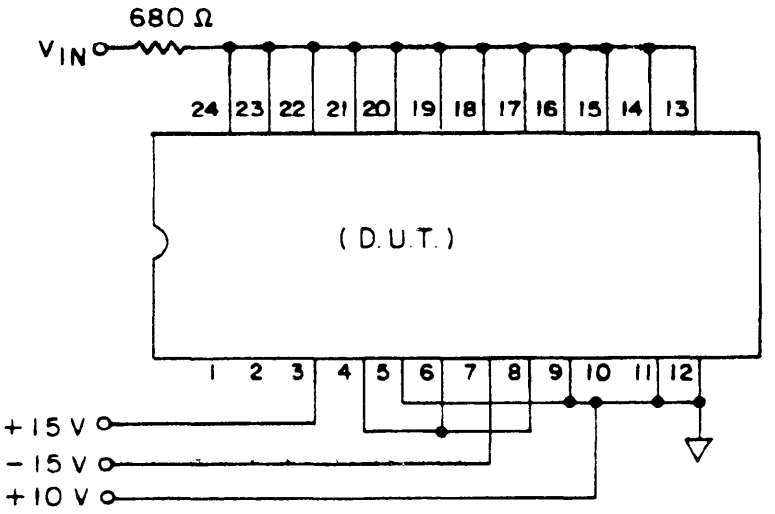


V_{IN} = 0 volt to 15 volts peak to peak square wave @ fo = 5 ±1 Hz.

FIGURE 3. Test circuit, burn-in and steady state test - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 03

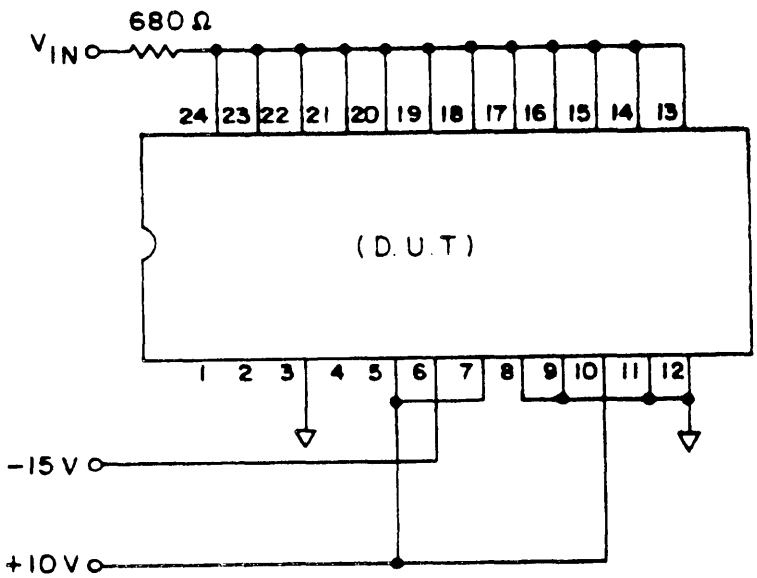


NOTE: For device type 03, V_{IN} = 0 V to 5.0 V peak to peak square wave at fo = 5 ±1 Hz.

FIGURE 3. Test circuit, burn-in, and operating life test - Continued.

Source: https://assist.dla.mil -- Downloaded: 202640-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 04

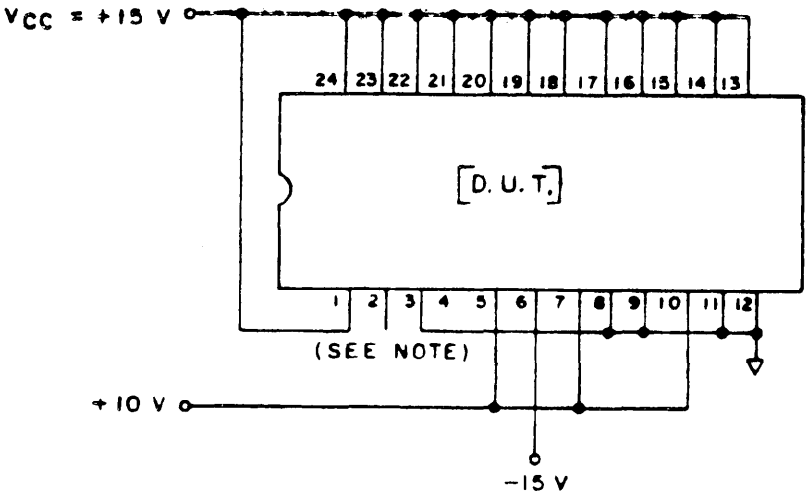


NOTE: For device type 04, V_{IN} = 0 V to 5.0 V peak to peak square wave at fo = 5 ±1 Hz.

FIGURE 3. Test circuit, burn-in, and operating life test - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 01



NOTE: For device type 01, pin 2 shall be V_{CC}

FIGURE 4. Test circuit, burn-in, (steady state power and reverse bias and accelerated) and steady state life.

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Device type 02

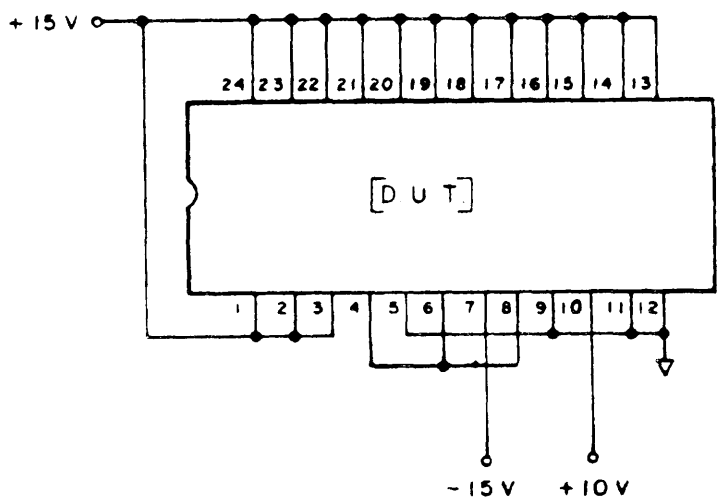


FIGURE 4. Test circuit, burn-in (steady state power and reverse bias and accelerated) and steady state life - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 03

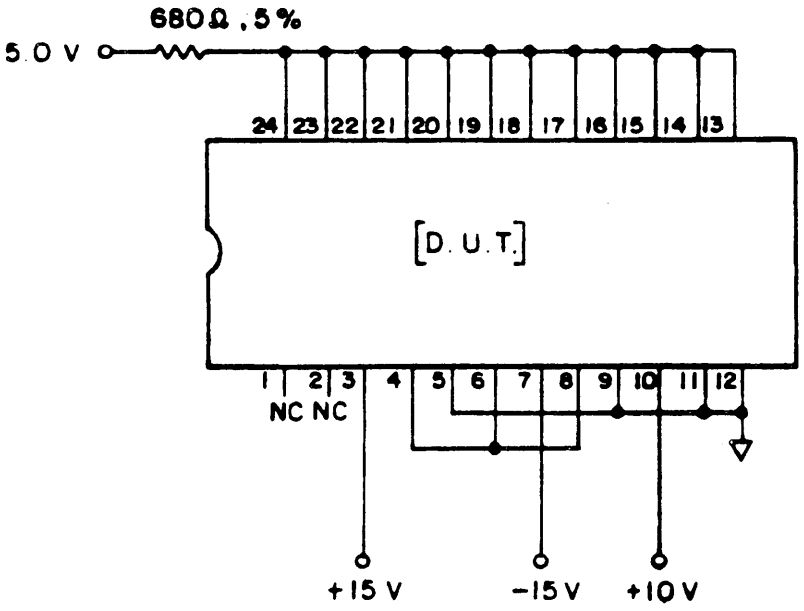


FIGURE 4. Test circuit, burn-in (steady state power and reverse bias) and operating life - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 04

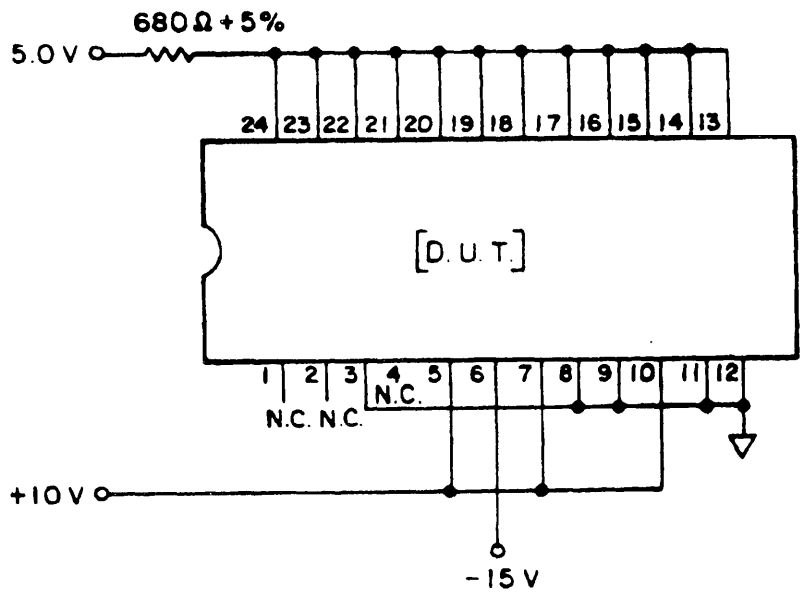


FIGURE 4. Test circuit, burn-in (steady state power and reverse bias) and operating life - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 01

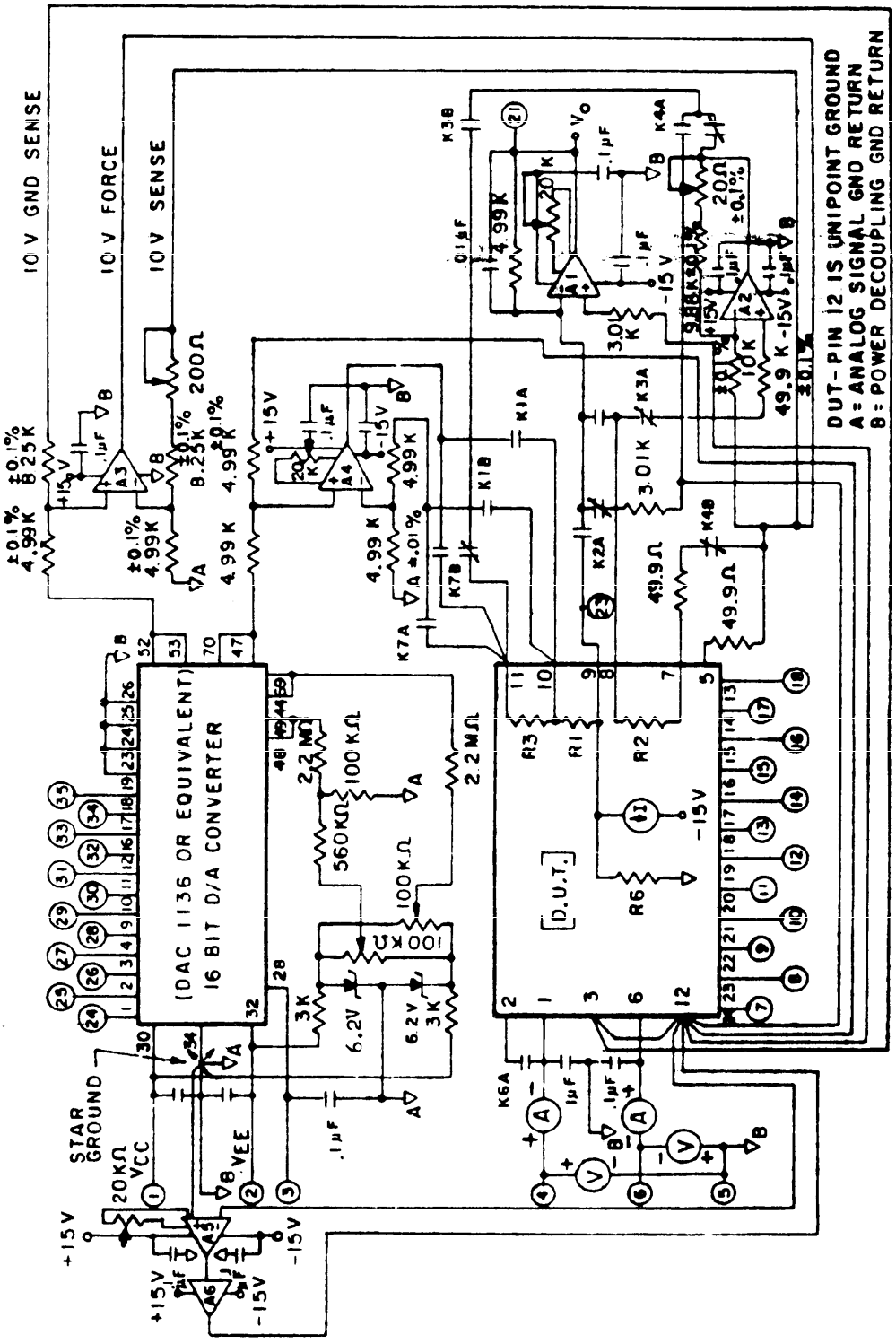


FIGURE 5. Static and dynamic tests circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:26Z
Check the source to verify that this is the current version before use.

De e type 02

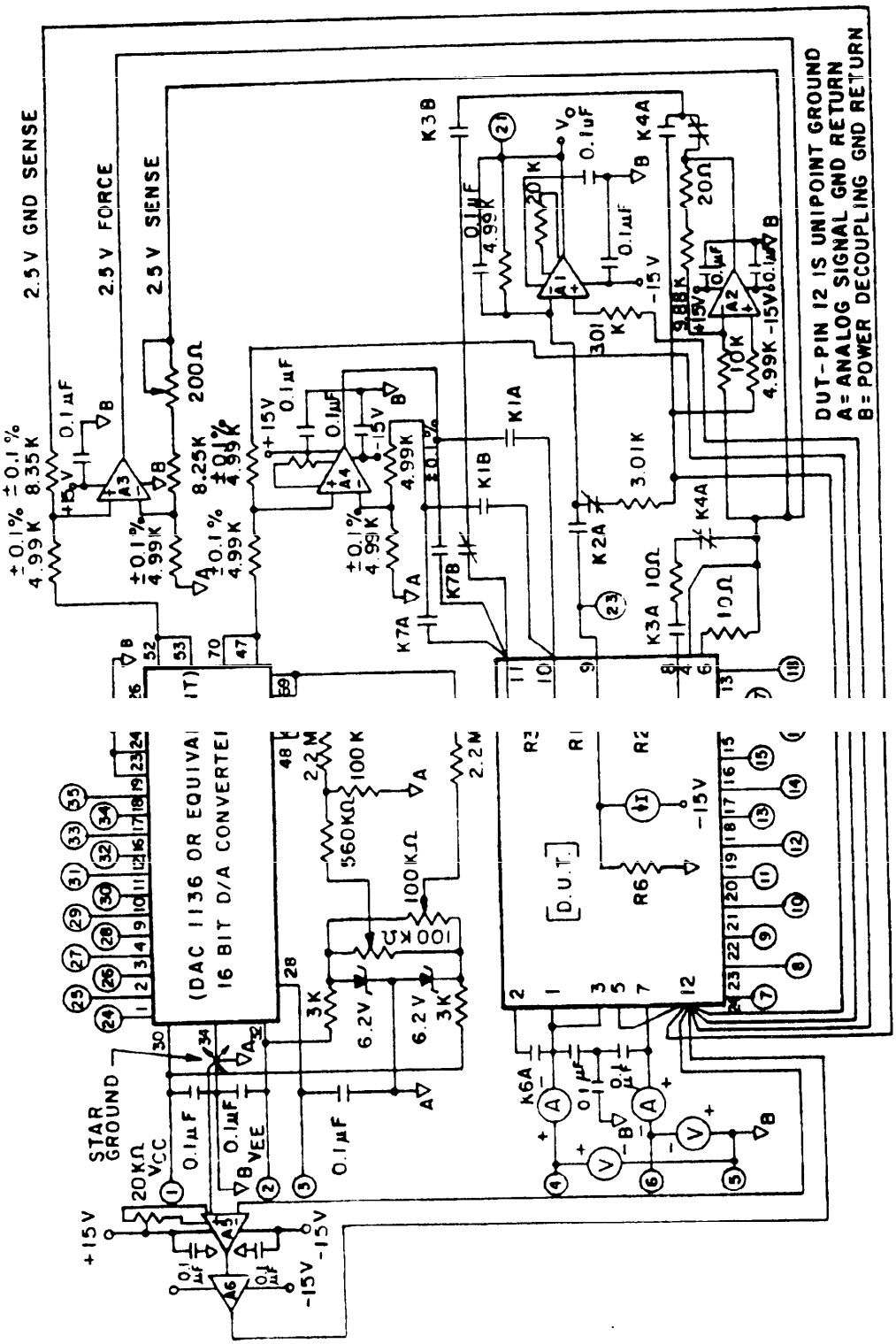


FIGURE 5. Static and dynamic tests circuit - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 03

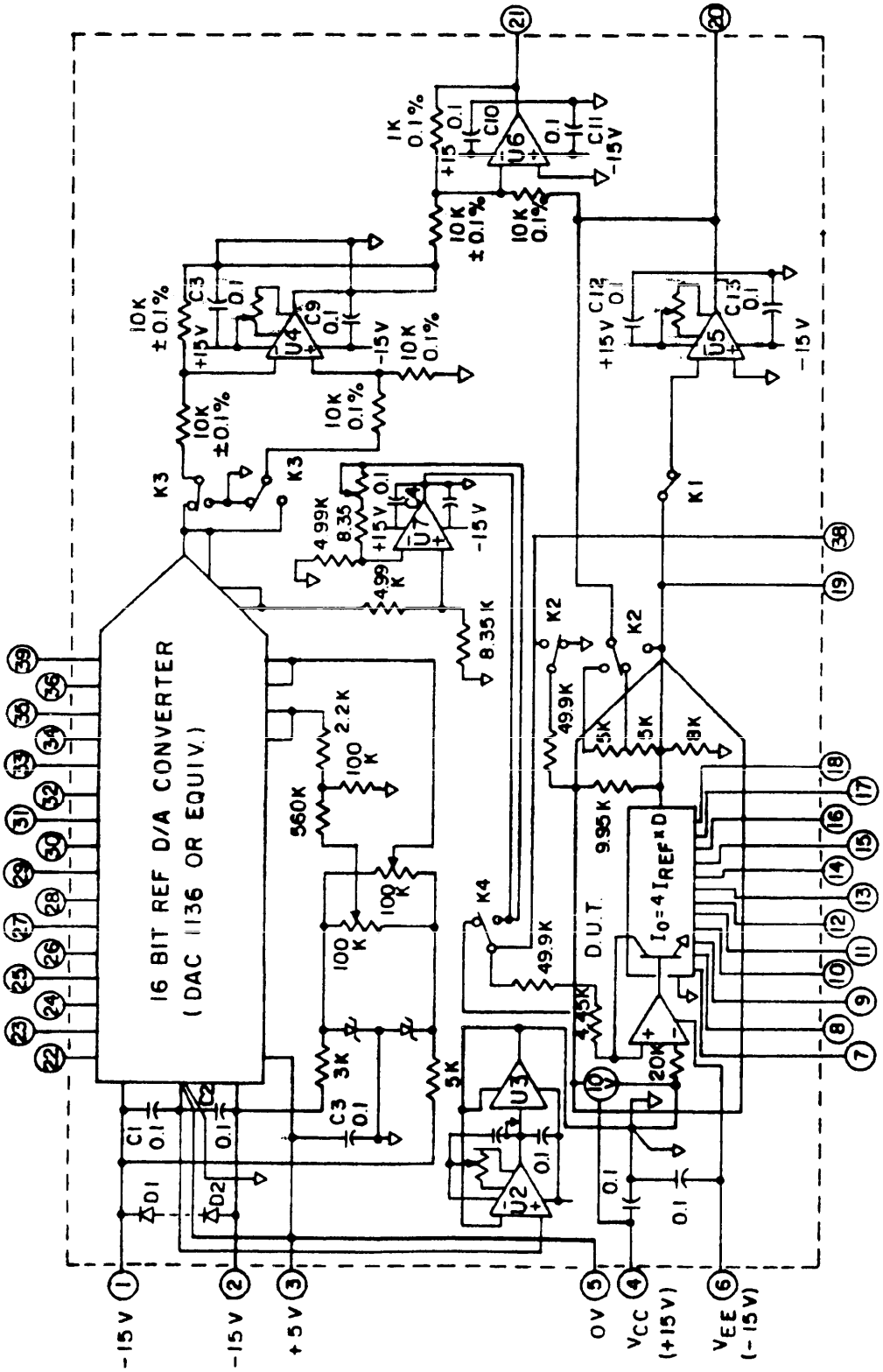


FIGURE 5. Test circuit for static tests - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 04

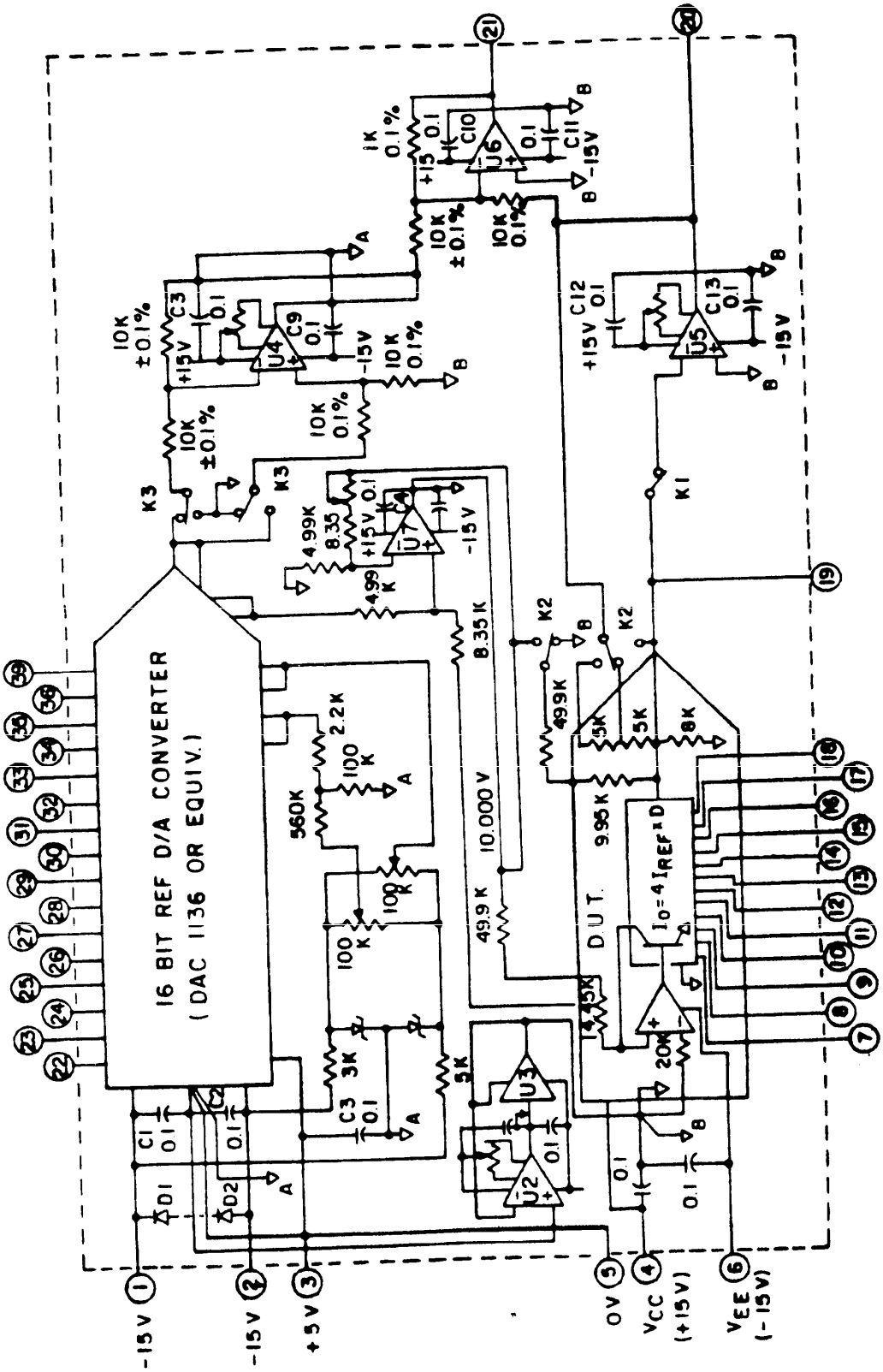
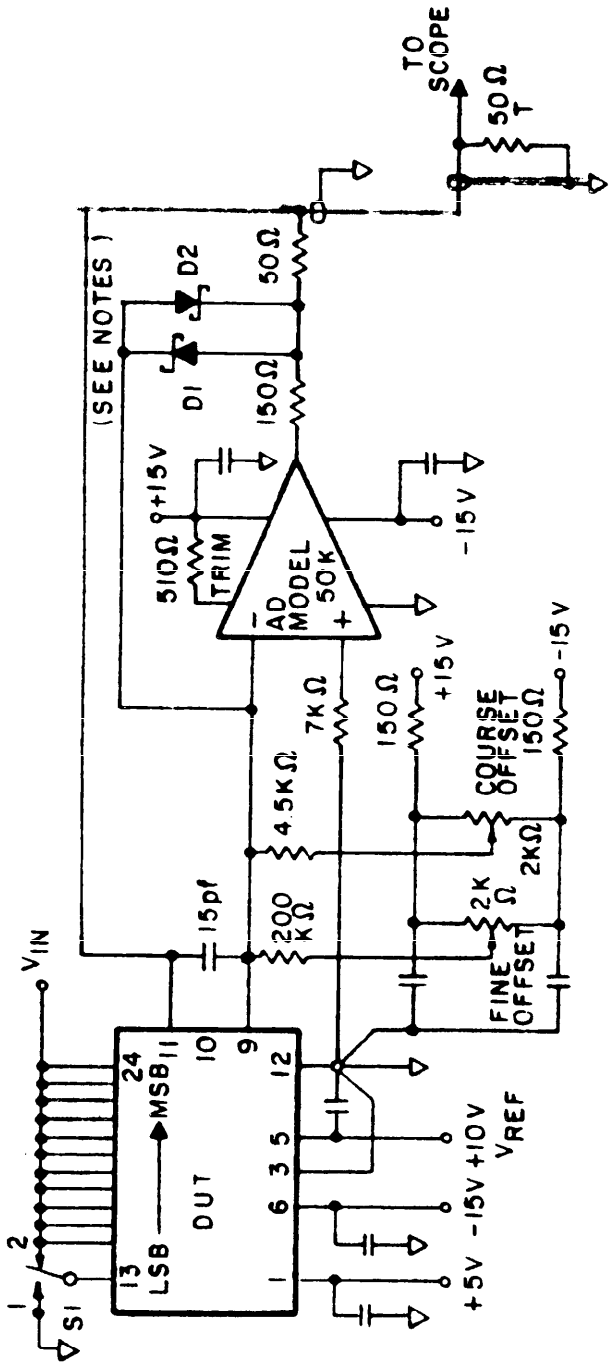


FIGURE 5. Test circuit for static tests - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 01

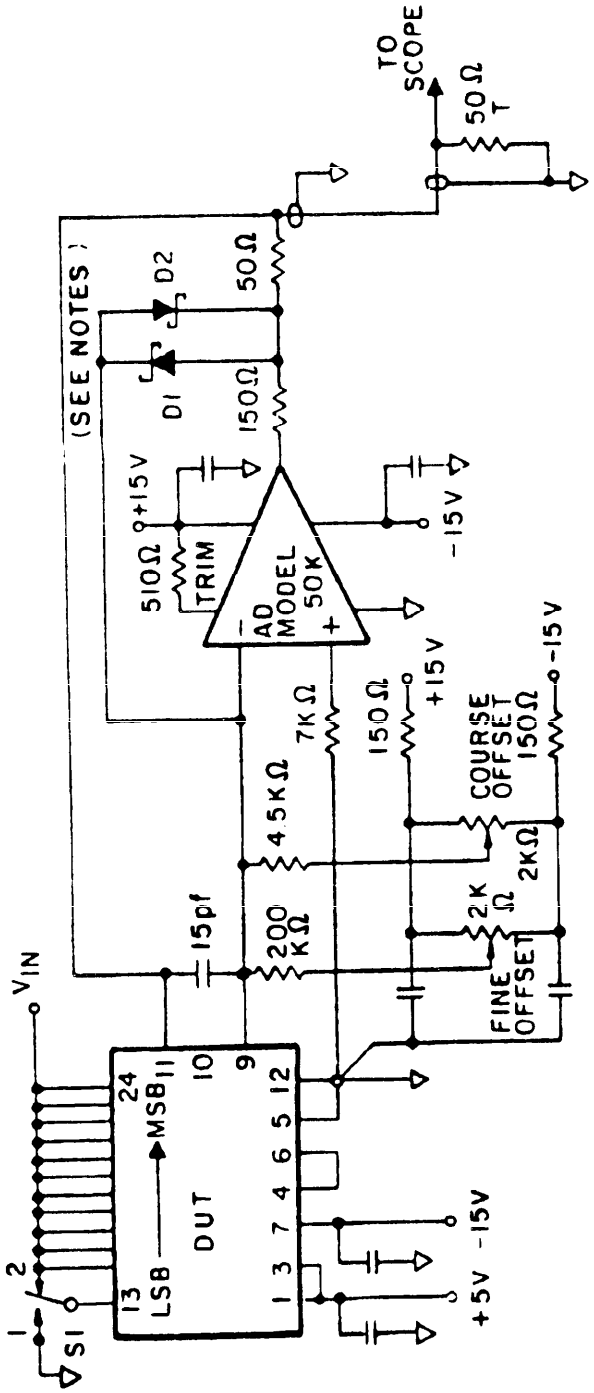


NOTES:
1. Unless otherwise defined, all capacitors are 47 µF tantalum paralleled with 0.1 µF ceramic.
2. Bandwidth of the oscilloscope shall be 50 MHz minimum; saturation of the preamp must be avoided. Use a voltage comparator preamp.
3. Adjust the course and fine offset controls to position the waveform final value on the oscilloscope center horizontal graticule.
4. While operating switch S-1, adjust the oscilloscope vertical gain for 1/2 LSB per cm.
5. D1 = D2 = HP 5082 - 2835 or EQUIVALENT.

FIGURE 6. Test circuit for settling time.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:26Z
Check the source to verify that this is the current version before use.

Device type 02

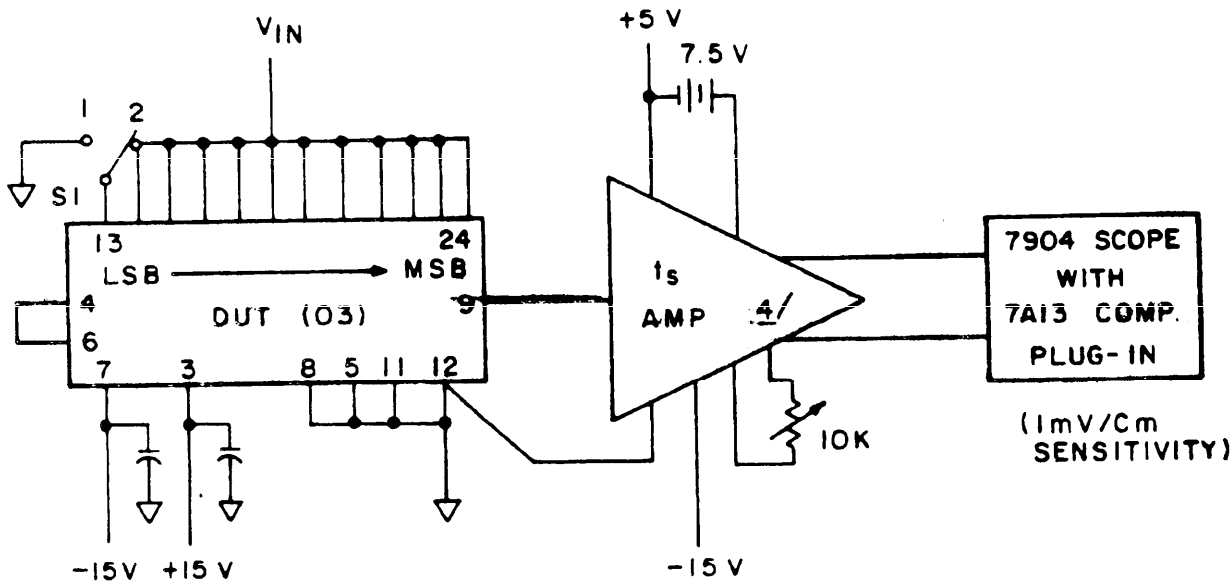


NOTES:
1. Unless otherwise defined, all capacitors are 47 µF tantalum paralleled with 0.1 µF ceramic.
2. Bandwidth of the oscilloscope shall be 50 MHz minimum; saturation of the preamp must be avoided. Use a voltage comparator preamp.
3. Adjust the course and fine offset controls to position the waveform final value of the oscilloscope center horizontal graticule.
4. While operating switch S-1, adjust the oscilloscope vertical gain for 1/2 LSB pre cm.
5. D1 = D2 = HP 5082 - 2835 or EQUIVALENT.

FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:26Z
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Device type 03



Device type 04

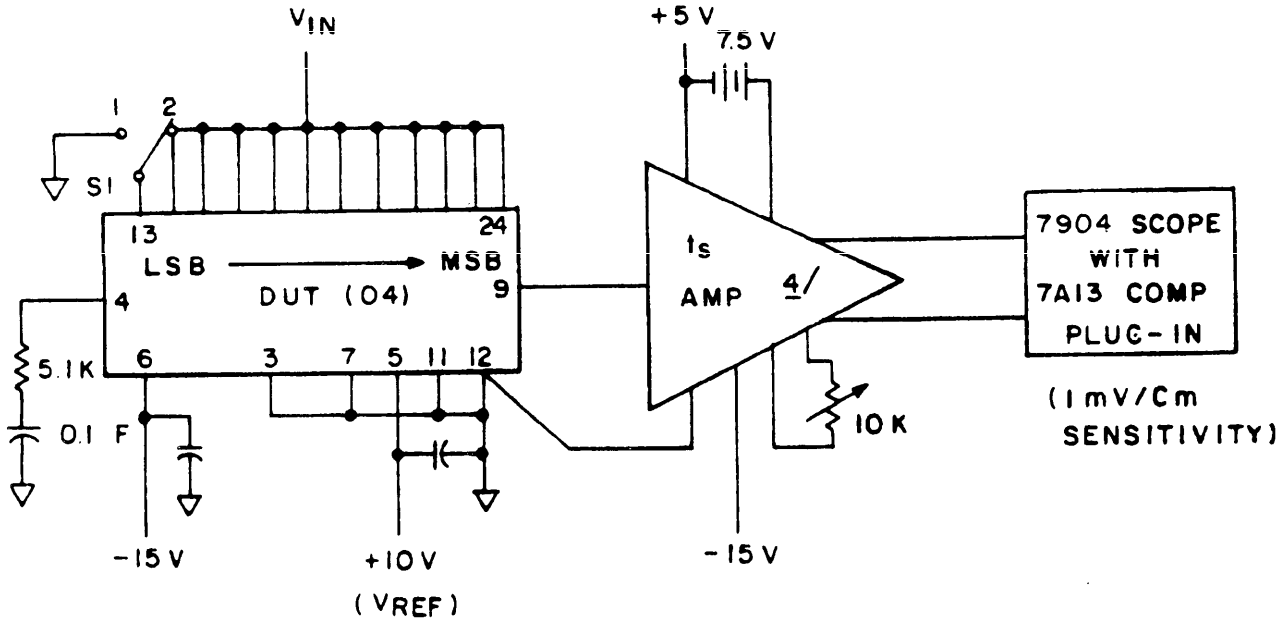
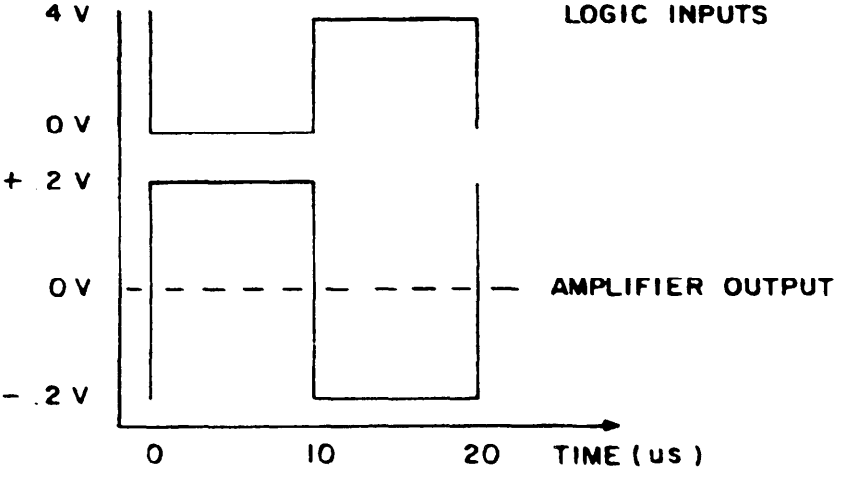


FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
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NOTES:
1. Unless otherwise specified, all capacitors are 47 µF tantalum in parallel with 0.1 µF ceramic.
2. The bandwidth of the oscilloscope main frame must be at least 500 MHz.
3. The plug-in, given a 400 mV to 0 input pulse, must recover to 0.5 mV on the 1 mV/cm gain range within 100 ns.
4. See figure 7A for details of the settling time amplifier. For proper readings, the input FET's (Q1, Q2) and the output FET's (Q4, Q5) must be balanced by the following procedure:
   a. With the DUT removed from the socket, measure the gate voltage of Q1 with a DVM while adjusting the 2 kΩ trim potentiometer for a reading of 0 V.
   b. Connect the (+) and (-) output jacks to a differential comparator scope preamp. Next, zero the scope display with the scope switch on "GND". Short the Schottky diodes out and adjust the 5 kΩ potentiometer so that the differential output display is in the ground reference position, with the scope switch on "DC".
5. Calibrate the scope display by installing a DUT and applying a 4 V, 100 kHz square wave to the logic inputs. With the plug-in controls set to 200 mV/cm and 20 µs/cm, adjust the 10 k pot until the amplifier output has its maximum amplitude. It should be a square wave approximately ±200 mV symmetrical about zero volts.

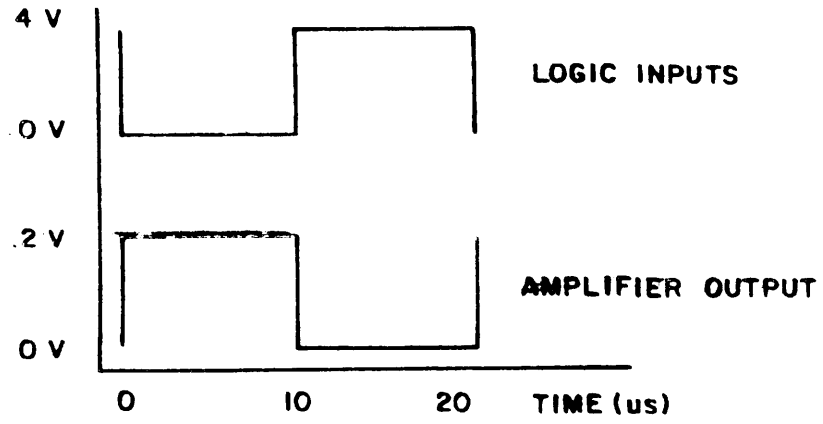


Adjust the 10 k pot such that the bottom of the square wave is at zero volts. Note that at 0 V the Schottky diodes are not clamping.

FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:26Z
Check the source to verify that this is the currentversion before use.

5. (Cont):



Increase the vertical plug-in gain to 1 mV/cm and the horizontal gain to 10 µs/cm. Toggle S1 and observe the shift in the amplifier output. This is the value of 1 LSB and should be approximately 1.2 mV. With S1 in position 2, observe the amplifier output level at 10 µs. It should be completely settled out and not changing. This level should be used for the output final value. Any variation in the zero volt output level such as may be caused by thermal changes must be corrected before proceeding to measure DUT settling time.

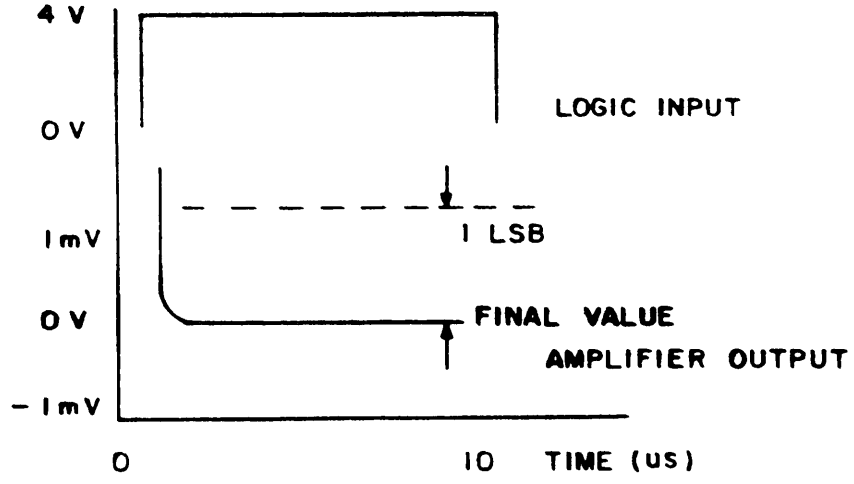
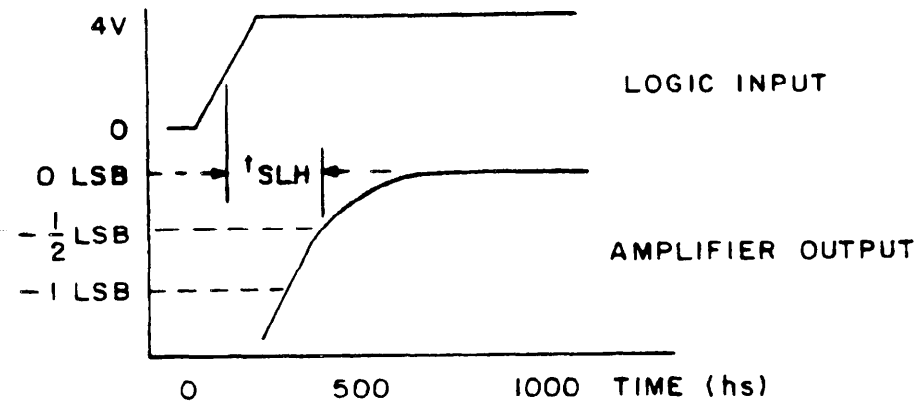


FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

6. Measure the settling time by changing the horizontal plug-in to 100 ns/cm and observing when the amplifier output gets within 1/2 LSB of the 10 µs final value.



7. Obtain t_{SLH} by adjusting the 10 K pot to zero the top of the amplifier output square wave and repeating tne measurement.

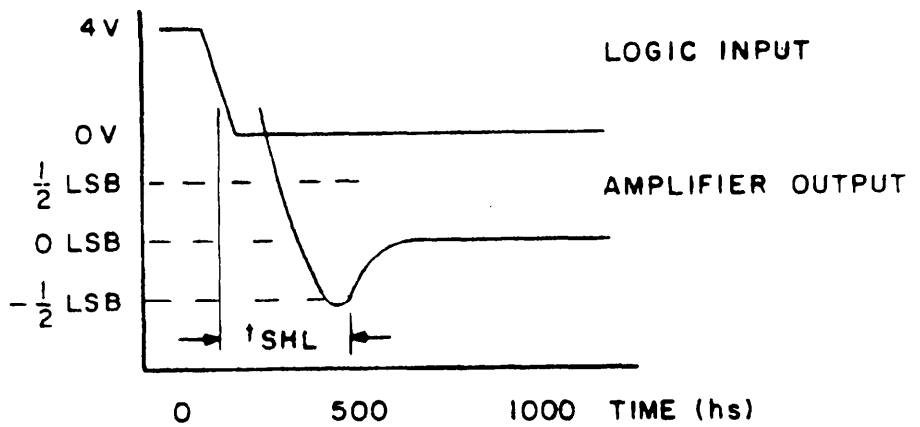


FIGURE 6. Test circuit for settling time - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
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Device types 03 and 04

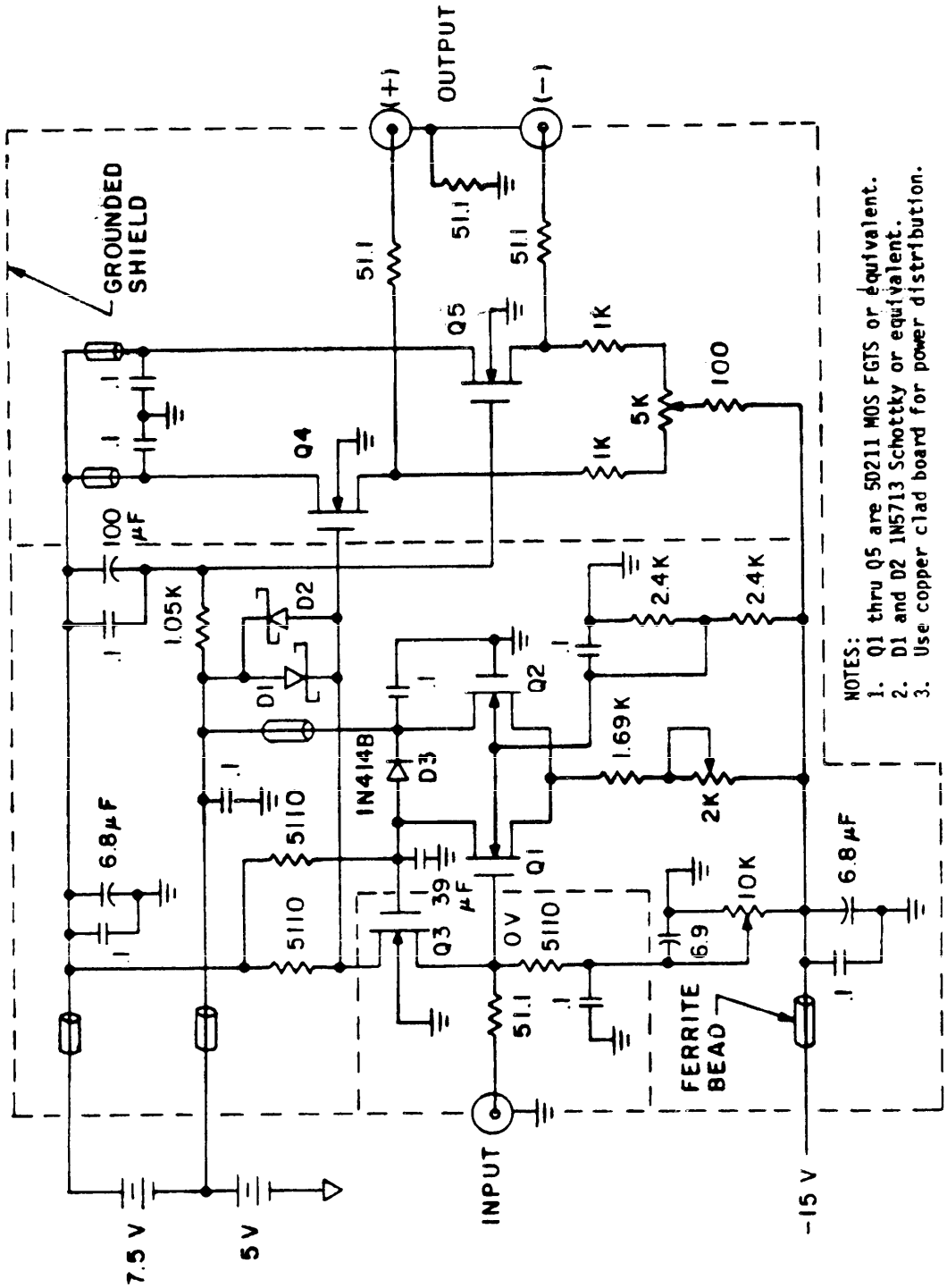


FIGURE 6. Amplifier for setting time

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Device types 01, 02, 03, and 04

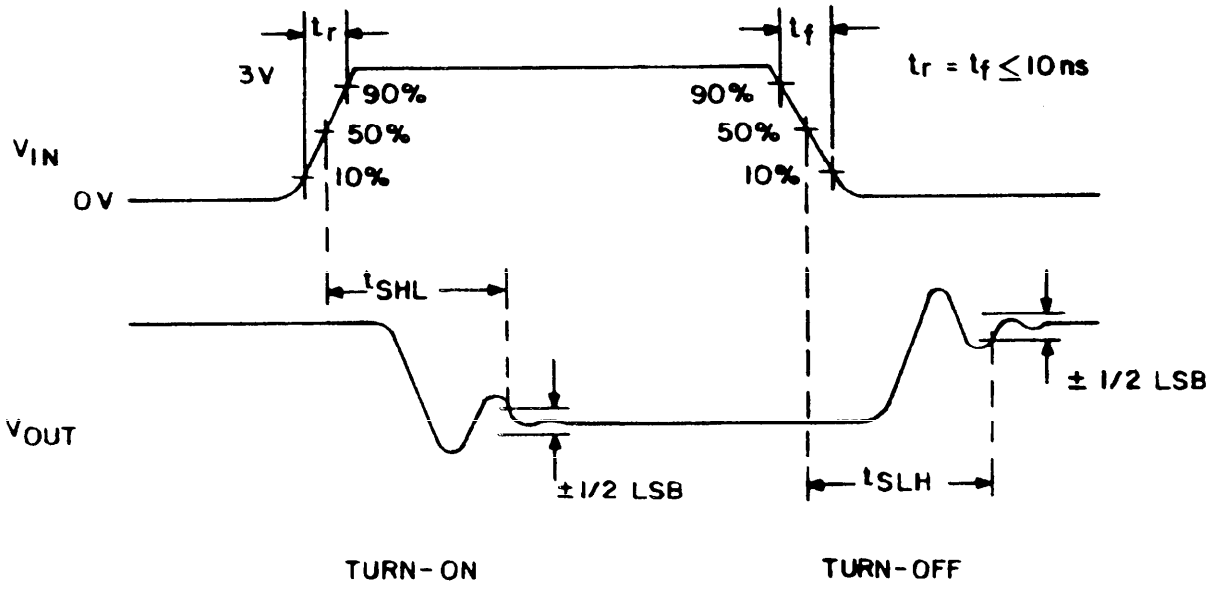
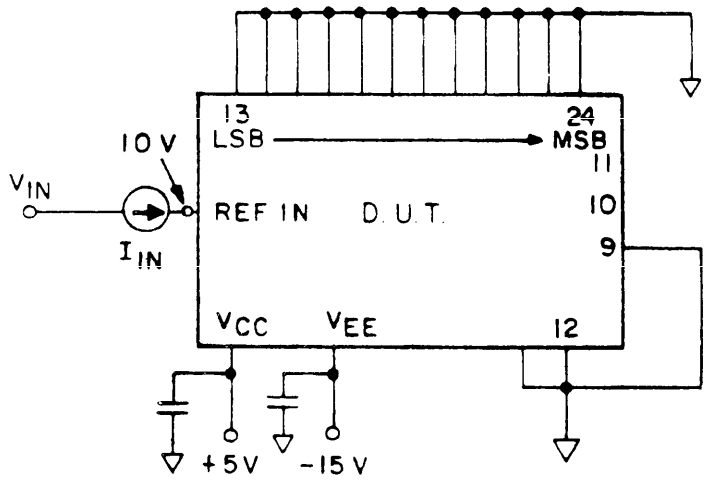


FIGURE 7. Test circuit waveforms for settling time.

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Check the source to verify that this is the current version before use.

Device types 01, 02, 03, and 04



NOTES:

1. All capacitors are 47 µF tantalum paralleled with 0.1 µF ceramic.
2. With 10.0 volts applied to DUT pin 5, measure the input current (I_{IN}).
3. The input resistance is

$$Ri = \frac{-10\ (v)}{I_{IN}\ (mA)} \quad k\Omega$$

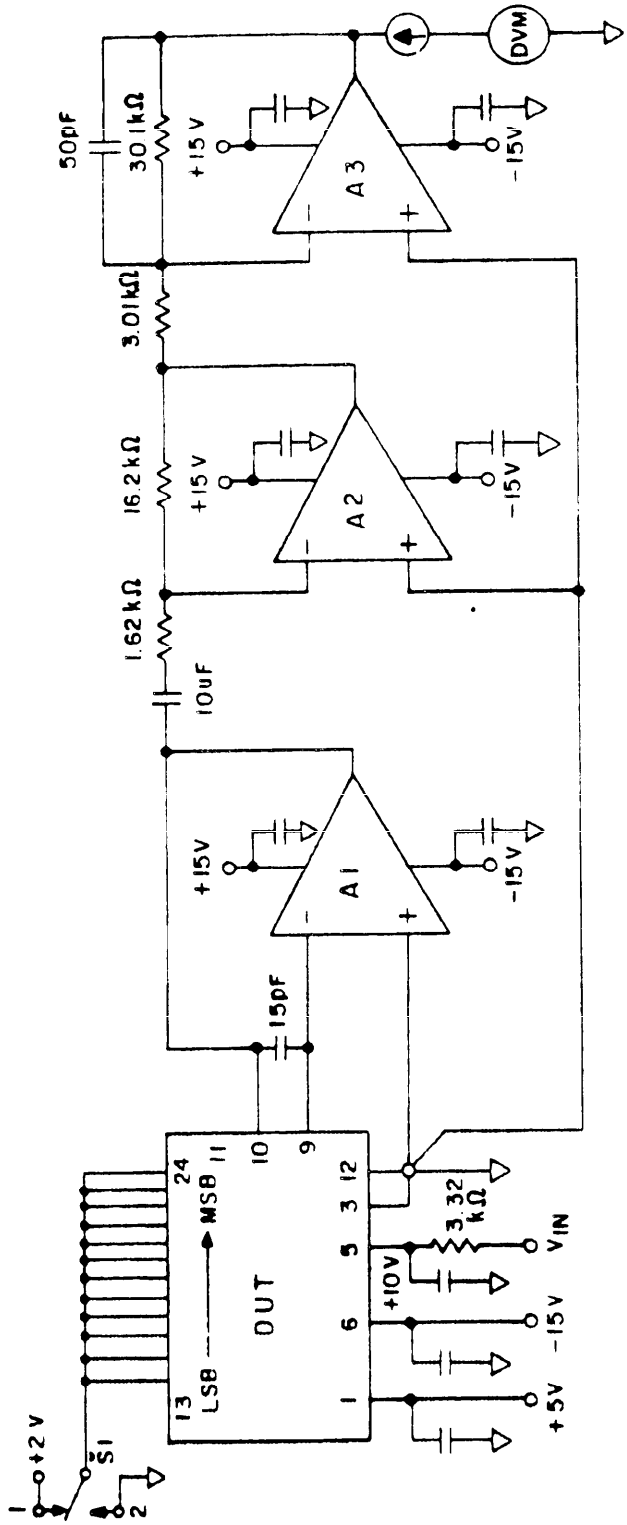
4. The following unique pin assignments apply:

Signal	Device type 01	02	03	04
Ref I_4	5	6	6	5
Ref GND	3	5	5	3
V_{CC}	1	1	NC	NC
V_{EE}	6	7	7	6

FIGURE 8. Reference input impedance test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:26Z
Check the source to verify that this is the current version before use.

Device types 01 and 04



NOTES:
1. Unless otherwise stated, all capacitors are 47 µF tantalum paralleled with 0.1 µF ceramic.
2. The volts rms meter bandwidth shall be greater than 100 kHz.
3. A_1, A_2, and A_3 shall have a gain bandwidth product greater than 10 MHz and an input noise density less than 10 nV/√Hz.
4. The noise voltage at OUT pins shall be less than 20 µVrms. (N_O for V_{ref})
5. With S-1 in position 2, the measured rms noise voltage shall be less than 10 mVrms. (N_O for all bits off).
6. With S-1 in position 1, the measured rms noise voltage shall be

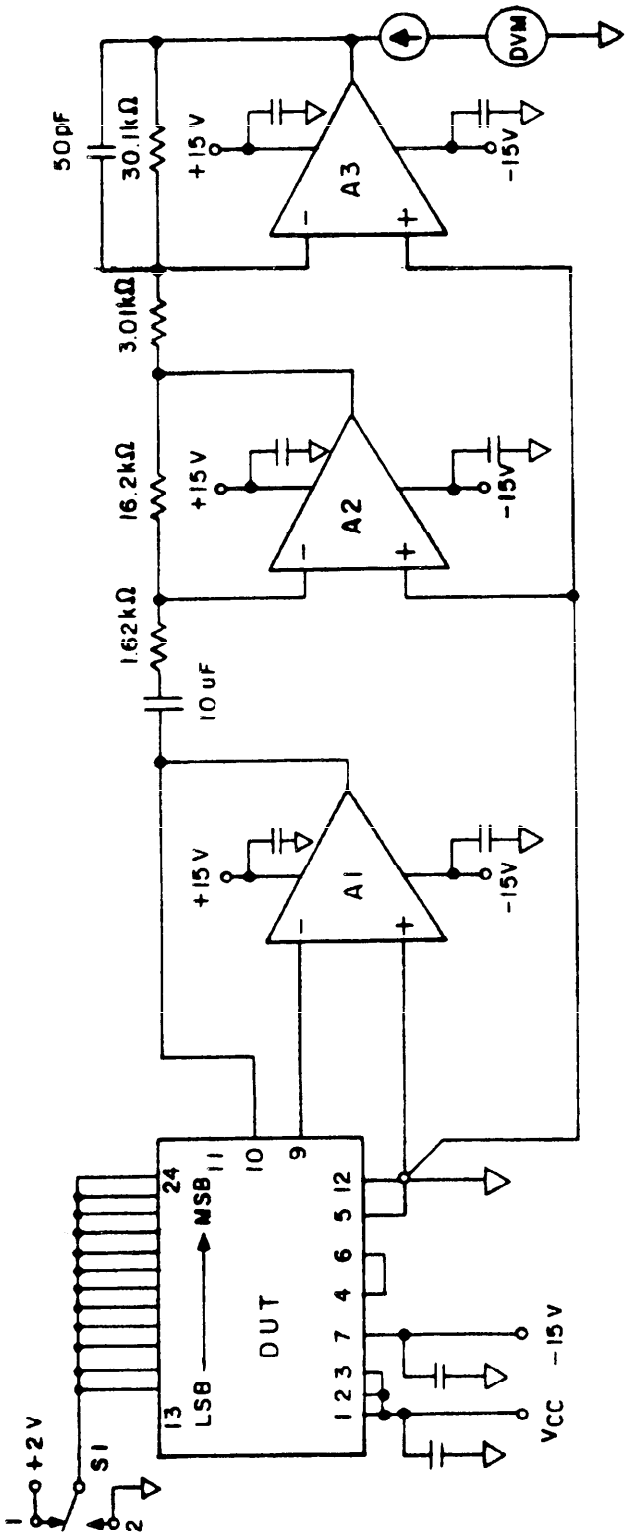
$$N_O \text{ for OUT} = \sqrt{(N_O)^2 \text{ for all bits on} - \frac{(N_O)^2 \text{ for all bits off} - (N_O)^2 \text{ for } V_{ref}}{100}}$$

7. There is no +5 V connection required for device type 04.

FIGURE 9. Output noise test circuit.

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Device types 02 and 03



NOTES:
1. Unless otherwise stated, all capacitors are 47 µF tantalum paralleled with 0.1 µF ceramic.
2. The volts rms meter bandwidth shall be greater than 100 kHz.
3. A_1, A_2, and A_3 shall have a gain bandwidth product greater than 1 MHz and an input noise density less than 10 nV/√Hz.
4. With S-1 in position 2, the measured rms noise voltage shall be less than 1.0 mVrms. ((N_o) for all bits off).
5. With S-1 in position 1, the measured rms noise voltage shall be

$$N_o \text{ for DUT} = \sqrt{(N_o)^2 \text{ for all bits on} - \frac{(N_o)^2 \text{ for all bits off} - (N_o)^2 \text{ for } V_{ref}}{100}}$$

6. V_{CC} is +5 V for device 02 and +15 V for device 03. Pins 1 and 2 are not connected for device 03.

FIGURE 9. Output noise test circuit - Continued.

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TABLE III. Group A inspection for device types 01 and 02.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/				Energized relays	Measured pin			Figure	Equation		Device types	Limits		Unit
			4	6	7 thru 18	24 thru 35		No.	Value	Unit					Min	Max	
1 T_A = 25°C	I_{CC}	1	15 V	-15 V	1111 1111 1111	1111 1111 1111	K_2,K_3 K_4,K_6	4	I_1	mA	5	$I_{CC} = I_1$		01 02	3 3	18 20	mA "
	I_{EE}	2	"	"	0000 0000 0000	1111 1111 1111	K_2,K_3 K_4,K_6	6	I_2	mA	"	$I_{EE} = I_2$		01, 02	-25	-5	"
	V_{REF}	3	"	"			K_4,K_6	36	V_1	V	"	$V_{REF} = V_1$		02	2.45	2.55	V
	I_{IH1}	4	"	"	1111 1111 1111	1111 1111 1111	K_6	7	I_3	µA	"	$I_{IH1} = I_3$	V_{IN} = 10.50 V	01, 02	-1	100	µA
	I_{IH2}	5	"	"	"	"	"	8	I_4	"	"	$I_{IH2} = I_4$	"	"	"	"	"
	I_{IH3}	6	"	"	"	"	"	9	I_5	"	"	$I_{IH3} = I_5$	"	"	"	"	"
	I_{IH4}	7	"	"	"	"	"	10	I_6	"	"	$I_{IH4} = I_6$	"	"	"	"	"
	I_{IH5}	8	"	"	"	"	"	11	I_7	"	"	$I_{IH5} = I_7$	"	"	"	"	"
	I_{IH6}	9	"	"	"	"	"	12	I_8	"	"	$I_{IH6} = I_8$	"	"	"	"	"
	I_{IH7}	10	"	"	"	"	"	13	I_9	"	"	$I_{IH7} = I_9$	"	"	"	"	"
	I_{IH8}	11	"	"	"	"	"	14	I_{10}	"	"	$I_{IH8} = I_{10}$	"	"	"	"	"
	I_{IH9}	12	"	"	"	"	"	15	I_{11}	"	"	$I_{IH9} = I_{11}$	"	"	"	"	"
	I_{IH10}	13	"	"	"	"	"	16	I_{12}	"	"	$I_{IH10} = I_{12}$	"	"	"	"	"
	I_{IH11}	14	"	"	"	"	"	17	I_{13}	"	"	$I_{IH11} = I_{13}$	"	"	"	"	"
	I_{IH12}	15	"	"	"	"	"	18	I_{14}	"	"	$I_{IH12} = I_{14}$	"	"	"	"	"
	I_{IL1}	16	"	"	0000 0000 0000	1111 1111 1111	"	7	I_{15}	"	"	$I_{IL1} = I_{15}$	V_{IN} = 0 V	01, 02	-200	1	"
	I_{IL2}	17	"	"	"	"	"	8	I_{16}	"	"	$I_{IL2} = I_{16}$	"	"	"	"	"
	I_{IL3}	18	"	"	"	"	"	9	I_{17}	"	"	$I_{IL3} = I_{17}$	"	"	"	"	"
	I_{IL4}	19	"	"	"	"	"	10	I_{18}	"	"	$I_{IL4} = I_{18}$	"	"	"	"	"
	I_{IL5}	20	"	"	"	"	"	11	I_{19}	"	"	$I_{IL5} = I_{19}$	"	"	"	"	"
	I_{IL6}	21	"	"	"	"	"	12	I_{20}	"	"	$I_{IL6} = I_{20}$	"	"	"	"	"
	I_{IL7}	22	"	"	"	"	"	13	I_{21}	"	"	$I_{IL7} = I_{21}$	"	"	"	"	"
	I_{IL8}	23	"	"	"	"	"	14	I_{22}	"	"	$I_{IL8} = I_{22}$	"	"	"	"	"
	I_{IL9}	24	"	"	"	"	"	15	I_{23}	"	"	$I_{IL9} = I_{23}$	"	"	"	"	"
	I_{IL10}	25	"	"	"	"	"	16	I_{24}	"	"	$I_{IL10} = I_{24}$	"	"	"	"	"
	I_{IL11}	26	"	"	"	"	"	17	I_{25}	"	"	$I_{IL11} = I_{25}$	"	"	"	"	"
	I_{IL12}	27	"	"	"	"	"	18	I_{26}	"	"	$I_{IL12} = I_{26}$	"	"	"	"	"
	I_{FS}	28	5 V	"	1111 1111 1111	1111 1111 1111	"	23	I_{27}	mA	"	$I_{FS} = I_{27}$		"	-2.7	-1.6	mA

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Measured pin Value	Measured pin Unit	Figure	Equation	Device types	Limits Min	Limits Max	unit
1 T_A = 25°C	$+P_{SS1}$		5 V	-15 V	1111 1111 1111	0000 0000 0000	K_1, K_2	21	V_2	V	5	$+P_{SS1}$ = (outputs on) $\left\{ \frac{\lvert V_3 - V_2 \rvert}{G_1} \times 41 \text{ for } -\Delta V_{CC} \right.$	01	-0.04	0.04	LSB/SPS
		29	4.5 V	"	1111 1111 1111	0000 0000 0000		"	V_3	"		$\left. \frac{\lvert V_4 - V_2 \rvert}{G_1} \times 41 \text{ for } +\Delta V_{CC} \right\}$	02	-0.15	0.15	"
			4.75 V	"												"
		30	5.5 V	"	1111 1111 1111	0000 0000 0000	"	"	V_4	"	"					"
			5.25 V	"												"
			2/									3/				"
			5 V	"	0000 0000 0000	1111 1111 1111	"	"	V_5	"	"	$+P_{SS1}$ = (outputs off) $\left\{ \frac{\lvert V_6 - V_5 \rvert}{G_1} \times 41 \text{ for } -\Delta V_{CC} \right.$	01	-0.04	0.04	"
		31	4.5 V	"	0000 0000 0000	1111 1111 1111	"	"	V_6	"	"		02	-0.15	0.15	"
			4.75 V	"								$\left. \frac{\lvert V_7 - V_5 \rvert}{G_1} \times 41 \text{ for } +\Delta V_{CC} \right\}$				"
		32	5.5 V	"	0000 0000 0000	1111 1111 1111	"	"	V_7	"	"					"
			5.25 V	"												"
	$+P_{SS2}$		15 V	"	1111 1111 1111	0000 0000 0000	K_1, K_2, K_6	"	V_8	"	"	$+P_{SS2}$ = (outputs on) $\left\{ \frac{\lvert V_9 - V_8 \rvert}{G_1} \times 41 \text{ for } -\Delta V_{CC} \right.$	01	-0.04	0.04	"
		33	13.5 V	"	1111 1111 1111	0000 0000 0000		"	V_9	"			02	-0.15	0.15	"
			14.25 V	"								$\left. \frac{\lvert V_{10} - V_8 \rvert}{G_1} \times 41 \text{ for } +\Delta V_{CC} \right\}$				"
		34	16.5 V	"	1111 1111 1111	0000 0000 0000	"	"	V_{10}	"	"					"
			15.75 V	"												"
			2/													"
			15 V	"	0000 0000 0000	1111 1111 1111	"	"	V_{11}	"	"	$+P_{SS2}$ = (outputs off) $\left\{ \frac{\lvert V_{12} - V_{11} \rvert}{G_1} \times 41 \text{ for } -\Delta V_{CC} \right.$	01	-0.04	0.04	"
		35	13.5 V	"	0000 0000 0000	1111 1111 1111	"	"	V_{12}	"	"		02	-0.15	0.15	"
			14.25 V	"								$\left. \frac{\lvert V_{13} - V_{11} \rvert}{G_1} \times 41 \text{ for } +\Delta V_{CC} \right\}$				"
		36	16.5 V	"	0000 0000 0000	1111 1111 1111	"	"	V_{13}	"	"					"
			15.75 V	"												"
	$-P_{SS}$		5 V	-15 V	1111 1111 1111	0000 0000 0000	K_1, K_2	"	V_{14}	"	"	$-P_{SS}$ = (outputs on) $\left\{ \frac{\lvert V_{15} - V_{14} \rvert}{G_1} \times 41 \text{ for } -\Delta V_{CC} \right.$	01	-0.07	0.07	"
		37	"	-13.5 V	1111 1111 1111	0000 0000 0000		"	V_{15}	"			02	-0.25	0.25	"
			"	-14.25 V								$\left. \frac{\lvert V_{16} - V_{14} \rvert}{G_1} \times 41 \text{ for } +\Delta V_{CC} \right\}$				"
		38	"	-16.5 V	1111 1111 1111	0000 0000 0000	"	"	V_{16}	"	"					"
			"	-15.75 V												"
			"	2/												"
			"	-15 V	0000 0000 0000	1111 1111 1111	"	"	V_{17}	"	"	$-P_{SS}$ = (outputs off) $\left\{ \frac{\lvert V_{18} - V_{17} \rvert}{G_1} \times 41 \text{ for } -\Delta V_{CC} \right.$	01	-0.07	0.07	"
		39	"	-13.5 V	0000 0000 0000	1111 1111 1111	"	"	V_{18}	"	"		02	-0.25	0.25	"
			"	-14.25 V								$\left. \frac{\lvert V_{19} - V_{17} \rvert}{G_1} \times 41 \text{ for } +\Delta V_{CC} \right\}$				"
		40	"	-16.5 V	0000 0000 0000	1111 1111 1111	"	"	V_{19}	"	"					"
			"	-15.75 V												"
2 T_A = 125°C	V_{REF}	41	15 V	-15 V			K_4, K_6	36	V_{20}	V	"	$V_{REF} = V_{20}$	02	2.45	2.55	V
	I_{ZS1}	42	5 V	"	0000 0000 0000	1111 1111 1111	K_6	23	I_{28}	µA	"	$I_{ZS} = (I_{28})(4095)/(I_{27}-I_{28})$	01, 02	-3	3	LSB
	I_{ZS2}	43	15 V	"	0000 0000 0000	1111 1111 1111	"	23	I_{29}	µA	"	$I_{ZS} = (I_{29})(4095)/(I_{27}-I_{29})$	01, 02	-3	3	LSB
	AE_1	44	5 V	"	0000 0000 0000	1111 1111 1111	K_1, K_2	21	V_{21}	V	"	$G_{1H} = (V_{22} - V_{21})/.\frac{10}{(4096)}$ (14)	01	-.20	.20	%
			"	"	0000 0000 0000	1111 1111 0001	"	"	V_{22}	"	"		02	-0.16	0.16	
			"	"	1111 1111 1111	0000 0000 0000	"	"	V_{23}	"	"	$A_{E1} = \left\{ \frac{V_{23} - V_{24}}{G_{1H}} \right\} \times 10$				
	B_{POE}	45	15 V	"	0000 0000 0000	1111 1111 1111	K_2, K_7	"	V_{24}	"	"	$G_{2H} = (V_{25} - V_{24}) / \frac{20}{(4096)}$ (14)	01	-8.0	8.0	LSB
			"	"	0000 0000 0000	1111 1111 0001	"	"	V_{25}	"	"		02	-6.5	6.5	
			"	"	0000 0000 0000	1111 1111 1111	"	"	V_{26}	"	"	$B_{POE} = 205 \times \frac{V_{26}}{G_{2H}}$				
	B_{ZE}	46	"	"	1000 0000 0000	0111 1111 1111	K_2, K_3	"	V_{27}	"	"	$B_{ZE} = 205 \times \frac{V_{27}}{G_{2H}}$	01	-6.5	6.5	LSB
													02	-6.5	6.5	

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Max	Unit
2 T_A = 125°C			5 V	-15 V	1111 1111 1111	0000 0000 0000	K_1, K_2	21	V_{28}	V	5	For each input code N, calculate the corresponding linearity error in LSB.	01, 02	-0.5	0.5	LSB
			"	"	0000 0000 0000	1111 1111 1111	"	"	V_{29}	"	"		"	"	"	"
	LE_1	47	"	"	0000 0000 0001	1111 1111 1110	"	"	V_{30}	"	"		"	"	"	"
		48	"	"	0000 0000 0010	1111 1111 1101	"	"	V_{31}	"	"		"	"	"	"
		49	"	"	0000 0000 0100	1111 1111 1011	"	"	V_{32}	"	"	$LE = \frac{410}{G_{1H}} \left\{ V_A - \left[\frac{V_{28} - V_{29}}{4095} \right] N - V_{29} \right\}$	"	"	"	"
		50	"	"	0000 0000 1000	1111 1111 0111	"	"	V_{33}	"	"		"	"	"	"
		51	"	"	0000 0001 0000	1111 1110 1111	"	"	V_{34}	"	"		"	"	"	"
		52	"	"	0000 0010 0000	1111 1101 1111	"	"	V_{35}	"	"		"	"	"	"
		53	"	"	0000 0100 0000	1111 1011 1111	"	"	V_{36}	"	"		"	"	"	"
	LE_8	54	"	"	0000 1000 0000	1111 0111 1111	"	"	V_{37}	"	"		"	"	"	"
					A. Record, (0000 BBBB BBBB), the sum of the code-words from tests 47 - 54 with positive linearity errors. B. Record, (0000 CCCC CCCC), the sum of the code-words from tests 47 - 54 with negative linearity errors.											
	LE_9	55	5 V	-15 V	0001 0000 0000	1110 1111 1111	K_1, K_2	21	V_{38}	V	5		01, 02	-0.5	0.5	LSB
		56	"	"	0010 0000 0000	1101 1111 1111	"	"	V_{39}	"	"		"	"	"	"
		57	"	"	0011 0000 0000	1100 1111 1111	"	"	V_{40}	"	"		"	"	"	"
		58	"	"	0100 0000 0000	1011 1111 1111	"	"	V_{41}	"	"		"	"	"	"
		59	"	"	0101 0000 0000	1010 1111 1111	"	"	V_{42}	"	"		"	"	"	"
		60	"	"	0110 0000 0000	1001 1111 1111	"	"	V_{43}	"	"		"	"	"	"
		61	"	"	0111 0000 0000	1000 1111 1111	"	"	V_{44}	"	"	$S = \left[\frac{V_{28} - V_{29}}{4095} \right]$	"	"	"	"
		62	"	"	1000 0000 0000	0111 1111 1111	"	"	V_{45}	"	"		"	"	"	"
		63	"	"	1001 0000 0000	0110 1111 1111	"	"	V_{46}	"	"		"	"	"	"
		64	"	"	1010 0000 0000	0101 1111 1111	"	"	V_{47}	"	"		"	"	"	"
		65	"	"	1011 0000 0000	0100 1111 1111	"	"	V_{48}	"	"		"	"	"	"
		66	"	"	1100 0000 0000	0011 1111 1111	"	"	V_{49}	"	"		"	"	"	"
		67	"	"	1101 0000 0000	0010 1111 1111	"	"	V_{50}	"	"		"	"	"	"
		68	"	"	1110 0000 0000	0001 1111 1111	"	"	V_{51}	"	"		"	"	"	"
	LE_{23}	69	"	"	1111 0000 0000	0000 1111 1111	"	"	V_{52}	"	"		"	"	"	"
					A. Record the DUT code-word, (XXXX 0000 0000), with the most positive error. B. Record the DUT code-word, (YYYY 0000 0000), with the most negative error.											
					Determine the sum of the recorded code-words obtained from tests 47 - 54 and from tests 55 - 69 for positive errors. Measure and calculate linearity error for the new DUT code-word (XXXX BBBB BBBB). (See Table I, note 6/).											
	LE_{24}	70	5 V	-15 V	XXXX BBBB BBBB XXXX BBBB BBBB 1111		K_1, K_2	21	V_{53}	V	5	$LE_{24} = \frac{410}{G_{1H}} \left(V_{53} - \left((\text{on-bits}) \times S \right) - V_{28} \right)$	01, 02	-0.5	0.5	LSB
					Determine the sum of the recorded code-words obtained from tests 47 - 54 and from tests 55 - 69 for negative errors. Measure and calculate linearity error for the new DUT code-word (YYYY CCCC CCCC). (See Table I, note 6/).											
	LE_{25}	71	5 V	-15 V	YYYY CCCC CCCC YYYY CCCC CCCC 1111		K_1, K_2	21	V_{54}	V	5	$LE_{25} = \frac{410}{G_{1H}} \left(V_{54} - \left((\text{on-bits}) \times S \right) - V_{28} \right)$	01, 02	-0.5	0.5	LSB

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Max	Unit
2 T_A = 125°C	MCE	72	5 V	-15 V	0111 1111 1111	1000 0000 0000	K_1,K_2	21	V_{55}	V	5	For each set of major carry codes N and N + 1, calculate the associated error in LSB's as follows:	01, 02	-1.0	1.0	LSB
			"	"	1000 0000 0000	0111 1111 1111	"	"	V_{56}	"	"		"	"	"	"
		73	"	"	0011 1111 1111	1100 0000 0000	"	"	V_{57}	"	"	$MCE = 410\left\{\left[V_m(N) - V_m(N-1)\right]/G_{1H} - \frac{10}{4096}\right\}$	"	"	"	"
			"	"	0100 0000 0000	1011 1111 1111	"	"	V_{58}	"	"		"	"	"	"
		74	"	"	0001 1111 1111	1110 0000 0000	"	"	V_{59}	"	"		"	"	"	"
			"	"	0010 0000 0000	1101 1111 1111	"	"	V_{60}	"	"		"	"	"	"
		75	"	"	0000 1111 1111	1111 0000 0000	"	"	V_{61}	"	"		"	"	"	"
			"	"	0001 0000 0000	1110 1111 1111	"	"	V_{62}	"	"		"	"	"	"
		76	"	"	0000 0111 1111	1111 1000 0000	"	"	V_{63}	"	"		"	"	"	"
			"	"	0000 1000 0000	1111 0111 1111	"	"	V_{64}	"	"		"	"	"	"
		77	"	"	0000 0011 1111	1111 1100 0000	"	"	V_{65}	"	"		"	"	"	"
			"	"	0000 0100 0000	1111 1011 1111	"	"	V_{66}	"	"		"	"	"	"
		78	"	"	0000 0001 1111	1111 1110 0000	"	"	V_{67}	"	"		"	"	"	"
			"	"	0000 0010 0000	1111 1101 1111	"	"	V_{68}	"	"		"	"	"	"
		79	"	"	0000 0000 1111	1111 1111 0000	"	"	V_{69}	"	"		"	"	"	"
			"	"	0000 0001 0000	1111 1110 1111	"	"	V_{70}	"	"		"	"	"	"
		80	"	"	0000 0000 0111	1111 1111 1000	"	"	V_{71}	"	"		"	"	"	"
			"	"	0000 0000 1000	1111 1111 0111	"	"	V_{72}	"	"		"	"	"	"
		81	"	"	0000 0000 0011	1111 1111 1100	"	"	V_{73}	"	"		"	"	"	"
			"	"	0000 0000 0100	1111 1111 1011	"	"	V_{74}	"	"		"	"	"	"
		82	"	"	0000 0000 0001	1111 1111 1110	"	"	V_{75}	"	"		"	"	"	"
			"	"	0000 0000 0010	1111 1111 1101	"	"	V_{76}	"	"		"	"	"	"
		83	"	"	0000 0000 0000	1111 1111 1111	"	"	V_{77}	"	"		"	"	"	"
			"	"	0000 0000 0001	1111 1111 1110	"	"	V_{78}	"	"		"	"	"	"
												Where V_m = measured value V_{55} through V_{78} "N = N + 1" are the decimal equivalents of the code words in column "7 thru 18" Example: For the first set N = 1000 0000 0000 = 2" = 2048 N - 1 = 0111 1111 1111 = 2" -1 = 2047 and $V_m(N) = V_{56}$ $V_m(N-1) = V_{55}$				
3 T_A = -55°C	V_{REF}	84	15 V	"			K_4,K_6	36	V_{79}	"	"	$V_{REF} = V_{79}$	02	2.45	2.55	V
	I_{ZS1}	85	5 V	"	0000 0000 0000	1111 1111 1111	K_6	23	I_{30}	µA	"	$I_{ZS} = (I_{30})\ (4095)/(I_{27}-I_{30})$	01, 02	-3	3	LSB
	I_{ZS2}	86	15 V	"	0000 0000 0000	1111 1111 1111	"	23	I_{31}	µA	"	$I_{ZS} = (I_{31})\ (4095)/(I_{27}-I_{31})$	01, 02	-3	3	LSB
	A_{E1}	87	5 V	"	0000 0000 0000	1111 1111 1111	K_1,K_2	21	V_{80}	V	"	$G_{1L} = (V_{81} - V_{80})/\left(\frac{10}{4096}\right)(14)$	01	-.20	.20	%
			"	"	0000 0000 0000	1111 1111 0001	"	"	V_{81}	"	"		02	-0.16	0.16	
			"	"	1111 1111 1111	0000 0000 0000	"	"	V_{82}	"	"	$A_{E1} = \left\{\frac{V_{82} - V_{80}}{G_{1L}}\right\} \times 10$				
	B_{POE}	88	15 V	"	0000 0000 0000	1111 1111 1111	K_2,K_7	"	V_{83}	"	"	$G_{2L} = (V_{84} - V_{83})/\frac{20}{4096}\ (14)$	01	-8.0	8.0	LSB
			"	"	0000 0000 0000	1111 1111 0001	K_2,K_7	"	V_{84}	"	"		02	-6.5	6.5	
			"	"	0000 0000 0000	1111 1111 1111	K_2,K_3	"	V_{85}	"	"	$B_{POE} = 205\left(\frac{V_{85}}{G_{2L}}\right) \times 10$				
	B_{ZE}	89	"	"	1000 0000 0000	0111 1111 1111	K_2,K_3	"	V_{86}	"	"	$B_{ZE} = 205\left(\frac{V_{86}}{G_{2L}}\right)$	01	-6.5	6.5	LSB
													02	-6.5	6.5	

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Max	Unit
3 T_A = -55°C	LE_1	90	5 V	-15 V	1111 1111 1111	0000 0000 0000	K_1,K_2	21	V_{87}	V	5	For each input code N, calculate the corresponding linearity error in LSB.	01, 02	-0.5	0.5	LSB
		91	"	"	0000 0000 0001	1111 1111 1110	"	"	V_{88}	"	"		"	"	"	"
		92	"	"	0000 0000 0010	1111 1111 1101	"	"	V_{89}	"	"		"	"	"	"
		93	"	"	0000 0000 0100	1111 1111 1011	"	"	V_{90}	"	"		"	"	"	"
		94	"	"	0000 0000 1000	1111 1111 0111	"	"	V_{91}	"	"		"	"	"	"
		95	"	"	0000 0001 0000	1111 1110 1111	"	"	V_{92}	"	"		"	"	"	"
		96	"	"	0000 0010 0000	1111 1101 1111	"	"	V_{93}	"	"		"	"	"	"
		97	"	"	0000 0100 0000	1111 1011 1111	"	"	V_{94}	"	"		"	"	"	"
	LE_8				0000 1000 0000	1111 0111 1111	"	"	V_{95}	"	"		"	"	"	"
								"	V_{96}	"	"		"	"	"	"

$$LE = \frac{410}{G_{1L}} \left\{ V_A - \left[\frac{V_{87} - V_{88}}{4095} \right] N - V_{88} \right\}$$

Where V_A = measured value from V_{89} through V_{111}
N = bit number (decimal)
$V_{89} \leq V_A \leq V_{111}$

A. Record, (0000 BBBB BBBB), the sum of the code-words from tests 90 - 97 with positive linearity errors.
B. Record, (0000 CCCC CCCC), the sum of the code-words from tests 90 - 97 with negative linearity errors.

	LE_9	98	5 V	-15 V	0001 0000 0000	1110 1111 1111	K_1,K_2	21	V_{97}	V	5		01, 02	-0.5	0.5	LSB
		99	"	"	0010 0000 0000	1101 1111 1111	"	"	V_{98}	"	"		"	"	"	"
		100	"	"	0011 0000 0000	1100 1111 1111	"	"	V_{99}	"	"		"	"	"	"
		101	"	"	0100 0000 0000	1011 1111 1111	"	"	V_{100}	"	"		"	"	"	"
		102	"	"	0101 0000 0000	1010 1111 1111	"	"	V_{101}	"	"		"	"	"	"
		103	"	"	0110 0000 0000	1001 1111 1111	"	"	V_{102}	"	"		"	"	"	"
		104	"	"	0111 0000 0000	1000 1111 1111	"	"	V_{103}	"	"		"	"	"	"
		105	"	"	1000 0000 0000	0111 1111 1111	"	"	V_{104}	"	"		"	"	"	"
		106	"	"	1001 0000 0000	0110 1111 1111	"	"	V_{105}	"	"		"	"	"	"
		107	"	"	1010 0000 0000	0101 1111 1111	"	"	V_{106}	"	"		"	"	"	"
		108	"	"	1011 0000 0000	0100 1111 1111	"	"	V_{107}	"	"		"	"	"	"
		109	"	"	1100 0000 0000	0011 1111 1111	"	"	V_{108}	"	"		"	"	"	"
		110	"	"	1101 0000 0000	0010 1111 1111	"	"	V_{109}	"	"		"	"	"	"
		111	"	"	1110 0000 0000	0001 1111 1111	"	"	V_{110}	"	"		"	"	"	"
	LE_{23}	112	"	"	1111 0000 0000	0000 1111 1111	"	"	V_{111}	"	"		"	"	"	"

$$S = \left[\frac{V_{87} - V_{88}}{4095} \right]$$

A. Record the DUT code-word, (XXXX 0000 0000), with the most positive error.
B. Record the DUT code-word, (YYYY 0000 0000), with the most negative error.

Determine the sum of the recorded code-words obtained from tests 90 - 97 and from tests 98 - 112 for positive errors.
Measure and calculate linearity error for the new DUT code-word (XXXX BBBB BBBB). (See Table I, note 6/).

	LE_{24}	113	5 V	-15 V	XXXX BBBB BBBB XXXX BBBB BBBB 1111		K_1,K_2	21	V_{112}	V	5	$LE_{24} = \frac{410}{G_{1L}} \left(V_{112} - \left((\text{on-bits}) \times S \right) - V_{87} \right)$	01, 02	-0.5	0.5	LSB

Determine the sum of the recorded code-words obtained from tests 90 - 97 and from tests 98 - 112 for negative errors.
Measure and calculate linearity error for the new DUT code-word (YYYY CCCC CCCC). (See Table I, note 6/).

	LE_{25}	114	5 V	-15 V	YYYY CCCC CCCC YYYY CCCC CCCC 1111		K_1,K_2	21	V_{113}	V	5	$LE_{25} = \frac{410}{G_{1L}} \left(V_{113} - \left((\text{on-bits}) \times S \right) - V_{87} \right)$	01, 02	-0.5	0.5	LSB

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Limits Max	Unit
3 T_A = -55°C	MC_1	115	5 V	-15 V	0111 1111 1111	1000 0000 0000	K_1,K_2	21	V_{114}	V	5	For each set of major carry codes N and N - 1, calculate the associated error in LSB's.	01, 02	-1.0	1.0	LSB
			"	"	1000 0000 0000	0111 1111 1111	"	"	V_{115}	"	"		"	"	"	"
	MC_2	116	"	"	0011 1111 1111	1100 0000 0000	"	"	V_{116}	"	"		"	"	"	"
			"	"	0100 0000 0000	1011 1111 1111	"	"	V_{117}	"	"	$MCE = 410\left(\left[V_m(N) - V_m(N-1)\right]/G_{1L} - \frac{10}{4096}\right)$	"	"	"	"
	MC_3	117	"	"	0001 1111 1111	1110 0000 0000	"	"	V_{118}	"	"		"	"	"	"
			"	"	0010 0000 0000	1101 1111 1111	"	"	V_{119}	"	"		"	"	"	"
	MC_4	118	"	"	0000 1111 1111	1111 0000 0000	"	"	V_{120}	"	"		"	"	"	"
			"	"	0001 0000 0000	1110 1111 1111	"	"	V_{121}	"	"		"	"	"	"
	MC_5	119	"	"	0000 0111 1111	1111 1000 0000	"	"	V_{122}	"	"		"	"	"	"
			"	"	0000 1000 0000	1111 0111 1111	"	"	V_{123}	"	"		"	"	"	"
	MC_6	120	"	"	0000 0011 1111	1111 1100 0000	"	"	V_{124}	"	"		"	"	"	"
			"	"	0000 0100 0000	1111 1011 1111	"	"	V_{125}	"	"		"	"	"	"
	MC_7	121	"	"	0000 0001 1111	1111 1110 0000	"	"	V_{126}	"	"		"	"	"	"
			"	"	0000 0010 0000	1111 1101 1111	"	"	V_{127}	"	"		"	"	"	"
	MC_8	122	"	"	0000 0000 1111	1111 1111 0000	"	"	V_{128}	"	"		"	"	"	"
			"	"	0000 0001 0000	1111 1110 1111	"	"	V_{129}	"	"		"	"	"	"
	MC_9	123	"	"	0000 0000 0111	1111 1111 1000	"	"	V_{130}	"	"		"	"	"	"
			"	"	0000 0000 1000	1111 1111 0111	"	"	V_{131}	"	"		"	"	"	"
	MC_{10}	124	"	"	0000 0000 0011	1111 1111 1100	"	"	V_{132}	"	"		"	"	"	"
			"	"	0000 0000 0100	1111 1111 1011	"	"	V_{133}	"	"		"	"	"	"
	MC_{11}	125	"	"	0000 0000 0001	1111 1111 1110	"	"	V_{134}	"	"		"	"	"	"
			"	"	0000 0000 0010	1111 1111 1101	"	"	V_{135}	"	"		"	"	"	"
	MC_{12}	126	"	"	0000 0000 0000	1111 1111 1111	"	"	V_{136}	"	"		"	"	"	"
			"	"	0000 0000 0001	1111 1111 1110	"	"	V_{137}	"	"		"	"	"	"
												Example:				
												N = 1000 0000 0000 = 2^{12} = 2048				
												N - 1 = 0111 1111 1111 = 2^{12}-1 = 2047				
												and $V_m(N) = V_{115}$				
												$V_m(N-1) = V_{114}$				
4 T_A = 25°C	B_{PAE}	127	"	"	0000 0000 0000	1111 1111 1111	K_2,K_7	"	V_{138}	"	"	$G_2 = (V_{139} - V_{138})/\left(\frac{20}{4096}\right)$ (14)	01	-.20	.20	%
			"	"	0000 0000 0000	1111 1111 0001	K_2,K_7	"	V_{139}	"	"		02	-.16	.16	
			"	"	0000 0000 0000	1111 1111 1111	K_2,K_3	"	V_{140}	"	"	$B_{PAE} = 5(V_{141} - V_{140})/G_2$				
			"	"	1111 1111 1111	0000 0000 0000	K_2,K_3	"	V_{141}	"	"					
	B_{POE}	128	"	"	0000 0000 0000	1111 1111 1111	"	"	V_{142}	"	"	$B_{POE} = 205 \times V_{142}/G_2$	01 02	-8 -6.5	8 6.5	LSB "
	B_{ZE}	129	"	"	1000 0000 0000	0111 1111 1111	"	"	V_{143}	"	"	$B_{ZE} = 205 \times V_{143}/G_2$	01, 02	-6.5	6.5	"
	I_{ZS1}	130	"	"	0000 0000 0000	1111 1111 1111	K_6	23	I_{32}	µA	"	$I_{ZS} = (I_{32})(4095)/(I_{27}-I_{32})$	01, 02	-2	2	"
	I_{ZS2}	131	15 V	"	0000 0000 0000	1111 1111 1111	"	23	I_{33}	µA	"	$I_{ZS} = (I_{33})(4095)/(I_{27}-I_{33})$	01, 02	-2	2	"
	A_{E1}	132	5 V	"	0000 0000 0000	1111 1111 1111	K_1,K_2	21	V_{144}	V	"	$G_1 = (V_{145} - V_{144})/\left(\frac{10}{4096}\right)$ (14)	01	-.20	.20	%
			"	"	0000 0000 0000	1111 1111 0001	"	"	V_{145}	"	"		02	-.16	.16	"
			"	"	1111 1111 1111	0000 0000 0000	"	"	V_{146}	"	"	$A_{E1} = 10(V_{146} - V_{144})/G_1$				
	A_{E2}	133	15 V	"	0000 0000 0000	1111 1111 1111	K_1,K_2,K_6	"	V_{147}	"	"	$A_{E2} = 10(V_{148} - V_{147})/G_1$	01	-.20	.20	"
				"	1111 1111 1111	0000 0000 0000	"	"	V_{148}	"	"		02	-.16	.16	"

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Max	Unit
4 T_A = 55°C			5 V	-15 V	1111 1111 1111	0000 0000 0000	K_1,K_2	21	V_{149}	V	5	For each input code N, calculate the corresponding linearity error in LSB.	01, 02	-0.5	0.5	LSB
			"	"	0000 0000 0000	1111 1111 1111	"	"	V_{150}	"	"		"	"	"	"
	LE_1	134	"	"	0000 0000 0001	1111 1111 1110	"	"	V_{151}	"	"	$LE = \frac{410}{G_1}\left\{V_A - \left[\frac{V_{149} - V_{150}}{4095}\right]N - V_{149}\right\}$	"	"	"	"
		135	"	"	0000 0000 0010	1111 1111 1101	"	"	V_{152}	"	"		"	"	"	"
		136	"	"	0000 0000 0100	1111 1111 1011	"	"	V_{153}	"	"		"	"	"	"
		137	"	"	0000 0000 1000	1111 1111 0111	"	"	V_{154}	"	"		"	"	"	"
		138	"	"	0000 0001 0000	1111 1110 1111	"	"	V_{155}	"	"		"	"	"	"
		139	"	"	0000 0010 0000	1111 1101 1111	"	"	V_{156}	"	"		"	"	"	"
		140	"	"	0000 0100 0000	1111 1011 1111	"	"	V_{157}	"	"	Where V_A = measured value from V_{151} through V_{173}	"	"	"	"
	LE_8	141	"	"	0000 1000 0000	1111 0111 1111	"	"	V_{158}	"	"	N = bit number (decimal)	"	"	"	"
												$V_{149} \leq V_A \leq V_{173}$				
					A. Record, (0000 BBBB BBBB), the sum of the code-words from tests 134 - 141 with positive linearity errors.											
					B. Record, (0000 CCCC CCCC), the sum of the code-words from tests 134 - 141 with negative linearity errors.											
	LE_9	142	5 V	-15 V	0001 0000 0000	1110 1111 1111	K_1,K_2	21	V_{159}	V	5		01, 02	-0.5	0.5	LSB
		143	"	"	0010 0000 0000	1101 1111 1111	"	"	V_{160}	"	"		"	"	"	"
		144	"	"	0011 0000 0000	1100 1111 1111	"	"	V_{161}	"	"		"	"	"	"
		145	"	"	0100 0000 0000	1011 1111 1111	"	"	V_{162}	"	"		"	"	"	"
		146	"	"	0101 0000 0000	1010 1111 1111	"	"	V_{163}	"	"		"	"	"	"
		147	"	"	0110 0000 0000	1001 1111 1111	"	"	V_{164}	"	"		"	"	"	"
		148	"	"	0111 0000 0000	1000 1111 1111	"	"	V_{165}	"	"	$S = \left[\frac{V_{149} - V_{150}}{4095}\right]$	"	"	"	"
		149	"	"	1000 0000 0000	0111 1111 1111	"	"	V_{166}	"	"		"	"	"	"
		150	"	"	1001 0000 0000	0110 1111 1111	"	"	V_{167}	"	"		"	"	"	"
		151	"	"	1010 0000 0000	0101 1111 1111	"	"	V_{168}	"	"		"	"	"	"
		152	"	"	1011 0000 0000	0100 1111 1111	"	"	V_{169}	"	"		"	"	"	"
		153	"	"	1100 0000 0000	0011 1111 1111	"	"	V_{170}	"	"		"	"	"	"
		154	"	"	1101 0000 0000	0010 1111 1111	"	"	V_{171}	"	"		"	"	"	"
		155	"	"	1110 0000 0000	0001 1111 1111	"	"	V_{172}	"	"		"	"	"	"
	LE_{23}	156	"	"	1111 0000 0000	0000 1111 1111	"	"	V_{173}	"	"		"	"	"	"
					A. Record the DUT code-word, (XXXX 0000 0000), with the most positive error.											
					B. Record the DUT code-word, (YYYY 0000 0000), with the most negative error.											
					Determine the sum of the recorded code-words obtained from tests 134 - 141 and from tests 142 - 156 for positive errors. Measure and calculate linearity error for the new DUT code-word (XXXX BBBB BBBB). (See Table I, note 6/).											
	LE_{24}	157	5 V	-15 V	XXXX BBBB BBBB XXXX BBBB BBBB 1111		K_1,K_2	21	V_{174}	V	5	$LE_{24} = \frac{410}{G_1}\left(V_{174} - \left((\text{on-bits}) \times S\right) - V_{149}\right)$	01, 02	-0.5	0.5	LSB
					Determine the sum of the recorded code-words obtained from tests 134 - 141 and from tests 142 - 156 for negative errors. Measure and calculate linearity error for the new DUT code-word (YYYY CCCC CCCC). (See Table I, note 6/).											
	LE_{25}	158	5 V	-15 V	YYYY CCCC CCCC YYYY CCCC CCCC 1111		K_1,K_2	21	V_{175}	V	5	$LE_{25} = \frac{410}{G_1}\left(V_{175} - \left((\text{on-bits}) \times S\right) - V_{149}\right)$	01, 02	-0.5	0.5	LSB

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Max	Unit
4 T_A = 25°C	MCE	159	5 V	-15 V	0111 1111 1111	1000 0000 0000	K_1,K_2	21	V_{176}	V	5	For each set of major carry codes N and N + 1, calculate the associated error in LSB's as follows:	01, 02	-1.0	1.0	LSB
			"	"	1000 0000 0000	0111 1111 1111	"	"	V_{177}	"	"		"	"	"	"
		160	"	"	0011 1111 1111	1100 0000 0000	"	"	V_{178}	"	"		"	"	"	"
			"	"	0100 0000 0000	1011 1111 1111	"	"	V_{179}	"	"		"	"	"	"
		161	"	"	0001 1111 1111	1110 0000 0000	"	"	V_{180}	"	"	$MCE = 410 \left\{ \left[V_m(N) - V_m(N-1) \right] / G_1 - \frac{10}{4096} \right\}$	"	"	"	"
			"	"	0010 0000 0000	1101 1111 1111	"	"	V_{181}	"	"		"	"	"	"
		162	"	"	0000 1111 1111	1111 0000 0000	"	"	V_{182}	"	"		"	"	"	"
			"	"	0001 0000 0000	1110 1111 1111	"	"	V_{183}	"	"		"	"	"	"
		163	"	"	0000 0111 1111	1111 1000 0000	"	"	V_{184}	"	"	Where V_m = measured value V_{176} through V_{199}	"	"	"	"
			"	"	0000 1000 0000	1111 0111 1111	"	"	V_{185}	"	"		"	"	"	"
		164	"	"	0000 0011 1111	1111 1100 0000	"	"	V_{186}	"	"		"	"	"	"
			"	"	0000 0100 0000	1111 1011 1111	"	"	V_{187}	"	"	"N and N-1" are the decimal equivalents of the code words in column "7 thru 18"	"	"	"	"
		165	"	"	0000 0001 1111	1111 1110 0000	"	"	V_{188}	"	"		"	"	"	"
			"	"	0000 0010 0000	1111 1101 1111	"	"	V_{189}	"	"		"	"	"	"
		166	"	"	0000 0000 1111	1111 1111 0000	"	"	V_{190}	"	"		"	"	"	"
			"	"	0000 0001 0000	1111 1110 1111	"	"	V_{191}	"	"	Example: For the first set	"	"	"	"
		167	"	"	0000 0000 0111	1111 1111 1000	"	"	V_{192}	"	"	N = 1000 0000 0000 = 2° = 2048	"	"	"	"
			"	"	0000 0000 1000	1111 1111 0111	"	"	V_{193}	"	"	N-1 = 0111 1111 1111 = 2° -1 = 2047	"	"	"	"
		168	"	"	0000 0000 0011	1111 1111 1100	"	"	V_{194}	"	"		"	"	"	"
			"	"	0000 0000 0100	1111 1111 1011	"	"	V_{195}	"	"	and	"	"	"	"
		169	"	"	0000 0000 0001	1111 1111 1110	"	"	V_{196}	"	"	$V_m(N) = V_{177}$	"	"	"	"
			"	"	0000 0000 0010	1111 1111 1101	"	"	V_{197}	"	"	$V_m(N-1) = V_{176}$	"	"	"	"
		170	"	"	0000 0000 0000	1111 1111 1111	"	"	V_{198}	"	"		"	"	"	"
			"	"	0000 0000 0001	1111 1111 1110	"	"	V_{199}	"	"		"	"	"	"
5 T_A = 125°C	I_{CC}	171	15 V	-15 V	1111 1111 1111	1111 1111 1111	K_2,K_3, K_4,K_6	4	I_{34}	mA	"	$I_{CC} = I_{34}$	01 02	3 3	18 20	mA "
	I_{EE}	172	"	"	0000 0000 0000	1111 1111 1111	K_2,K_3, K_4,K_6	6	I_{35}	mA	"	$I_{EE} = I_{35}$	01, 02	-25	-5	"
	V_{REF}	173	"	"			K_4,K_6	36	V_{200}	V	"	$V_{REF} = V_{200}$	02	2.45	2.55	"
	I_{IH1}	174	"	"	1111 1111 1111	1111 1111 1111	K_6	7	I_{36}	µA	"	$I_{IH1} = I_{36}$ V_{IN} = 10.50 V	01, 02	-1	100	µA
	I_{IH2}	175	"	"	"	"	"	8	I_{37}	"	"	$I_{IH2} = I_{37}$ "	"	"	"	"
	I_{IH3}	176	"	"	"	"	"	9	I_{38}	"	"	$I_{IH3} = I_{38}$ "	"	"	"	"
	I_{IH4}	177	"	"	"	"	"	10	I_{39}	"	"	$I_{IH4} = I_{39}$ "	"	"	"	"
	I_{IH5}	178	"	"	"	"	"	11	I_{40}	"	"	$I_{IH5} = I_{40}$ "	"	"	"	"
	I_{IH6}	179	"	"	"	"	"	12	I_{41}	"	"	$I_{IH6} = I_{41}$ "	"	"	"	"
	I_{IH7}	180	"	"	"	"	"	13	I_{42}	"	"	$I_{IH7} = I_{42}$ "	"	"	"	"
	I_{IH8}	181	"	"	"	"	"	14	I_{43}	"	"	$I_{IH8} = I_{43}$ "	"	"	"	"
	I_{IH9}	182	"	"	"	"	"	15	I_{44}	"	"	$I_{IH9} = I_{44}$ "	"	"	"	"
	I_{IH10}	183	"	"	"	"	"	16	I_{45}	"	"	$I_{IH10} = I_{45}$ "	"	"	"	"
	I_{IH11}	184	"	"	"	"	"	17	I_{46}	"	"	$I_{IH11} = I_{46}$ "	"	"	"	"
	I_{IH12}	185	"	"	"	"	"	18	I_{47}	"	"	$I_{IH12} = I_{47}$ "	"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Max	Unit
5 T_A = 125°C	I_{IL1}	186	15 V	-15 V	0000 0000 0000	1111 1111 1111	K_6	7	I_{48}	µA	5	$I_{IL1} = I_{48}$ V_{IN} = 0 V	01, 02	-200	1	µA
	I_{IL2}	187	"	"	"	"	"	8	I_{49}	"	"	$I_{IL2} = I_{49}$ "	"	"	"	"
	I_{IL3}	188	"	"	"	"	"	9	I_{50}	"	"	$I_{IL3} = I_{50}$ "	"	"	"	"
	I_{IL4}	189	"	"	"	"	"	10	I_{51}	"	"	$I_{IL4} = I_{51}$ "	"	"	"	"
	I_{IL5}	190	"	"	"	"	"	11	I_{52}	"	"	$I_{IL5} = I_{52}$ "	"	"	"	"
	I_{IL6}	191	"	"	"	"	"	12	I_{53}	"	"	$I_{IL6} = I_{53}$ "	"	"	"	"
	I_{IL7}	192	"	"	"	"	"	13	I_{54}	"	"	$I_{IL7} = I_{54}$ "	"	"	"	"
	I_{IL8}	193	"	"	"	"	"	14	I_{55}	"	"	$I_{IL8} = I_{55}$ "	"	"	"	"
	I_{IL9}	194	"	"	"	"	"	15	I_{56}	"	"	$I_{IL9} = I_{56}$ "	"	"	"	"
	I_{IL10}	195	"	"	"	"	"	16	I_{57}	"	"	$I_{IL10} = I_{57}$ "	"	"	"	"
	I_{IL11}	196	"	"	"	"	"	17	I_{58}	"	"	$I_{IL11} = I_{58}$ "	"	"	"	"
	I_{IL12}	197	"	"	"	"	"	18	I_{59}	"	"	$I_{IL12} = I_{59}$ "	"	"	"	"
	I_{FS}	198	5 V	"	1111 1111 1111	1111 1111 1111	"	23	I_{60}	mA	"	$I_{FS} = I_{60}$	"	-2.7	-1.6	mA
	$+P_{SS1}$	199	5 V 4.5 V 4.75 V	"	1111 1111 1111	0000 0000 0000	K_1, K_2	21 "	V_{201} V_{202}	V "	"	$+P_{SS1}$ = (outputs on) $\left\{ \left\| \frac{V_{202} - V_{201}}{G_{1H}} \right\| \times 41 \text{ for } - \Delta V_{CC} \right.$; $\left. \left\| \frac{V_{203} - V_{201}}{G_{1H}} \right\| \times 41 \text{ for } + \Delta V_{CC} \right\}$	01 02	-0.07 -0.25	0.07 0.25	LSB/%PS "
		200	5.5 V 5.25 V 2/	"	1111 1111 1111	0000 0000 0000	"	"	V_{203}	"	"					"
		201	5 V 4.5 V 4.75 V	"	0000 0000 0000	1111 1111 1111	"	" "	V_{204} V_{205}	" "	" "	$+P_{SS1}$ = (outputs off) $\left\{ \left\| \frac{V_{205} - V_{204}}{G_{1H}} \right\| \times 41 \text{ for } - \Delta V_{CC} \right.$; $\left. \left\| \frac{V_{206} - V_{204}}{G_{1H}} \right\| \times 41 \text{ for } + \Delta V_{CC} \right\}$	01 02	-0.07 -0.25	0.07 0.25	"
		202	5.5 V 5.25 V 2/	"	0000 0000 0000	1111 1111 1111	"	"	V_{206}	"	"					"
	$+P_{SS2}$	203	15 V 13.5 V 14.25 V	"	1111 1111 1111	0000 0000 0000	K_1, K_2, K_6	" "	V_{207} V_{208}	" "	"	$+P_{SS2}$ = (outputs on) $\left\{ \left\| \frac{V_{208} - V_{207}}{G_{1H}} \right\| \times 41 \text{ for } - \Delta V_{CC} \right.$; $\left. \left\| \frac{V_{209} - V_{207}}{G_{1H}} \right\| \times 41 \text{ for } + \Delta V_{CC} \right\}$	01 02	-0.07 -0.25	0.07 0.25	"
		204	16.5 V 15.75 V 2/	"	1111 1111 1111	0000 0000 0000	"	"	V_{209}	"	"					"
		205	15 V 13.5 V 14.25 V	"	0000 0000 0000	1111 1111 1111	"	" "	V_{210} V_{211}	" "	" "	$+P_{SS2}$ = (outputs off) $\left\{ \left\| \frac{V_{211} - V_{210}}{G_{1H}} \right\| \times 41 \text{ for } - \Delta V_{CC} \right.$; $\left. \left\| \frac{V_{212} - V_{210}}{G_{1H}} \right\| \times 41 \text{ for } + \Delta V_{CC} \right\}$	01 02	-0.07 -0.25	0.07 0.25	"
		206	16.5 V 15.75 V 2/	"	0000 0000 0000	1111 1111 1111	"	"	V_{212}	"	"					"
	$-P_{SS}$	207	5 V "	-15 V -13.5 V -14.25 V	1111 1111 1111	0000 0000 0000	K_1, K_2	" "	V_{213} V_{214}	" "	"	$-P_{SS}$ = (outputs on) $\left\{ \left\| \frac{V_{214} - V_{213}}{G_{1H}} \right\| \times 41 \text{ for } - \Delta V_{CC} \right.$; $\left. \left\| \frac{V_{215} - V_{213}}{G_{1H}} \right\| \times 41 \text{ for } + \Delta V_{CC} \right\}$	01 02	-0.15 -0.5	0.15 0.5	"
		208	" "	-16.5 V -15.75 V 2/	1111 1111 1111	0000 0000 0000	"	"	V_{215}	"	"					"
		209	" "	-15 V -13.5 V -14.25 V	0000 0000 0000	1111 1111 1111	"	" "	V_{216} V_{217}	" "	" "	$-P_{SS}$ = (outputs off) $\left\{ \left\| \frac{V_{217} - V_{216}}{G_{1H}} \right\| \times 41 \text{ for } - \Delta V_{CC} \right.$; $\left. \left\| \frac{V_{218} - V_{216}}{G_{1H}} \right\| \times 41 \text{ for } + \Delta V_{CC} \right\}$	01 02	-0.15 -0.5	0.15 0.5	"
		210	" "	-16.5 V -15.75 V 2/	0000 0000 0000	1111 1111 1111	"	"	V_{218}	"	"					"

See footnotes at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/				Energized relays	Measured pin			Figure	Equation		Device types	Limits		Unit
			4	6	7 thru 18	24 thru 35		No.	Value	Unit					Min	Max	
6 T_A = -55°C	I_{CC}	211	15 V "	-15 V "	1111 1111 1111	1111 1111 1111	K_2,K_3, K_4,K_6	4	I_{61}	mA	5	$I_{CC} = I_{61}$		01 02	3 3	18 20	mA "
	I_{EE}	212	"	"	0000 0000 0000	1111 1111 1111	K_2,K_3, K_4,K_6	6	I_{62}	mA	"	$I_{EE} = I_{62}$		01, 02	-25	-5	"
	V_{REF}	213	"	"			K_4,K_6		V_{219}	V	"	$V_{REF} = V_{219}$		02	2.45	2.55	V
	I_{IH1}	214	"	"	1111 1111 1111	1111 1111 1111	K_6	7	I_{63}	µA	"	$I_{IH1} = I_{63}$	V_{IH} = 10.50 V	01, 02	-1	100	µA
	I_{IH2}	215	"	"	"	"	"	8	I_{64}	"	"	$I_{IH2} = I_{64}$	"	"	"	"	"
	I_{IH3}	216	"	"	"	"	"	9	I_{65}	"	"	$I_{IH3} = I_{65}$	"	"	"	"	"
	I_{IH4}	217	"	"	"	"	"	10	I_{66}	"	"	$I_{IH4} = I_{66}$	"	"	"	"	"
	I_{IH5}	218	"	"	"	"	"	11	I_{67}	"	"	$I_{IH5} = I_{67}$	"	"	"	"	"
	I_{IH6}	219	"	"	"	"	"	12	I_{68}	"	"	$I_{IH6} = I_{68}$	"	"	"	"	"
	I_{IH7}	220	"	"	"	"	"	13	I_{69}	"	"	$I_{IH7} = I_{69}$	"	"	"	"	"
	I_{IH8}	221	"	"	"	"	"	14	I_{70}	"	"	$I_{IH8} = I_{70}$	"	"	"	"	"
	I_{IH9}	222	"	"	"	"	"	15	I_{71}	"	"	$I_{IH9} = I_{71}$	"	"	"	"	"
	I_{IH10}	223	"	"	"	"	"	16	I_{72}	"	"	$I_{IH10} = I_{72}$	"	"	"	"	"
	I_{IH11}	224	"	"	"	"	"	17	I_{73}	"	"	$I_{IH11} = I_{73}$	"	"	"	"	"
	I_{IH12}	225	"	"	"	"	"	18	I_{74}	"	"	$I_{IH12} = I_{74}$	"	"	"	"	"
	I_{IL1}	226	"	"	0000 0000 0000	1111 1111 1111	"	7	I_{75}	"	"	$I_{IL1} = I_{75}$	V_{IN} = 0 V	01, 02	-200	1	"
	I_{IL2}	227	"	"	"	"	"	8	I_{76}	"	"	$I_{IL2} = I_{76}$	"	"	"	"	"
	I_{IL3}	228	"	"	"	"	"	9	I_{77}	"	"	$I_{IL3} = I_{77}$	"	"	"	"	"
	I_{IL4}	229	"	"	"	"	"	10	I_{78}	"	"	$I_{IL4} = I_{78}$	"	"	"	"	"
	I_{IL5}	230	"	"	"	"	"	11	I_{79}	"	"	$I_{IL5} = I_{79}$	"	"	"	"	"
	I_{IL6}	231	"	"	"	"	"	12	I_{80}	"	"	$I_{IL6} = I_{80}$	"	"	"	"	"
	I_{IL7}	232	"	"	"	"	"	13	I_{81}	"	"	$I_{IL7} = I_{81}$	"	"	"	"	"
	I_{IL8}	233	"	"	"	"	"	14	I_{82}	"	"	$I_{IL8} = I_{82}$	"	"	"	"	"
	I_{IL9}	234	"	"	"	"	"	15	I_{83}	"	"	$I_{IL9} = I_{83}$	"	"	"	"	"
	I_{IL10}	235	"	"	"	"	"	16	I_{84}	"	"	$I_{IL10} = I_{84}$	"	"	"	"	"
	I_{IL11}	236	"	"	"	"	"	17	I_{85}	"	"	$I_{IL11} = I_{85}$	"	"	"	"	"
	I_{IL12}	237	"	"	"	"	"	18	I_{86}	"	"	$I_{IL12} = I_{86}$	"	"	"	"	"
	I_{FS}	238	5 V	"	1111 1111 1111	1111 1111 1111	"	23	I_{87}	mA	"	$I_{FS} = I_{87}$		"	-2.7	-1.6	mA

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Limits Max	Unit
6 T_A = -55°C	$+P_{SS1}$	239	5 V 4.5 V 4.75 V	-15 V "	1111 1111 1111	0000 0000 0000 0000 0000 0000	K_1, K_2	21 "	V_{220} V_{221}	V "	5 "	$+P_{SS1}$ (outputs on) = $\left\{ \left\| \frac{V_{221} - V_{220}}{G_{1L}} \right\| \times 41 \text{ for } -\Delta V_{CC} \right.$ $\left. \left\| \frac{V_{222} - V_{220}}{G_{1L}} \right\| \times 41 \text{ for } +\Delta V_{CC} \right\}$	01 02	-0.07 -0.25	0.07 0.25	LSB/%PS "
		240	5.5 V 5.25 V 2/	"	1111 1111 1111	0000 0000 0000	"	"	V_{222}	"	"					"
		241	5 V 4.5 V 4.75 V	"	0000 0000 0000 0000 0000 0000	1111 1111 1111 1111 1111 1111	"	" "	V_{223} V_{224}	" "	" "	$+P_{SS1}$ (outputs off) = $\left\{ \left\| \frac{V_{224} - V_{223}}{G_{1L}} \right\| \times 41 \text{ for } -\Delta V_{CC} \right.$ $\left. \left\| \frac{V_{225} - V_{223}}{G_{1L}} \right\| \times 41 \text{ for } +\Delta V_{CC} \right\}$	01 02	-0.07 -0.25	0.07 0.25	" "
		242	5.5 V 5.25 V 2/	"	0000 0000 0000	1111 1111 1111	"	"	V_{225}	"	"					"
	$+P_{SS2}$	243	15 V 13.5 V 14.25 V	"	1111 1111 1111	0000 0000 0000 0000 0000 0000	K_1, K_2, K_6	" "	V_{226} V_{227}	" "	" "	$+P_{SS2}$ (outputs on) = $\left\{ \left\| \frac{V_{227} - V_{226}}{G_{1L}} \right\| \times 41 \text{ for } -\Delta V_{CC} \right.$ $\left. \left\| \frac{V_{228} - V_{226}}{G_{1L}} \right\| \times 41 \text{ for } +\Delta V_{CC} \right\}$	01 02	-0.07 -0.25	0.07 0.25	" "
		244	16.5 V 15.75 V 2/	"	1111 1111 1111	0000 0000 0000	"	"	V_{228}	"	"					"
		245	15 V 13.5 V 14.25 V	"	0000 0000 0000	1111 1111 1111 1111 1111 1111	"	" "	V_{229} V_{230}	" "	" "	$+P_{SS2}$ (outputs off) = $\left\{ \left\| \frac{V_{230} - V_{229}}{G_{1L}} \right\| \times 41 \text{ for } -\Delta V_{CC} \right.$ $\left. \left\| \frac{V_{231} - V_{229}}{G_{1L}} \right\| \times 41 \text{ for } +\Delta V_{CC} \right\}$	01 02	-0.07 -0.25	0.07 0.25	" "
		246	16.5 V 15.75 V	"	0000 0000 0000	1111 1111 1111	"	"	V_{231}	"	"					"
	$-P_{SS}$	247	5 V "	-15 V -13.5 V	1111 1111 1111	0000 0000 0000 0000 0000 0000	K_1, K_2	" "	V_{232} V_{233}	" "	" "	$-P_{SS}$ (outputs on) = $\left\{ \left\| \frac{V_{233} - V_{232}}{G_{1L}} \right\| \times 41 \text{ for } -\Delta V_{CC} \right.$ $\left. \left\| \frac{V_{234} - V_{232}}{G_{1L}} \right\| \times 41 \text{ for } +\Delta V_{CC} \right\}$	01 02	-0.15 -0.5	0.15 0.5	" "
		248	" "	-14.25 V -16.5 V	1111 1111 1111	0000 0000 0000	"	"	V_{234}	"	"					"
		249	" "	-15.75 V 2/ -15 V -13.5 V	0000 0000 0000	1111 1111 1111 1111 1111 1111	"	" "	V_{235} V_{236}	" "	" "	$-P_{SS}$ (outputs off) = $\left\{ \left\| \frac{V_{236} - V_{235}}{G_{1L}} \right\| \times 41 \text{ for } -\Delta V_{CC} \right.$ $\left. \left\| \frac{V_{237} - V_{235}}{G_{1L}} \right\| \times 41 \text{ for } +\Delta V_{CC} \right\}$	01 02	-0.15 -0.5	0.15 0.5	" "
		250	" "	-14.25 V -16.5 V -15.75 V 2/	0000 0000 0000	1111 1111 1111	"	"	V_{237}	"	"					"
7 T_A = 25°C	LE		"	-15 V	1111 1111 1111	0000 0000 0000	"	"	V_{238}	"	"	$LE = \frac{410}{G_1} \left\{ V_A - \left[\frac{V_{238} - V_{239}}{4095} \right] N - V_{238} \right\}$	01, 02			
			"	"	0000 0000 0000	1111 1111 1111	"	"	V_{239}	"	"		"			
			"	"	0000 0000 0001	1111 1111 1110	"	"	V_{240}	"	"		"			
			"	"	0000 0000 0010	1111 1111 1101	"	"	V_{241}	"	"		"			
			"	"	0000 0000 0011	1111 1111 1100	"	"	V_{242}	"	"		"			
			"	"	0000 0000 0100	1111 1111 1011	"	"	V_{243}	"	"		"			
			"	"			"	"	$V_{...}$	"	"	Where V_A = measured value from V_{240} through V_{4333}	"			
			"	"			"	"	$V_{...}$	"	"		"			
			"	"	1111 1111 1100	0000 0000 0011	"	"	V_{4331}	"	"		"			
			"	"	1111 1111 1101	0000 0000 0010	"	"	V_{4332}	"	"		"			
			"	"	1111 1111 1110	0000 0000 0001	"	"	V_{4333}	"	"		"			
		251	"	"			"	"		"	"	Worst positive error	"	0	0.5	LSB
			"	"			"	"		"	"	Worst positive code	"	-	---	OCT
			"	"			"	"		"	"	Worst negative error	"	0	0.5	LSB
		252	"	"			"	"		"	"	Worst negative code	"	-	---	OCT

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TABLE III. Group A inspection for types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	24 thru 35	Energized relays	Measured pin No.	Value	Unit	Figure	Equation	Device types	Limits Min	Max	Unit
8 T_A = 125°C	LE		5 V	-15 V	1111 1111 1111	0000 0000 0000	K_1,K_2	21	V_{4334}	V	5	For each input code N, calculate the corresponding linearity error in LSB's.	01, 02			
			"	"	0000 0000 0000	1111 1111 1111	"	"	V_{4335}	"	"		"			
			"	"	0000 0000 0001	1111 1111 1110	"	"	V_{4336}	"	"		"			
			"	"	0000 0000 0010	1111 1111 1101	"	"	V_{4337}	"	"		"			
			"	"	0000 0000 0011	1111 1111 1100	"	"	V_{4338}	"	"		"			
			"	"	0000 0000 0100	1111 1111 1011	"	"	V_{4339}	"	"		"			
			"	"			"	"		"	"		"			
			"	"			"	"		"	"		"			
			"	"			"	"		"	"		"			
			"	"			"	"		"	"		"			
			"	"	1111 1111 1100	0000 0000 0011	"	"	V_{8427}	"	"		"			
			"	"	1111 1111 1101	0000 0000 0010	"	"	V_{8428}	"	"		"			
			"	"	1111 1111 1110	0000 0000 0001	"	"	V_{8429}	"	"		"			
		253	"	"			"	"		"	"	Worst positive error	"	0	0.5	LSB
			"	"			"	"		"	"	Worst positive code	"	-	---	OCT
		254	"	"			"	"		"	"	Worst negative error	"	0	0.5	LSB
			"	"			"	"		"	"	Worst negative code	"	-	---	OCT
	$\frac{dI_{ZS}}{dt}$	255	Calculate dI_{ZS}/dt = (I_{ZS1} at 125°C - I_{ZS1} at 25°C)/100°C from tests 42 and 130 respectively.										01, 02	-0.01	0.01	LSB/°C
	$\frac{dA_{E1}}{dt}$	256	Calculate dA_E/dt = (A_E at 125°C - A_E at 25°C)/100°C from tests 44 and 132 respectively.										01 02	-5.0 -30.0	5.0 30.0	PPM/°C PPM/°C
	$\frac{dB_{POE}}{dt}$	257	Calculate dB_{POE}/dt = (B_{POE} at 125°C - B_{POE} at 25°C)/100°C from tests 45 and 128 respectively.										01 02	-0.02 -0.12	0.2 0.12	LSB/°C "
	$\frac{dB_{ZE}}{dt}$	258	Calculate dB_{ZE}/dt = (B_{ZE} at 125°C - B_{ZE} at 25°C)/100°C from test 46 and 129 respectively.										01 02	-0.02 -0.04	0.02 0.04	" "
8 T_A = -55°C	LE	259 thru 260	Same tests, terminal conditions and limits as in subgroup 7, tests 251 thru 252 except T_A = -55°C (V_{OS} and V_{FS} are the values determined in subgroup 3.)										01, 02	-0.5 - 0 -	0.5 - 0.5 -	LSB OCT LSB OCT
	$\frac{dI_{ZS}}{dt}$	261	Calculate dI_{ZS}/dt = (I_{ZS1} at -55°C - I_{ZS1} at 25°C)/80°C from tests 85 and 130 respectively.										01, 02	-0.1	0.1	LSB/°C
	$\frac{dA_{E1}}{dt}$	262	Calculate dA_E/dt = (A_E at -55°C - A_E at 25°C)/80°C from tests 87 and 132 respectively.										01 02	-5.0 -30.0	5.0 30.0	PPM/°C PPM/°C
	$\frac{dB_{POE}}{dt}$	263	Calculate dB_{POE}/dt = (B_{POE} at -55°C - B_{POE} at 25°C)/80°C from tsts 88 and 128 respectively.										01 02	-0.02 -0.12	0.02 0.12	LSB/°C "
	$\frac{dB_{ZE}}{dt}$	264	Calculate dB_{ZE}/dt = (B_{ZE} at -55°C - B_{ZE} at 25°C)/80°C from tests 89 and 129 respectively.										01 02	-0.02 -0.04	0.02 0.04	" "

$$LE = \frac{410}{G_{IH}} \left\{ V_A - \left[\frac{V_{4334} - V_{4335}}{4095} \right] N - V_{4334} \right\}$$

Where V_A = measured from V_{4334} through V_{8429}

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	Test no.	Test condition	Measured pin			Equation	Device type	Limits		Unit
				No.	Value	Unit			Min	Max	
12 T_A = 25°C	t_{SLH}	265	V_{CC} = +5 V V_{EE} = -15 V V_{IN} switched from all inputs at logic "0" to all inputs at logic "1" figure 6	9	t_1	µs	$t_{SLH} = t_1$	01, 02	---	3.0	µs
	t_{SHL}	266	V_{CC} = +5 V V_{EE} = -15 V V_{IN} switched from all inputs at logic "1" to all inputs at logic "0" figure 6	9	t_2	µs	$t_{SHL} = t_2$	01, 02	---	3.0	µs
	R1	267	V_{CC} = +5 V V_{EE} = -15 V V_{ref} = 10 V figure 7	5	I_{88}	mA	$R1 = \frac{10}{I_{88}}$	01	15	25	kΩ
	N_{02}	268	V_{CC} = +5 V V_{EE} = -15 V V_{IN} = all inputs at logic "1"	9	V_{12526}	mV	$N_{02} = \frac{V_{12526}}{100}$	01 02	--- ---	60 100	µVrms

1/ Apply +15 Vdc to adapter pin 1.
Apply -15 Vdc adapter pin 2.
Connect power supply ground to adapter pins 3 and 5.

$$\frac{1}{01/4096} = 410$$

2/ The power supplies for device type 02 are varied by 5% (ie 15 x .05 = .75).

3/ The calculation of G1 can be found in test no. 132 in this table.

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TABLE III. Group A inspection for device types 03 and 04.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/				Energized relays	Measured pin			Equation 2/		Device types	Limits		Unit
			4	6	7 thru 18	22 thru 37		No.	Value	Unit				Min	Max	
Calibration		1	Adjust P6 with all bits at 0 V				None	21	E_1	µV	$E_{cal} = E_1$		03, 04	-	100	µV
1/ $T_A = 25°C$	I_{CC}	2	+15 V	-15 V	1111 1111 1111	1111 1111 1111 1111	"	4	I_1	mA	$I_{CC} = I_1$	(Digital 1 = 2.4 V)	03	1	5	mA
	I_{EE}	3	"	"	0000 0000 0000	1111 1111 1111 1111	"	6	I_2	mA	$I_{EE} = I_2$	(Digital 0 = 0.8 V)	03, 04	-18	-5	mA
	V_{REF}	4	"	"	0000 0000 0000	1111 1111 1111 1111	"	2	E_2	V	$V_{REF} = E_2$	Iref = 2 mA	03	9.90	10.1	V
	I_{IL}	5	"	"	0000 0000 0000	1111 1111 1111 1111	"	7	I_3	µA	$I_{IL} = I_3$	(Digital 0 = 0 V)	03, 04	5	100	µA
		6	"	"	"	"	"	8	I_4	"	$I_{IL} = I_4$	"	"	"	"	"
		7	"	"	"	"	"	9	I_5	"	$I_{IL} = I_5$	"	"	"	"	"
		8	"	"	"	"	"	10	I_6	"	$I_{IL} = I_6$	"	"	"	"	"
		9	"	"	"	"	"	11	I_7	"	$I_{IL} = I_7$	"	"	"	"	"
		10	"	"	"	"	"	12	I_8	"	$I_{IL} = I_8$	"	"	"	"	"
		11	"	"	"	"	"	13	I_9	"	$I_{IL} = I_9$	"	"	"	"	"
		12	"	"	"	"	"	14	I_{10}	"	$I_{IL} = I_{10}$	"	"	"	"	"
		13	"	"	"	"	"	15	I_{11}	"	$I_{IL} = I_{11}$	"	"	"	"	"
		14	"	"	"	"	"	16	I_{12}	"	$I_{IL} = I_{12}$	"	"	"	"	"
		15	"	"	"	"	"	17	I_{13}	"	$I_{IL} = I_{13}$	"	"	"	"	"
		16	"	"	"	"	"	18	I_{14}	"	$I_{IL} = I_{14}$	"	"	"	"	"
	I_{IH}	17	"	"	1111 1111 1111	1111 1111 1111 1111	"	7	I_{15}	"	$I_{IH} = I_{15}$	(Digital 1 = 5.5 V)	"	145	300	"
		18	"	"	"	"	"	8	I_{16}	"	$I_{IH} = I_{16}$	"	"	"	"	"
		19	"	"	"	"	"	9	I_{17}	"	$I_{IH} = I_{17}$	"	"	"	"	"
		20	"	"	"	"	"	10	I_{18}	"	$I_{IH} = I_{18}$	"	"	"	"	"
		21	"	"	"	"	"	11	I_{19}	"	$I_{IH} = I_{19}$	"	"	"	"	"
		22	"	"	"	"	"	12	I_{20}	"	$I_{IH} = I_{20}$	"	"	"	"	"
		23	"	"	"	"	"	13	I_{21}	"	$I_{IH} = I_{21}$	"	"	"	"	"
		24	"	"	"	"	"	14	I_{22}	"	$I_{IH} = I_{22}$	"	"	"	"	"
		25	"	"	"	"	"	15	I_{23}	"	$I_{IH} = I_{23}$	"	"	"	"	"
		26	"	"	"	"	"	16	I_{24}	"	$I_{IH} = I_{24}$	"	"	"	"	"
		27	"	"	"	"	"	17	I_{25}	"	$I_{IH} = I_{25}$	"	"	"	"	"
		28	"	"	"	"	"	18	I_{26}	"	$I_{IH} = I_{26}$	"	"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/				Energized relays	Measured pin			Equation 2/	Device types	Limits		Unit
			4	6	7 thru 18	22 thru 37		No.	Value	Unit			Min	Max	
1 T_A = 25°C	I_{FS}	29	+15 V	-15 V	1111 1111 1111	1111 1111 1111 1111	K_1	19	I_{27}	mA	$I_{FS} = I_{27}$	03, 04	-2.7	-1.6	mA
	$+P_{SS}$	30	+13.5 V	"	"	0000 0000 0000 1111	None	21	E_3	V	$+P_{SS} = 0.41\ (E_4 - E_3)$	03	-0.04	0.04	LSB/%PS
		31	+15 V	"	"	"	"	"	E_4	"		"	"	"	"
		32	+16.5 V	"	"	"	"	"	E_5	"	$+P_{SS} = 0.41\ (E_4 - E_5)$	"	"	"	"
	$-P_{SS}$	33	+5 V	-13.5 V	"	"	"	"	E_6	"	$-P_{SS} = 0.41\ (E_6 - E_7)$	03, 04	-0.1	0.1	"
		34	"	-15 V	"	"	"	"	E_7	"		"	"	"	"
		35	"	-10.5 V	"	"	"	"	E_8	"	$-P_{SS} = 0.41\ (E_6 - E_8)$	"	"	"	"
2 T_A = 125°C	V_{REF}	36	+15 V	-15 V	0000 0000 0000	1111 1111 1111 1111	"	2	E_9	"	$V_{REF} = E_9 \qquad I_{REF} = 2$ mA	03	9.90	10.0	V
	I_{ZS1}	37	"	"	0000 0000 0000	1111 1111 1111 1111	"	19	I_{28}	µA	$I_{ZS1} = (I_{28})\ (4096)/(I_{56} - I_{28})$	03, 04	-3	3	LSB
	A_E	38	"	"	1111 1111 1111	0000 0000 0000 1111	"	21	E_{10}	V	$A_E = 0.1\ (E_{10} - E_{11})$	"	-.20	.20	%
			"	"	0000 0000 0000	1111 1111 1111 1111	"	"	E_{11}	"					
	B_{POE}	39	"	"	0000 0000 0000	1111 1111 1111 1111	K_2,K_3	"	E_{12}	"	$B_{POE} = (-2.05)\ (E_{12})$	"	-8	8	LSB
	B_{PAE}	40	"	"	1111 1111 1111	0000 0000 0000 1111	K_2	"	E_{13}	"	$B_{PAE} = 0.05\ (E_{12} - E_{13})$	"	-.20	.20	%
	B_{7E}	41	"	"	1000 0000 0000	0111 1111 1111 1111	K_2	"	E_{14}	"	$B_{7E} = (-2.05)\ (E_{14})$	"	-6.0	6.0	LSB
	LE_1	42	"	"	0001 0000 0000	1110 1111 1111 1111	None	"	E_{15}	"	For each input code N calculate the corresponding linearity error in LSB. $LE = 4.1\left[(E_{10} - E_{11})\ (N/4095) + E_{11} - EA\right]$ where EA = measured value E_{15} through E_{37} N = DUT input digital code word in decimal form. Thus for 12 bits N = (A1)(2 exp 11)+(A2)(2 exp 10)+ ... +(A12)(2 exp 0) A1, A2, A12 are 0 or 1. E_3 and E_4 correspond to the error amplifier outputs at full scale and zero scale respectively.	"	-0.5	+0.5	"
		43	"	"	0010 0000 0000	1101 "	"	"	E_{16}	"		"	"	"	"
		44	"	"	0011 0000 0000	1100 "	"	"	E_{17}	"		"	"	"	"
		45	"	"	0100 0000 0000	1011 "	"	"	E_{18}	"		"	"	"	"
		46	"	"	0101 0000 0000	1010 "	"	"	E_{19}	"		"	"	"	"
		47	"	"	0110 0000 0000	1001 "	"	"	E_{20}	"		"	"	"	"
		48	"	"	0111 0000 0000	1000 "	"	"	E_{21}	"		"	"	"	"
		49	"	"	1000 0000 0000	0111 "	"	"	E_{22}	"		"	"	"	"
		50	"	"	1001 0000 0000	0110 "	"	"	E_{23}	"		"	"	"	"
		51	"	"	1010 0000 0000	0101 "	"	"	E_{24}	"		"	"	"	"
		52	"	"	1011 0000 0000	0100 "	"	"	E_{25}	"		"	"	"	"
		53	"	"	1100 0000 0000	0011 "	"	"	E_{26}	"		"	"	"	"
		54	"	"	1101 0000 0000	0010 "	"	"	E_{27}	"		"	"	"	"
		55	"	"	1110 0000 0000	0001 "	"	"	E_{28}	"		"	"	"	"
	LE_{15}	56	"	"	1111 0000 0000	0000 "	"	"	E_{29}	"		"	"	"	"
						A. Record the DUT code-word, (XXXX 0000 0000), with the most positive error.						"	"	"	"
						B. Record the DUT code-word, (YYYY 0000 0000), with the most negative error.						"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	22 thru 37	Energized relays	Measured pin No.	Value	Unit	Equation 2/	Device types	Limits Min	Max	Unit
2/ T_A = 125°C	LE_{16}	57	+15 V	-15 V	0000 0000 0001	1111 1111 1111 1111	None	21	E_{30}	V	$S = \left[\frac{E_{10} - E_{11}}{4095}\right]$	03, 04	-0.5	0.5	LSB
		58	"	"	0000 0000 0010	1111 1111 1101 1111	"	"	E_{31}	"		"	"	"	"
		59	"	"	0000 0000 0100	1111 1111 1011 1111	"	"	E_{32}	"		"	"	"	"
	LE_{19}	60	"	"	0000 0000 1000	1111 1111 0111 1111	"	"	E_{33}	"		"	"	"	"
	LE_{20}	61	"	"	0000 0001 0000	1111 1110 1111 1111	"	"	E_{34}	"		"	"	"	"
		62	"	"	0000 0010 0000	1111 1101 1111 1111	"	"	E_{35}	"		"	"	"	"
		63	"	"	0000 0100 0000	1111 1011 1111 1111	"	"	E_{36}	"		"	"	"	"
	LE_{23}	64	"	"	0000 1000 0000	1111 0111 1111 1111	"	"	E_{37}	"		"	"	"	"
			"	"	A. Record, (0000 BBBB BBBB), the sum of the codewords from tests 57 - 64 with positive linearity errors.							"	"	"	"
			"	"	B. Record, (0000 CCCC CCCC), the sum of the codewords from tests 57 - 64 with negative linearity errors.							"	"	"	"
			"	"	Determine the sum of the recorded code-words obtained from tests 42 - 56 and from tests 57 - 64 for positive errors.							"	"	"	"
			"	"	Measure and calculate linearity error for the new DUT code-word (XXXX BBBB BBBB). (See Table I, Note 6/)							"	"	"	"
	LE_{24}	65	"	"	XXXX BBBB BBBB XXXX BBBB BBBB 1111		None	21	E_{38}	V	$LE_{24} = 4.1\left(E_{11} + \left((\text{on-bits}) \times S\right) - E_{38}\right)$	"	"	"	"
			"	"	Determine the sum of the recorded code-words obtained from tests 42 - 56 and from tests 57 - 64 for negative errors.							"	"	"	"
			"	"	Measure and calculate linearity error for the new DUT code-word (YYYY CCCC CCCC). (See Table I, Note 6/)							"	"	"	"
	LE_{25}	66	"	"	YYYY CCCC CCCC YYYY CCCC CCCC 1111		None	21	E_{39}	V	$LE_{25} = 4.1\left(E_{11} + \left((\text{on-bits}) \times S\right) - E_{39}\right)$	"	"	"	"
	MCE	67	"	"	0111 1111 1111	1000 0000 0000 1111	"	"	E_{40}	"	For each set of major carry codes N and N + 1 calculate the associated error in LSB's as follows:	"	-1.0	1.0	"
			"	"	1000 0000 0000	0111 1111 1111 1111	"	"	E_{41}	"		"	"	"	"
		68	"	"	0011 1111 1111	1100 0000 0000 1111	"	"	E_{42}	"		"	"	"	"
			"	"	0100 0000 0000	1011 1111 1111 1111	"	"	E_{43}	"	MCE = 4.1 (EA - EB)	"	"	"	"
		69	"	"	0001 1111 1111	1110 0000 0000 1111	"	"	E_{44}	"		"	"	"	"
			"	"	0010 0000 0000	1101 1111 1111 1111	"	"	E_{45}	"	where	"	"	"	"
		70	"	"	0000 1111 1111	1111 0000 0000 1111	"	"	E_{46}	"	EB = measured value from E_{40}, E_{42}, E_{44}, etc.	"	"	"	"
			"	"	0001 0000 0000	1110 1111 1111 1111	"	"	E_{47}	"		"	"	"	"
		71	"	"	0000 0111 1111	1111 1000 0000 1111	"	"	E_{48}	"	EA = measured value from E_{41}, E_{43}, E_{45}, etc.	"	"	"	"
			"	"	0000 1000 0000	1111 0111 1111 1111	"	"	E_{49}	"		"	"	"	"
		72	"	"	0000 0011 1111	1111 1100 0000 1111	"	"	E_{50}	"		"	"	"	"
			"	"	0000 0100 0000	1111 1011 1111 1111	"	"	E_{51}	"		"	"	"	"
		73	"	"	0000 0001 1111	1111 1110 0000 1111	"	"	E_{52}	"		"	"	"	"
			"	"	0000 0010 0000	1111 1101 1111 1111	"	"	E_{53}	"		"	"	"	"
		74	"	"	0000 0000 1111	1111 1111 0000 1111	"	"	E_{54}	"		"	"	"	"
			"	"	0000 0001 0000	1111 1110 1111 1111	"	"	E_{55}	"		"	"	"	"
		75	"	"	0000 0000 0111	1111 1111 1000 1111	"	"	E_{56}	"		"	"	"	"
			"	"	0000 0000 1000	1111 1111 0111 1111	"	"	E_{57}	"		"	"	"	"
		76	"	"	0000 0000 0011	1111 1111 1100 1111	"	"	E_{58}	"		"	"	"	"
			"	"	0000 0000 0100	1111 1111 1011 1111	"	"	E_{59}	"		"	"	"	"
		77	"	"	0000 0000 0001	1111 1111 1110 1111	"	"	E_{60}	"		"	"	"	"
			"	"	0000 0000 0010	1111 1111 1101 1111	"	"	E_{61}	"		"	"	"	"
		78	"	"	0000 0000 0000	1111 1111 1111 1111	"	"	E_{62}	"		"	"	"	"
			"	"	0000 0000 0001	1111 1111 1110 1111	"	"	E_{63}	"		"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/				Energized relays	Measured pin			Equation 2/	Device types	Limits		Unit
			4	6	7 thru 18	22 thru 37		No.	Value	Unit			Min	Max	
3 T_A = -55°C	V_{REF}	79	+15 V	-15 V	0000 0000 0000	1111 1111 1111 1111	None	2	E_{64}	V	$V_{REF} = E_{64}$ I_{REF} = 2 mA	03	9.90	10.1	V
	I_{ZS}	80	"	"	0000 0000 0000	1111 1111 1111 1111	K_1	19	I_{Z9}		$I_{ZS} = (I_{Z9})(4096)/(I_{B3} - I_{Z9})$	03, 04	-3	3	LSB
	A_E	81	" "	" "	1111 1111 1111 0000 0000 0000	0000 0000 0000 1111 1111 1111 1111 1111	" "	21 "	E_{65} E_{66}	V "	$A_E = 0.1 (E_{65} - E_{66})$	"	-.20	.20	%
	B_{POE}	82	"	"	0000 0000 0000	1111 1111 1111 1111	K_2,K_3	"	E_{67}	"	$B_{POE} = (-2.05)(E_{67})$	"	-8.0	8.0	LSB
	B_{PAE}	83	"	"	1111 1111 1111	0000 0000 0000 1111	K_2	"	E_{68}	"	$B_{PAE} = 0.05 (E_{68})$	"	-.20	.20	%
	B_{ZE}	84	"	"	1000 0000 0000	0111 1111 1111 1111	K_2	"	E_{69}	"	$B_{ZE} = (-2.05)(E_{69})$	"	-6.0	6.0	LSB
	LE_1	85	"	"	0001 0000 0000	1110 1111 1111 1111	None	"	E_{70}	"	For each input code N calculate the corresponding linearity error in LSB.	"	-0.5	+0.5	"
		86	"	"	0010 0000 0000	1101 "	"	"	E_{71}	"		"	"	"	"
		87	"	"	0011 0000 0000	1100 "	"	"	E_{72}	"		"	"	"	"
		88	"	"	0100 0000 0000	1011 "	"	"	E_{73}	"	$LE = 4.1 \left[(E_{65} - E_{66})(N/4095) + E_{66} - EA \right]$	"	"	"	"
		89	"	"	0101 0000 0000	1010 "	"	"	E_{74}	"		"	"	"	"
		90	"	"	0110 0000 0000	1001 "	"	"	E_{75}	"		"	"	"	"
		91	"	"	0111 0000 0000	1000 "	"	"	E_{76}	"	where	"	"	"	"
		92	"	"	1000 0000 0000	0111 "	"	"	E_{77}	"	EA = measured value E_{70} through E_{92}	"	"	"	"
		93	"	"	1001 0000 0000	0110 "	"	"	E_{78}	"	N = DUT input digital code word in decimal form.	"	"	"	"
		94	"	"	1010 0000 0000	0101 "	"	"	E_{79}	"	Thus for 12 bits N = (A1)(2 exp 11)+(A2)	"	"	"	"
		95	"	"	1011 0000 0000	0100 "	"	"	E_{80}	"	(2 exp 10)+ ... +(A12)(2 exp 0)	"	"	"	"
		96	"	"	1100 0000 0000	0011 "	"	"	E_{81}	"	A1, A2, A12 are 0 or 1.	"	"	"	"
		97	"	"	1101 0000 0000	0010 "	"	"	E_{82}	"	E_{65} and E_{66} correspond to the error amplifier	"	"	"	"
		98	"	"	1110 0000 0000	0001 "	"	"	E_{83}	"	outputs at full scale and zero scale respectively.	"	"	"	"
	LE_{15}	99	"	"	1111 0000 0000	0000 "	"	"	E_{84}	"		"	"	"	"
					A. Record the DUT code-word, (XXXX 0000 0000), with the most positive error. 2/							"	"	"	"
					B. Record the DUT code-word, (YYYY 0000 0000), with the most negative error. 2/							"	"	"	"
	LE_{16}	100	+15 V	-15 V	0000 0000 0001	1111 1111 1110 1111	None	21	E_{85}	V		03, 04	-0.5	0.5	LSB
		101	"	"	0000 0000 0010	1111 1111 1101 1111	"	"	E_{86}	"		"	"	"	"
		102	"	"	0000 0000 0100	1111 1111 1011 1111	"	"	E_{87}	"		"	"	"	"
	LE_{19}	103	"	"	0000 0000 1000	1111 1111 0111 1111	"	"	E_{88}	"		"	"	"	"
	LE_{20}	104	"	"	0000 0001 0000	1111 1110 1111 1111	"	"	E_{89}	"	$S = \left[\frac{E_{65} - E_{66}}{4095} \right]$	"	"	"	"
		105	"	"	0000 0010 0000	1111 1101 1111 1111	"	"	E_{90}	"		"	"	"	"
		106	"	"	0000 0100 0000	1111 1011 1111 1111	"	"	E_{91}	"		"	"	"	"
	LE_{23}	107	"	"	0000 1000 0000	1111 0111 1111 1111	"	"	E_{92}	"		"	"	"	"
			"	"	A. Record, (0000 BBBB BBBB), the sum of the codewords from tests 100 - 107 with positive linearity errors.							"	"	"	"
			"	"	B. Record, (0000 CCCC CCCC), the sum of the codewords from tests 100 - 107 with negative linearity errors.							"	"	"	"
			"	"	Determine the sum of the recorded code-words obtained from tests 85 - 99 and from tests 100 - 107 for positive errors.							"	"	"	"
			"	"	Measure and calculate linearity error for the new DUT code-word (XXXX BBBB BBBB). (See Table I, Note 6/)							"	"	"	"
	LE_{24}	108	"	"	XXXX BBBB BBBB XXXX BBBB BBBB 1111		None	21	E_{93}	V	$LE_{24} = 4.1 \left(E_{66} + \left((\text{on-bits}) \times S \right) - E_{93} \right)$	"	"	"	"
			"	"	Determine the sum of the recorded code-words obtained from tests 85 - 99 and from tests 100 - 107 for negative errors.							"	"	"	"
			"	"	Measure and calculate linearity error for the new DUT code-word (YYYY CCCC CCCC). (See Table I, Note 6/)							"	"	"	"
	LE_{25}	109	"	"	YYYY CCCC CCCC YYYY CCCC CCCC 1111		None	21	E_{94}	V	$LE_{25} = 4.1 \left(E_{66} + \left((\text{on-bits}) \times S \right) - E_{94} \right)$	"	"	"	"

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TABLE III. Group A inspection vice types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	22 thru 37	Energized relays	Measured pin No.	Value	Unit	Equation 2/	Device types	Limits Min	Max	Unit
3 T_A = -55°C	MCE	110	+15 V	-15 V	0111 1111 1111	1000 0000 0000 1111	None	21	E_{95}	V	For each set of major carry codes N and N + 1 calculate the associated error in LSB's as follows:	03, 04	-1.0	1.0	LSB
			"	"	1000 0000 0000	0111 1111 1111 1111	"	"	E_{96}	"		"	"	"	"
		111	"	"	0011 1111 1111	1100 0000 0000 1111	"	"	E_{97}	"		"	"	"	"
			"	"	0100 0000 0000	1011 1111 1111 1111	"	"	E_{98}	"	MCE = 4.1 (EA - EB)	"	"	"	"
		112	"	"	0001 1111 1111	1110 0000 0000 1111	"	"	E_{99}	"		"	"	"	"
			"	"	0010 0000 0000	1101 1111 1111 1111	"	"	E_{100}	"	where	"	"	"	"
		113	"	"	0000 1111 1111	1111 0000 0000 1111	"	"	E_{101}	"	EB = measured value from E_{95}, E_{97}, E_{99}, etc.	"	"	"	"
			"	"	0001 0000 0000	1110 1111 1111 1111	"	"	E_{102}	"		"	"	"	"
		114	"	"	0000 0111 1111	1111 1000 0000 1111	"	"	E_{103}	"	EA = measured value from E_{96}, E_{98}, E_{100}, etc.	"	"	"	"
			"	"	0000 1000 0000	1111 0111 1111 1111	"	"	E_{104}	"		"	"	"	"
		115	"	"	0000 0011 1111	1111 1100 0000 1111	"	"	E_{105}	"		"	"	"	"
			"	"	0000 0100 0000	1111 1011 1111 1111	"	"	E_{106}	"		"	"	"	"
		116	"	"	0000 0001 1111	1111 1110 0000 1111	"	"	E_{107}	"		"	"	"	"
			"	"	0000 0010 0000	1111 1101 1111 1111	"	"	E_{108}	"		"	"	"	"
		117	"	"	0000 0000 1111	1111 1111 0000 1111	"	"	E_{109}	"		"	"	"	"
			"	"	0000 0001 0000	1111 1110 1111 1111	"	"	E_{110}	"		"	"	"	"
		118	"	"	0000 0000 0111	1111 1111 1000 1111	"	"	E_{111}	"		"	"	"	"
			"	"	0000 0000 1000	1111 1111 0111 1111	"	"	E_{112}	"		"	"	"	"
		119	"	"	0000 0000 0011	1111 1111 1100 1111	"	"	E_{113}	"		"	"	"	"
			"	"	0000 0000 0100	1111 1111 1011 1111	"	"	E_{114}	"		"	"	"	"
		120	"	"	0000 0000 0001	1111 1111 1110 1111	"	"	E_{115}	"		"	"	"	"
			"	"	0000 0000 0010	1111 1111 1101 1111	"	"	E_{116}	"		"	"	"	"
		121	"	"	0000 0000 0000	1111 1111 1111 1111	"	"	E_{117}	"		"	"	"	"
			"	"	0000 0000 0001	1111 1111 1110 1111	"	"	E_{118}	"		"	"	"	"
4 T_A = 25°C	I_{ZS}	122	"	"	0000 0000 0000	1111 1111 1111 1111	K_1	19	I_{30}		$I_{ZS} = (I_{30})(4096)/(I_{27} - I_{30})$	03, 04	-2	2	LSB
	A_E	123	"	"	1111 1111 1111	0000 0000 0000 1111	None	21	E_{119}	V	$A_E = 0.1 (E_{119} - E_{120})$	"	-.20	.20	%
			"	"	0000 0000 0000	1111 1111 1111 1111		"	E_{120}	"					
	B_{POE}	124	"	"	0000 0000 0000	1111 1111 1111 1111	K_2,K_3	"	E_{121}	"	$B_{POE} = (-2.05)(E_{121})$	"	-8	8	LSB
	B_{PAE}	125	"	"	1111 1111 1111	0000 0000 0000 1111	K_2	"	E_{122}	"	$B_{PAE} = 0.05 (E_{122})$	"	-.20	.20	%
	B_{ZE}	126	"	"	1000 0000 0000	0111 1111 1111 1111	K_2	"	E_{123}	"	$B_{ZE} = (-2.05)(E_{123})$	"	-6.0	6.0	LSB
	LE_1	127	"	"	0001 0000 0000	1110 1111 1111 1111	None	"	E_{124}	"	For each input code N calculate the corresponding linearity error in LSB.	"	-0.5	+0.5	"
		128	"	"	0010 0000 0000	1101 "	"	"	E_{125}	"		"	"	"	"
		129	"	"	0011 0000 0000	1100 "	"	"	E_{126}	"		"	"	"	"
		130	"	"	0100 0000 0000	1011 "	"	"	E_{127}	"	$LE = 4.1 \left[(E_{119} - E_{120})(N/4095) + E_{120} - EA \right]$	"	"	"	"
		131	"	"	0101 0000 0000	1010 "	"	"	E_{128}	"		"	"	"	"
		132	"	"	0110 0000 0000	1001 "	"	"	E_{129}	"		"	"	"	"
		133	"	"	0111 0000 0000	1000 "	"	"	E_{130}	"	where	"	"	"	"
		134	"	"	1000 0000 0000	0111 "	"	"	E_{131}	"	EA = measured value E_{124} through E_{146}	"	"	"	"
		135	"	"	1001 0000 0000	0110 "	"	"	E_{132}	"	N = DUT input digital code word in decimal form.	"	"	"	"
		136	"	"	1010 0000 0000	0101 "	"	"	E_{133}	"	Thus for 12 bits N = (A1)(2 exp 11)+(A2)	"	"	"	"
		137	"	"	1011 0000 0000	0100 "	"	"	E_{134}	"	(2 exp 10)+ ... +(A12)(2 exp 0)	"	"	"	"
		138	"	"	1100 0000 0000	0011 "	"	"	E_{135}	"	A1, A2, A12 are 0 or 1.	"	"	"	"
		139	"	"	1101 0000 0000	0010 "	"	"	E_{136}	"	E_{119} and E_{120} correspond to the error amplifier	"	"	"	"
		140	"	"	1110 0000 0000	0001 "	"	"	E_{137}	"	outputs at full scale and zero scale respectively.	"	"	"	"
	LE_{15}	141	"	"	1111 0000 0000	0000 "	"	"	E_{138}	"		"	"	"	"
						A. Record the DUT code-word, (XXXX 0000 0000), with the most positive error. 2/						"	"	"	"
						B. Record the DUT code-word, (YYYY 0000 0000), with the most negative error. 2/						"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	22 thru 37	Energized relays	Measured pin No.	Value	Unit	Equation 2/	Device types	Limits Min	Max	Unit
4 T_A = 25°C	LE_{16}	142	+15 V	-15 V	0000 0000 0001	1111 1111 1110 1111	None	21	E_{139}	V		03, 04	-0.5	0.5	LSB
		143	"	"	0000 0000 0010	1111 1111 1101 1111	"	"	E_{140}	"		"	"	"	"
		144	"	"	0000 0000 0100	1111 1111 1011 1111	"	"	E_{141}	"		"	"	"	"
	LE_{19}	145	"	"	0000 0000 1000	1111 1111 0111 1111	"	"	E_{142}	"		"	"	"	"
	LE_{20}	146	"	"	0000 0001 0000	1111 1110 1111 1111	"	"	E_{143}	"		"	"	"	"
		147	"	"	0000 0010 0000	1111 1101 1111 1111	"	"	E_{144}	"		"	"	"	"
		148	"	"	0000 0100 0000	1111 1011 1111 1111	"	"	E_{145}	"		"	"	"	"
	LE_{23}	149	"	"	0000 1000 0000	1111 0111 1111 1111	"	"	E_{146}	"	$S = \left[\frac{E_{110} - E_{120}}{4095}\right]$	"	"	"	"
			"	"	A. Record, (0000 BBBB BBBB), the sum of the codewords from tests 142 - 149 with positive linearity errors.							"	"	"	"
			"	"	B. Record, (0000 CCCC CCCC), the sum of the codewords from tests 142 - 149 with negative linearity errors.							"	"	"	"
			"	"	Determine the sum of the recorded code-words obtained from tests 127 - 141 and from tests 142 - 149 for positive errors.							"	"	"	"
			"	"	Measure and calculate linearity error for the new DUT code-word (XXXX BBBB BBBB). (See Table I, Note 6/)							"	"	"	"
	LE_{24}	150	"	"	XXXX BBBB BBBB XXXX BBBB BBBB 1111		None	21	E_{147}	V	$LE_{24} = 4.1\left(E_{120} + \left((\text{on-bits}) \times S\right) - E_{147}\right)$	"	"	"	"
			"	"	Determine the sum of the recorded code-words obtained from tests 127 - 141 and from tests 142 - 149 for negative errors.							"	"	"	"
			"	"	Measure and calculate linearity error for the new DUT code-word (YYYY CCCC CCCC). (See Table I, Note 6/)							"	"	"	"
	LE_{25}	151	"	"	YYYY CCCC CCCC YYYY CCCC CCCC 1111		None	21	E_{148}	V	$LE_{25} = 4.1\left(E_{120} + \left((\text{on-bits}) \times S\right) - E_{148}\right)$	"	"	"	"
	MCE	152	"	"	0111 1111 1111	1000 0000 0000 1111	"	21	E_{149}	"	For each set of major carry codes N and N + 1	"	-1.0	1.0	"
			"	"	1000 0000 0000	0111 1111 1111 1111	"	"	E_{150}	"	calculate the associated error in LSB's as follows:	"	"	"	"
		153	"	"	0011 1111 1111	1100 0000 0000 1111	"	"	E_{151}	"		"	"	"	"
			"	"	0100 0000 0000	1011 1111 1111 1111	"	"	E_{152}	"	MCE = 4.1 (EA - EB)	"	"	"	"
		154	"	"	0001 1111 1111	1110 0000 0000 1111	"	"	E_{153}	"		"	"	"	"
			"	"	0010 0000 0000	1101 1111 1111 1111	"	"	E_{154}	"	where	"	"	"	"
		155	"	"	0000 1111 1111	1111 0000 0000 1111	"	"	E_{155}	"	EB = measured value from E_{149}, E_{151}, E_{153}, etc.	"	"	"	"
			"	"	0001 0000 0000	1110 1111 1111 1111	"	"	E_{156}	"		"	"	"	"
		156	"	"	0000 0111 1111	1111 1000 0000 1111	"	"	E_{157}	"	EA = measured value from E_{150}, E_{152}, E_{154}, etc.	"	"	"	"
			"	"	0000 1000 0000	1111 0111 1111 1111	"	"	E_{158}	"		"	"	"	"
		157	"	"	0000 0011 1111	1111 1100 0000 1111	"	"	E_{159}	"		"	"	"	"
			"	"	0000 0100 0000	1111 1011 1111 1111	"	"	E_{160}	"		"	"	"	"
		158	"	"	0000 0001 1111	1111 1110 0000 1111	"	"	E_{161}	"		"	"	"	"
			"	"	0000 0010 0000	1111 1101 1111 1111	"	"	E_{162}	"		"	"	"	"
		159	"	"	0000 0000 1111	1111 1111 0000 1111	"	"	E_{163}	"		"	"	"	"
			"	"	0000 0001 0000	1111 1110 1111 1111	"	"	E_{164}	"		"	"	"	"
		160	"	"	0000 0000 0111	1111 1111 1000 1111	"	"	E_{165}	"		"	"	"	"
			"	"	0000 0000 1000	1111 1111 0111 1111	"	"	E_{166}	"		"	"	"	"
		161	"	"	0000 0000 0011	1111 1111 1100 1111	"	"	E_{167}	"		"	"	"	"
			"	"	0000 0000 0100	1111 1111 1011 1111	"	"	E_{168}	"		"	"	"	"
		162	"	"	0000 0000 0001	1111 1111 1110 1111	"	"	E_{169}	"		"	"	"	"
			"	"	0000 0000 0010	1111 1111 1101 1111	"	"	E_{170}	"		"	"	"	"
		163	"	"	0000 0000 0000	1111 1111 1111 1111	"	"	E_{171}	"		"	"	"	"
			"	"	0000 0000 0001	1111 1111 1110 1111	"	"	E_{172}	"		"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/				Energized relays	Measured pin			Equation 2/		Device types	Limits		Unit
			4	6	7 thru 18	22 thru 37		No.	Value	Unit				Min	Max	
5 T_A = 125°C	I_{CC}	164	+15 V	-15 V	1111 1111 1111	1111 1111 1111 1111	None	4	I_{30}	mA	$I_{CC} = I_{30}$	(Digital 1 = 2.4 V)	03	1	5	mA
	I_{EE}	165	"	"	0000 0000 0000	1111 1111 1111 1111	"	6	I_{31}	mA	$I_{EE} = I_{31}$	(Digital 0 = 0.8 V)	03, 04	-18	-5	mA
	I_{IL}	166	"	"	0000 0000 0000	1111 1111 1111 1111	"	7	I_{32}	µA	$I_{IL} = I_{32}$	(Digital 0 = 0 V)	"	5	100	µA
		167	"	"	"	"	"	8	I_{33}	"	$I_{IL} = I_{33}$	"	"	"	"	"
		168	"	"	"	"	"	9	I_{34}	"	$I_{IL} = I_{34}$	"	"	"	"	"
		169	"	"	"	"	"	10	I_{35}	"	$I_{IL} = I_{35}$	"	"	"	"	"
		170	"	"	"	"	"	11	I_{36}	"	$I_{IL} = I_{36}$	"	"	"	"	"
		171	"	"	"	"	"	12	I_{37}	"	$I_{IL} = I_{37}$	"	"	"	"	"
		172	"	"	"	"	"	13	I_{38}	"	$I_{IL} = I_{38}$	"	"	"	"	"
		173	"	"	"	"	"	14	I_{39}	"	$I_{IL} = I_{39}$	"	"	"	"	"
		174	"	"	"	"	"	15	I_{40}	"	$I_{IL} = I_{40}$	"	"	"	"	"
		175	"	"	"	"	"	16	I_{41}	"	$I_{IL} = I_{41}$	"	"	"	"	"
		176	"	"	"	"	"	17	I_{42}	"	$I_{IL} = I_{42}$	"	"	"	"	"
		177	"	"	"	"	"	18	I_{43}	"	$I_{IL} = I_{43}$	"	"	"	"	"
	I_{IH}	178	"	"	1111 1111 1111	1111 1111 1111 1111	"	7	I_{44}	"	$I_{IH} = I_{44}$	(Digital 1 = 5.5 V)	"	145	300	"
		179	"	"	"	"	"	8	I_{45}	"	$I_{IH} = I_{45}$	"	"	"	"	"
		180	"	"	"	"	"	9	I_{46}	"	$I_{IH} = I_{46}$	"	"	"	"	"
		181	"	"	"	"	"	10	I_{47}	"	$I_{IH} = I_{47}$	"	"	"	"	"
		182	"	"	"	"	"	11	I_{48}	"	$I_{IH} = I_{48}$	"	"	"	"	"
		183	"	"	"	"	"	12	I_{49}	"	$I_{IH} = I_{49}$	"	"	"	"	"
		184	"	"	"	"	"	13	I_{50}	"	$I_{IH} = I_{50}$	"	"	"	"	"
		185	"	"	"	"	"	14	I_{51}	"	$I_{IH} = I_{51}$	"	"	"	"	"
		186	"	"	"	"	"	15	I_{52}	"	$I_{IH} = I_{52}$	"	"	"	"	"
		187	"	"	"	"	"	16	I_{53}	"	$I_{IH} = I_{53}$	"	"	"	"	"
		188	"	"	"	"	"	17	I_{54}	"	$I_{IH} = I_{54}$	"	"	"	"	"
		189	"	"	"	"	"	18	I_{55}	"	$I_{IH} = I_{55}$	"	"	"	"	"

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TABLE III. Group A inspection for device types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/				Energized relays	Measured pin			Equation 2/	Device types	Limits		Unit
			4	6	7 thru 18	22 thru 37		No.	Value	Unit			Min	Max	
5 T_A = 125°C	I_{FS}	190	+15 V	-15 V	1111 1111 1111	1111 1111 1111 1111	K_1	19	I_{56}	mA	$I_{FS} = I_{56}$	03, 04	-2.7	-1.6	mA
	$+P_{SS}$	191	+13.5 V	"	1111 1111 1111	0000 0000 0000 1111	None	21	E_{173}	V	$+P_{SS} = 0.41 (E_{174} - E_{173})$	03	-0.08	0.08	LSB/%PS
		192	+15 V	"	"	"	"	"	E_{174}	"		"	"	"	"
		193	+16.5 V	"	"	"	"	"	E_{175}	"	$+P_{SS} = 0.41 (E_{174} - E_{175})$	"	"	"	"
	$-P_{SS}$	194	+5 V	-13.5 V	"	"	"	"	E_{176}	"	$-P_{SS} = 0.41 (E_{177} - E_{176})$	03, 04	-0.2	0.2	"
		195	"	-15 V	"	"	"	"	E_{177}	"		"	"	"	"
		196	"	-16.5 V	"	"	"	"	E_{178}	"	$-P_{SS} = 0.41 (E_{177} - E_{178})$	"	"	"	"
6 T_A = -55°C	I_{CC}	197	+15 V	-15 V	1111 1111 1111	1111 1111 1111 1111	"	4	I_{57}	mA	$I_{CC} = I_{57}$ (Digital 1 = 2.4 V)	03	1	5	mA
	I_{EE}	198	"	"	0000 0000 0000	"	"	6	I_{58}	mA	$I_{EE} = I_{58}$ (Digital 0 = 0.8 V)	03, 04	-18	-5	mA
	I_{IL}	199	"	"	"	"	"	7	I_{59}	µA	$I_{IL} = I_{59}$ (Digital 0 = 0 V)	03, 04	5	100	µA
		200	"	"	"	"	"	8	I_{60}	"	$I_{IL} = I_{60}$ "	"	"	"	"
		201	"	"	"	"	"	9	I_{61}	"	$I_{IL} = I_{61}$ "	"	"	"	"
		202	"	"	"	"	"	10	I_{62}	"	$I_{IL} = I_{62}$ "	"	"	"	"
		203	"	"	"	"	"	11	I_{63}	"	$I_{IL} = I_{63}$ "	"	"	"	"
		204	"	"	"	"	"	12	I_{64}	"	$I_{IL} = I_{64}$ "	"	"	"	"
		205	"	"	"	"	"	13	I_{65}	"	$I_{IL} = I_{65}$ "	"	"	"	"
		206	"	"	"	"	"	14	I_{65}	"	$I_{IL} = I_{66}$ "	"	"	"	"
		207	"	"	"	"	"	15	I_{67}	"	$I_{IL} = I_{67}$ "	"	"	"	"
		208	"	"	"	"	"	16	I_{68}	"	$I_{IL} = I_{68}$ "	"	"	"	"
		209	"	"	"	"	"	17	I_{69}	"	$I_{IL} = I_{69}$ "	"	"	"	"
		210	"	"	"	"	"	18	I_{70}	"	$I_{IL} = I_{70}$ "	"	"	"	"
	I_{IH}	211	"	"	1111 1111 1111	"	"	7	I_{71}	"	$I_{IH} = I_{71}$ (Digital 1 = 5.5 V)	"	145	300	"
		212	"	"	"	"	"	8	I_{72}	"	$I_{IH} = I_{72}$ "	"	"	"	"
		213	"	"	"	"	"	9	I_{73}	"	$I_{IH} = I_{73}$ "	"	"	"	"
		214	"	"	"	"	"	10	I_{74}	"	$I_{IH} = I_{74}$ "	"	"	"	"
		215	"	"	"	"	"	11	I_{75}	"	$I_{IH} = I_{75}$ "	"	"	"	"
		216	"	"	"	"	"	12	I_{76}	"	$I_{IH} = I_{76}$ "	"	"	"	"
		217	"	"	"	"	"	13	I_{77}	"	$I_{IH} = I_{77}$ "	"	"	"	"
		218	"	"	"	"	"	14	I_{78}	"	$I_{IH} = I_{78}$ "	"	"	"	"
		219	"	"	"	"	"	15	I_{79}	"	$I_{IH} = I_{79}$ "	"	"	"	"
		220	"	"	"	"	"	16	I_{80}	"	$I_{IH} = I_{80}$ "	"	"	"	"
		221	"	"	"	"	"	17	I_{81}	"	$I_{IH} = I_{81}$ "	"	"	"	"
		222	"	"	"	"	"	18	I_{82}	"	$I_{IH} = I_{82}$ "	"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection/ vice types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	22 thru 37	Energized relays	Measured pin No.	Value	Unit	Equation 2/	Device types	Limits Min	Max	Unit
6 T_A = -55°C	I_{FS}	223	+15 V	-15 V	1111 1111 1111	1111 1111 1111 1111	K_1	19	Y_{83}	mA	$I_{FS} = I_{83}$	03, 04	-2.7	-1.6	mA
	$+P_{SS}$	224 225 226	+13.5 V +15 V +16.5 V	" " "	" " "	0000 0000 0000 1111 " "	None " "	21 " "	E_{179} E_{180} E_{181}	V " "	$+P_{SS} = 0.41\ (E_{180} - E_{179})$ $+P_{SS} = 0.41\ (E_{180} - E_{179})$	03 "	-0.08 "	0.08 "	LSB/%PS "
	$-P_{SS}$	227 228 229	+5 V " "	-13.5 V -15 V -16.5 V	" " "	" " "	" " "	" " "	E_{182} E_{183} E_{184}	" " "	$-P_{SS} = 0.41\ (E_{183} - E_{182})$ $-P_{SS} = 0.41\ (E_{183} - E_{184})$	03, 04 "	-0.2 "	0.2 "	" "
7 T_A = 25°C 8	LE		+15 V " " " " " " " "	-15 V " " " " " " " "	0000 0000 0001 0000 0000 0010 0000 0000 0011 1111 1111 1100 1111 1111 1101 1111 1111 1110 This subgroup checks all digital input codes, except zero code (VOS) and fullscale code (VFS) which were determined in Test No. 123.	1111 1111 1110 1111 1111 1111 1101 1111 1111 1111 1100 1111 0000 0000 0011 1111 0000 0000 0010 1111 0000 0000 0001 1111	" " " " " " " " "	" " " " " " " " "	E_{185} E_{186} E_{187} $E_{...}$ $E_{...}$ E_{4278} E_{4279} E_{4280}	" " " " " " " " "	For each input code N calculate the corresponding linearity error in LSB. $LE = 4.1\left((E_{119} - E_{120})(N/4095) + E_{120} - EA\right)$ where EA = measured value E_{185} through E_{4280} N = DUT input digital code word in decimal form. Thus for 12 bits N = (A1)(2 exp 11)+(A2)(2 exp 10)+ ... +(A12)(2 exp 0) A1, A2, ... A12 are 0 or 1. E_{119} and E_{120} correspond to the error amplifier outputs at full scale and zero scale respectively.	" "			
		230									Worst positive error Worst positive code	" "	0 -	.5 -	LSB OCT
		231									Worst negative error Worst negative code	" "	-.5 -	0 -	LSB OCT
8 T_A = 125°C	LE	232 233	Same tests, terminal conditions and limits as in subgroup 7, tests 232 and 233 except T_A = 125°C. V_{OS} and V_{FS} are the values determined in subgroup 2.									" " " "	0 - -.5 -	.5 - 0 -	LSB OCT LSB OCT
	dI_{ZS}/dt	234	Calculate dI_{ZS}/dt = (I_{ZS} at 125°C - I_{ZS} at 25°C)/100°C from tests 37 and 122 respectively.									"	-0.01	0.01	LSB/°C
	dA_E/dt	235	Calculate dA_E/dt = (A_E at 125°C - A_E at 25°C)/100 from tests 38 and 123 respectively.									03 04	-30 -10	30 10	PPM/°C PPM/°C
	dB_{POE}/dt	236	Calculate dB_{POE}/dt = (B_{POE} at 125°C - B_{POE} at 25°C)/100°C from tests 39 and 124 respectively.									03 04	-0.12 -0.04	0.12 0.04	LSB/°C LSB/°C
	dB_{ZE}/dt	237	Calculate dB_{ZE}/dt = (B_{ZE} at 125°C - B_{ZE} at 25°C)/100°C from tests 41 and 126 respectively.									03, 04	-0.04	0.04	LSB/°C

See footnotes at end of table.

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TABLE III. Group A inspection for device types 03 and 04 - Continued.

Subgroup	Symbol	Test no.	Adapter pin numbers 1/ 4	6	7 thru 18	22 thru 37	Energized relays	Measured pin No.	Value	Unit	Equation 2/	Device types	Limits Min	Max	Unit
8 T_A = -55°C	LE	238 239	Same tests, terminal conditions and limits as in subgroup 7, tests 238 and 239 except T_A = -55°C. V_{OS} and V_{FS} are the values determined in subgroup 3.									03, 04 " "	0 - -.5 -	0.5 - 0 -	LSB OCT LSB OCT
	dI_{ZS}/dt	240	Calculate dI_{ZS}/dt = (I_{ZS} at -55°C - I_{ZS} at +25°C)/80°C from tests 80 and 122 respectively.									"	-0.01	0.01	LSB/°C
	d_{AE}/dt	241	Calculate d_{AE}/dt = (A_E at -55°C - A_E at +25°C)/125 from tests 81 and 123 respectively.									03 04	-30 -10	30 10	PPM/°C PPM/°C
	dB_{POE}/dt	242	Calculate dB_{POE}/dt = (B_{POE} at -55°C - B_{POE} at +25°C)/80°C from tests 82 and 124 respectively.									03 04	-0.12 -0.04	0.12 0.04	LSB/°C LSB/°C
	d_{BZE}/dt	243	Calculate d_{BZE}/dt = (B_{ZE} at -55°C - B_{ZE} at 25°C)/80°C from tests 84 and 126 respectively.									03, 04	-0.04	0.04	LSB/°C
12 T_A = 25°C	t_{SLH}	244	All digital inputs switched simultaneously from logic low to logic high. For each device type output settling is to within 1/2 LSB of the rated accuracy. See figure 6.				None	6	t_1		$t_{SLH} = t_1$	03 04	- -	400 600	ns "
	t_{SHL}	245	All digital inputs switched simultaneously from logic high to logic low. For each device type output settling is to within 1/2 LSB of the rated accuracy. See figure 6.				"	6	t_2		$t_{SHL} = t_2$	03 04	- -	400 600	" "
	Rf	256	+15 V	-15 V	See figure 8.		"	6 5	I_{84} I_{85}	mA	Rf = 10/I_{84} V_{REF} = +10 V Rf = 10/I_{85}	03 04	15 15	25 25	Kohms Kohms
	No	257	+15 V	-15 V	All bits off All bits on See figure 9.		" "	9 9	e1 e2	mVrms	No = Sqr(e1 exp2-e2 exp2)	03 04	- -	80 60	mVrms mVrms

1/ +15 V is not required for device type 04.

2/ The equations take into account the error amplifier gain of 100 V/V and other circuit contacts so that the calculated value is in Table I units.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and method 5005 of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) For class S devices: Test condition D using the circuit shown on figure 3.

(2) For class B devices: Test condition D using the circuit shown on figure 3, or test condition C or F using the circuit shown on figure 4.

NOTE: If accelerated high-temperture test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). For class S devices only using the circuit shown on figure 4.

c. Interim electrical parameters shall be performed for class S and B devices and shall be as specified in table II herein.

d. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II, herein.

b. Subgroups 9, 10, and 11 shall be omitted.

c. A special subgroup shall be added to group A inspection for class S only and it shall consist of the conditions and limits as specified in table III, subgroup 12. LTPD shall be 15.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups and LTPD values shown in table IIb of method 5005 of MIL-STD-883, class B, and as follows:

a. Electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 5 of group B inspection.

b. Steady-state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 3. If the alternate burn-in conditions are used, the circuit shown on figure 4 shall be used.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II, herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

b. A special subgroup shall be added to the group C inpection requirements for class B only and shall consist of the tests, conditions, and limits specified for subgroup 12 of group A. The LTPD for the special subgroup shall be 15 for all classes.

c. Life test (method 1005 of MIL-STD-883) for class B devices; test condition D using the circuit shown on figure 3, or test condition F using the circuit shown on figure 5 (see 4.2a.2 note).

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given, except the input offset voltage (or differential voltage) are referenced to the external zero reference level of the supply voltage. Current values given are for conventional current and are positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to an ambient temperature of 25°C ±3°C, and the electrical parameter end-point measurements shall then be performed.

TABLE IV. Group C end-point electrical parameters (T_A = 25°C).

Device type	Test no.	Symbol	Limits Min	Limits Max	Delta	Unit
01, 02, 03, 04	130	I_{ZS1}	-2	2	±1	LSB
01, 03, 04	132	A_{E1}	-0.20	0.20	±0.05	%
02			-0.16	1.16	±0.04	%
01, 03, 04	128	B_{POE}	-8	8	±2	LSB
02			-6.5	6.5	±2	LSB
01, 02, 03, 04	129	B_{ZE}	-6.5 -6.0	6.5 6.0	±2 ±2	LSB LSB

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product to the contracting activity, in addition to notification to qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance provisions.

g. Requirement for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

Bipolar mode:	The D/A converter operation mode that provides both positive and negative output voltage or currents in response to an offset binary input code.
Bipolar gain:	The ratio of the output voltage change to the input code change in the bipolar mode. In this specification, it is measured as the output voltage change between the input codes of zero and full-scale.
Bipolar gain error (B_{PAE}):	The difference between the actual static gain and the ideal static gain in the bipolar mode. It is measured between zero and full-scale and therefore expressed as a percent of full-scale range.
Bipolar offset error (B_{POE}):	The difference between the actual and the ideal output voltage with the device operating in the bipolar mode and the digital inputs at zero. It is expressed in LSB units.
Bipolar offset drift (dB_{POE}/dT):	The change of the bipolar offset with temperature with the device operating in the bipolar voltage output mode. It is expressed in LSB/°C.

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Bipolar zero error (B_{ZE}):	The difference between the actual and the ideal (0 volts) output with the device operating in the bipolar mode and with the input digital code at 1000 0000 0000. It is expressed in LSB units.
Bipolar zero drift (dB_{ZE}/dT):	The change of the bipolar zero error with temperature with the device operating in the bipolar voltage output mode. It is expressed in LSB/°C.
Differential linearity error:	The difference between the actual separation of two adjacent output levels and the ideal separation of one LSB. It is expressed in LSB.
Full-scale current (I_{FS}):	The total current output of the device with all digital inputs at logic 1. It is expressed in mA.
Full-scale output voltage range (V_{FSR}):	The output voltage change between the input codes of all zeros to all ones plus the voltage equivalent to one additional LSB.
Gain:	The ratio of the output voltage change to an input code change with the device operating in the unipolar mode. In this specification it is measured between the codes of zero to full-scale.
Gain drift:	The change of gain with temperature. It is expressed in PPM/°C.
Gain error:	The difference between the actual static gain and the ideal static gain. It is expressed as a percent of the ideal.
Least significant bit (LSB):	That bit of a digital code word that causes the least change to the output level. When the LSB changes state, the ideal output level changes by the value of the full-scale range divided by 4096.
Major carry (MC):	A major carry (transition) is the output level change corresponding to the input transition between the digital code word defined by turning one bit on and all other bits off and the preceding adjacent digital code word. There are "N" such transitions for an N-bit converter.
Major carry error (MCE):	The differential linearity error between the output levels corresponding to the digital code word defined by turning one bit on and all other bits off and the preceding adjacent digital code word. It is expressed in LSB.
Monotonicity:	A DAC transfer function is monotonic if the ratio of the incremental change in the input does not change sign over the entire full-scale output range.
Most significant bit (MSB):	That bit of the digital code word that causes the most change to the outut level. When the MSB changes state, the output level changes by one-half of the full-scale output range.
Offset:	The output voltage for an input code of all zeros.

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Offset binary code:	The natural binary code representing the full-scale range but offset by one-half of the full-scale range. Offset binary code represents analog values between minus full-scale at all zeros, to plus full-scale for all ones, with the MSB turned on at analog zero.
Output compliance voltage range:	The range of output voltages over which a current output DAC can operate within a specified error band.
Power supply sensitivity at full-scale (P_{SS}):	The change in the full-scale output level with a change in the power supply voltage. It is generally expressed in mV/%.
Resolution:	The smallest ideal increment between two adjacent output levels over the full-scale range of the converter.
Settling time:	The elapsed time from the application of a full-scale step input to the DAC to the time when the output has entered and remained within a 1/2 LSB error band around its final value.
End-point linearity error:	The difference between the actual output level and the ideal output level for the same digital code word input after any rotation or translation to make the actual and the ideal zero and full-scale end-points coincide.
Unipolar mode:	The converter operation mode that includes output voltages or currents of one polarity only.
Zero scale current (I_{ZS}):	The output current with the device operating in the unipolar current output mode and with the digital inputs at zero. It is expressed in LSB.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military-device type	Generic-industry type
01	562
02	563
03	565
04	566

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  Army - ER
  Navy - EC
  Air Force - 17
  NASA - NA

Review activities:
  Army - AR, MI
  Navy - OS, SH
  Air Force - 11, 19, 80, 85, 99
  DLA - ES

User activities:
  Army - SM
  Navy - AS, CG, MC

Agent:
  DLA - ES

Preparing activity:
  Air Force - 17

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